p-ADIC HEIGHTS ON SHIMURA CURVES AND HIGHER p-ADIC GREEN FUNCTIONS

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ABSTRACT. Let X be a Shimura curve with Cerednik–Drinfeld uniformization at a prime p and let $\mathcal{M}_n$ be the motive attached to modular forms of even weight $n+2$. We study Nekovář's local p-adic height of Heegner cycles on $\mathcal{M}_n$. The central result expresses this height as a Coleman integral of a canonical vector valued meromorphic differential on the p-adic upper half plane. We then construct this differential by means of Poincaré series and obtain an explicit formula in terms of p-adic logarithms. The final section proves that sum of higher p-adic Green functions over all the Heegner points is equal to the log of a rational number.

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1. INTRODUCTION

The aim of this paper is to relate the local p-adic height of Heegner cycles on Shimura curves to higher p-adic Green functions. The point of view is both arithmetic and analytic. On the arithmetic side, Heegner cycles define extensions of semistable Galois representations and therefore biextensions in the category of filtered (φ, N)-modules. On the analytic side, the Čerednik–Drinfeld uniformization identifies the Shimura curve at p with a quotient of the p-adic upper half plane. This permits us to describe the relevant filtered modules, their monodromy, and their Poincaré pairing in terms of harmonic cocycles and Coleman primitives.

Let $N = pN^-N^+$, where N^- is squarefree with an odd number of prime divisors and N^+ is relatively prime to pN^-. Let X be the Shimura curve attached to the indefinite quaternion algebra of discriminant pN^-. We fix an even integer $n = 2m$ and consider the motive $\mathcal{M}_n$ over X which realizes the local system of modular forms of weight $n+2$. If K is an imaginary quadratic field satisfying the Heegner hypothesis of Section 3.8, CM points on X give cycles in $\mathrm{CH}^{m+1}(\mathcal{M}_n)$. Their p-adic Abel–Jacobi images determine extensions of the Galois representation

$$V = H^1_{\text{ét}}(X_M \otimes \overline{\mathbf{Q}}, \mathbb{L}_n)^G.$$

Pairing two such extensions gives the biextension whose local height we compute.

The construction belongs to the framework introduced by Nekovář in [Nek93, Nek95]. Our geometric input is the description of the motive, the semistable realization, and the monodromy module developed by Iovita and Spieß in [IS03]. The relation between Nekovář's height and

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integration on curves with semistable reduction is close in spirit to the work of Besser and of Besser and Zerbes [Bes22, BZ23]. The analytic construction is motivated by the treatment of higher Green functions by Brown and Fonseca [BF25], while the geometry of Green functions on Mumford curves is developed by Wei in [Wei18].

Relation with previous work. The seminal work of Gross–Zagier on the BSD conjecture [GZ86] and the following work [GKZ87] have shed light on interesting connections between automorphic functions, intersection theory and modular objects in general. The higher Green functions $G_k(z, w)$ are one of those cornerstone objects in the proof and subsequent works since they connect the hyperbolic Laplace operator to intersection theory of Heegner cycles. The value of these functions at CM points has been conjectured, in the latter paper, to be equal to the logarithm of an algebraic number. This has sparked research in several directions and has now recently been fully resolved in many instances by fairly different methods. The nature of the object has indeed attracted researchers both from a more analytic side [BEY21], [Via11] and [Li23] via the theory of Borcherds lifts and analytic Maass forms and from a geometric one [Zha97], [Mel08] and [BF25] using a more motivic approach. Our paper has been inspired by this latter point of view; it gave us the idea of bridging the p-adic heights of Heegner cycles with explicit analytic description of the higher Green functions using cohomological tools.

The local formula belongs first to the theory of Coleman and Gross [CG89]. Their height on a curve is described by intersection theory away from p and by Coleman integration of normalized differentials of the third kind above p. Nekovář recast p-adic heights in terms of mixed extensions of Galois representations and applied this formalism to Heegner cycles [Nek93, Nek95]. Besser proved that the Coleman–Gross and Nekovář constructions agree for curves [Bes04]; his later work extends this comparison to semistable reduction by means of Vologodsky integration [Bes22]. Our construction uses Nekovář's mixed extension formalism, but the resulting local formula has the shape of a Coleman–Gross integral with coefficients in $\mathcal{V}_n$.

The closest higher weight antecedent is Nekovář's formula for Heegner cycles [Nek95]. In that work p does not divide the level and the modular form is ordinary at p, so the local height is defined using the ordinary splitting. Here p divides the level exactly once, the local representation is Steinberg, and the Shimura curve is described by Čerednik–Drinfeld uniformization. The purpose of Sections 4 through 6 is to replace the ordinary calculation at p by a semistable calculation in terms of Frobenius, monodromy, Bruhat–Tits cocycles, and Coleman primitives. The monodromy module calculations of Iovita and Spieß and their extension by Masdeu provide the essential precedent for this passage [IS03, Mas11].

The formulas of Perrin–Riou, Nekovář, and Shnidman relate p-adic heights of Heegner points or generalized Heegner cycles to derivatives of p-adic L-functions [PR87, Nek95, Shn16]. Disegni places this circle of ideas on Shimura curves with potentially ordinary, including semistable, reduction and subsequently treats primes which are nonsplit in the CM field [Dis17, Dis23]. The aim here is different and complementary. We compute the local height at the Steinberg prime directly and identify it with a higher p-adic Green function. We do not construct a p-adic L-function or prove a height formula for its derivative. In the prospective global statement of Section 7, the relevant height is the cyclotomic component. The Hecke operator is expected to annihilate the middle representation and isolate a Kummer extension, leading to an algebraicity statement rather than to a p-adic Gross–Zagier formula.

On the analytic side, Bertolini, Darmon, Iovita, and Spieß relate the Čerednik–Drinfeld uniformization and boundary distributions to Coleman integration [BDIS02], while Iovita and Spieß use monodromy modules to study Heegner cycles [IS03]. Brown and Fonseca interpret higher Green functions through motivic biextensions in the complex setting [BF25], and Wei develops Green functions on Mumford curves in the p-adic setting [Wei18]. The contribution sought here is the direct bridge between these two aspects: the mixed extension attached to higher weight Heegner cycles is evaluated by an explicit vector valued Poincaré series on the p-adic upper half plane.

The new feature needed here is the simultaneous treatment of nontrivial coefficients, semistable reduction at p, and relative cohomology for complements of CM divisors. Let D_1 and D_2 be disjoint effective divisors and let Q_i be the residue data determined by the corresponding Heegner cycles.

The relative de Rham groups

$$H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) \quad \text{and} \quad H^1_{\mathrm{dR}}(X_\Gamma - D_2, D_1, \mathcal{V}_n)$$

carry compatible Frobenius, monodromy, filtration, and Poincaré pairings. The slope decomposition determines a canonical complement to the relevant Hodge filtration. This is the complement used in Nekovář's construction of the local height.

For residue data Q we denote by Ψ_Q the unique class with prescribed residue, lying in the top filtered piece and having trivial monodromy. We call it the Coleman–Gross differential associated to Q. The main result of the paper is the following formula.

Theorem A (Local height formula)**.** *Let D_1 and D_2 be disjoint reduced divisors defined over a field H and let Q_1 and Q_2 be residue classes defining the corresponding Heegner extension e_1, e_2 and a biextension $E = E(Q_1, Q_2)$. Assume that the points in D_1 reduce to one irreducible component and that*

$$\sum_{z \in D_1} Q_{1,z}/\#\Gamma_z = 0.$$

Then the local height at $v|p$ is given by

$$h(e_1, e_2)_{E,v} = \mathrm{Tr}_{H_v/\mathbb{Q}_p}\left(\int_{D_1} (\Psi_{Q_2}, Q_1)\right).$$

Equivalently, after choosing lifts to the p-adic upper half plane and a Coleman primitive $F_{\Psi_{Q_2}}$, the right hand side is the sum of the values

$$\mathrm{Tr}_{H_v/\mathbb{Q}_p}\left(\sum_{z \in D_1} \frac{(F_{\Psi_{Q_2}}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}\right).$$

The proof identifies the semistable realization of the biextension, computes its Bloch–Kato class, and compares the Frobenius and monodromy splitting with the splitting defined by the Hodge filtration. Poincaré duality then reduces the height to the displayed Coleman integral. This is carried out in Section 5.

The second result gives an analytic realization of the differential in Theorem A. For two points w_1, w_2 on the p-adic upper half plane and $i + j = n$, consider the Poincaré series

$$\Psi^{i,j}_\Gamma(z; w_1, w_2) = \sum_{\gamma \in \Gamma} \frac{1}{j(\gamma, z)^{n+2}} \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}}.$$

We prove that this series converges to a rigid meromorphic modular form. The associated vector valued differential has the prescribed residues and trivial monodromy. It is therefore the Coleman–Gross differential entering the height formula. Its Coleman primitive can be written in terms of derivatives of

$$\frac{(z - w_1)^j (z - w_2)^i}{(w_1 - w_2)^n} \log_{\mathscr{L}}\left(\frac{z - w_1}{z - w_2}\right).$$

As a consequence, the local height is expressed by a convergent Poincaré sum of explicit p-adic logarithmic terms. This is the *higher p-adic Green function* studied in Section 6.

The global conclusion is a rationality statement in the spirit of Gross, Kohnen, and Zagier [GKZ87]. If a finite combination of Hecke operators annihilates the middle Galois representation V, the Heegner biextension is reduced to a Kummer extension. The sum of all the local heights above p is consequently the p-adic logarithm of a rational number. This is explained and proven in Section 7.

Organization of the paper. Section 2 recalls semistable representations and proves the extension formulas used later. Section 3.1 introduces the Shimura curves and their Čerednik–Drinfeld uniformization. Sections 3.2 through 3.8 describe the motive $\mathcal{M}_n$, its p-adic and de Rham realizations, and the Heegner cycles. Section 4 studies relative de Rham cohomology through Bruhat–Tits cocycles. Section 5 specializes Nekovář's height machinery and proves Theorem A. Section 6 constructs the Poincaré series and derives the analytic formula. Section 7 proves that the sum of the Green functions over Heegner points is the p-adic logarithm of a rational number.

2. Filtered (φ, N)-modules and Galois representations

Before delving into the p-adic aspects of Shimura curves and their cohomology, we will start recalling some basic facts about p-adic Galois representations and the associated category of abstract linear algebra objects introduced by [Fon79]. This will help us set up the notation and already establish the categories where the realizations of certain motives will live. We will follow the exposition of [Nek93].

2.1. **Filtered (φ, N)-modules.** Let F be a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}_F$ and residue field k. Denote by $F_0 \subseteq F$ the maximal unramified subextension that can be identified with the fraction field of the ring of Witt vectors $W(k)$. Let $\sigma : F_0 \to F_0$ the absolute Frobenius automorphism. Write G_F for the absolute Galois group $G_F = \mathrm{Gal}(\bar{F}/F)$.

Definition 2.1. *Define the category* $\mathrm{MF}_F(\varphi, N)$ *of* filtered (φ, N)-modules *over F to be the category of finite dimensional F_0-vector spaces D equipped with:*

- *a σ-semilinear bijective map $\varphi : D \to D$ called* Frobenius,
- *a F_0-linear map $N : D \to D$ satisfying $N\varphi = p\varphi N$ called* monodromy operator,
- *a decreasing* filtration $\mathrm{Fil}^i D_F$ *on the vector space $D_F = D \otimes_{F_0} F$ satisfying $\bigcup \mathrm{Fil}^i D_F = D_F$ and $\bigcap \mathrm{Fil}^i D_F = 0$.*

The morphisms are defined to be compatible with φ, N and $\mathrm{Fil}^\bullet$.

The category $\mathrm{MF}_F(\varphi, N)$ is an additive tensor category admitting kernels and cokernels. Given a filtered (φ, N)-module $D = (D, \mathrm{Fil}^\bullet_D, \varphi_D, N_D)$ we can define its j-th *Tate twist* to be the object

$$D[j] = (D, \mathrm{Fil}^{\bullet - j}_D, p^j \varphi_D, N_D). \tag{2.1}$$

Consider D a filtered (φ, N)-module. For a rational number $\lambda = n/m$ with $n, m \in \mathbb{Z}$ and $(n, m) = 1$, denote D_λ the largest subspace of D which has an $\mathcal{O}_{F_0}$-stable lattice M satisfying

$$\varphi^m(M) = p^n(M). \tag{2.2}$$

The subspace D_λ is called *isotypical component* of D of *slope* λ. By definition of the monodromy operator we have $N(D_\lambda) \subseteq D_{\lambda - 1}$ for all $\lambda \in \mathbb{Q}$.

A *p-adic representation* V of G_F is a finite dimensional vector space over $\mathbb{Q}_p$ endowed with a continuous linear action of G_F. We denote by $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F)$ the category of such representations with obvious morphisms. It is an abelian tensor category with twists given by tensoring with Tate representation $\mathbb{Q}_p(1)$.

Fontaine [Fon94] constructed topological rings endowed with continuous G_F action

$$B_{\mathrm{cris}} \subset B_{\mathrm{st}} \subset B_{\mathrm{dR}} \tag{2.3}$$

and certain extra structure. The embedding $B_{\mathrm{st}} \hookrightarrow B_{dR}$ depends on the choice of a branch of a p-adic logarithm.

Consider the following functors

$$\begin{aligned} D_{\mathrm{cris}} : \underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F) &\to \mathrm{MF}_F(\varphi, N) \\ V &\mapsto D_{\mathrm{cris}}(V) := (V \otimes B_{\mathrm{cris}})^{G_F}, \end{aligned} \tag{2.4}$$

$$\begin{aligned} D_{\mathrm{st}} : \underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F) &\to \mathrm{MF}_F(\varphi, N) \\ V &\mapsto D_{\mathrm{st}}(V) := (V \otimes B_{\mathrm{st}})^{G_F}, \end{aligned} \tag{2.5}$$

$$\begin{aligned} \mathrm{DR} : \underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F) &\to \mathrm{FilVec}_F \\ V &\mapsto D_{\mathrm{dR}}(V) := (V \otimes B_{\mathrm{dR}})^{G_F} \end{aligned} \tag{2.6}$$

where FilVec_F denotes the category of *filtered F-vector spaces*.

Definition 2.2. *A p-adic representation V is called*

- crystalline *if* $\dim_{F_0} D_{\mathrm{cris}}(V) = \dim_{\mathbb{Q}_p}(V)$,

- semistable *if* $\dim_{F_0} D_{\mathrm{st}}(V) = \dim_{\mathbb{Q}_p}(V)$,
- de Rham *if* $\dim_F \mathrm{DR}(V) = \dim_{\mathbb{Q}_p}(V)$,

The category of crystalline (resp. semistable, resp. de Rham) representations, denoted $\underline{\mathrm{Rep}}_{\mathrm{cris}}(G_F)$ *(resp.* $\underline{\mathrm{Rep}}_{\mathrm{st}}(G_F)$*, resp.* $\underline{\mathrm{Rep}}_{\mathrm{dR}}(G_F)$*) is the full subcategory of* $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F)$ *of crystalline (resp. semistable, resp. de Rham) objects.*

The Tate twist $F_0[n]$ is given by

$$F_0[n] := D_{\mathrm{cris}}(\mathbb{Q}_p(-n)) = D_{\mathrm{st}}(\mathbb{Q}_p(-n)) = (F_0, \mathrm{Fil}^n, p^n\sigma, 0) \tag{2.7}$$

with the filtration concentrated in degree $[n]$. The de Rham realization consists in the filtered vector space $\mathrm{DR}(\mathbb{Q}_p(n)) = (F, \mathrm{Fil}^{\bullet -n})$. The effect of tensoring by the Tate twist in $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(G_F)$ corresponds to taking the tensor product by (2.7) in the category of filtered (φ, N)-modules. In particular, according to our notation, for every p-adic representation V we have the following relation:

$$D_{\mathrm{st}}(V(n)) = D_{\mathrm{st}}(V)[-n]. \tag{2.8}$$

Definition 2.3. *A filtered (φ, N)-module D is called* admissible *if is isomorphic to $D_{\mathrm{st}}(V)$ for some semistable representation V of G_F. The full subcategory of admissible filtered (φ, N)-modules is denoted* $\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)$.

Theorem 2.1 ([CF00])**.** *The functor D_{st} gives an equivalence of categories between* $\underline{\mathrm{Rep}}_{\mathrm{st}}(G_F)$ *and* $\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)$*, which is compatible with exact sequences, tensor product and duality.*

This theorem will be fundamental in the construction of the p-adic height of pairing à la Nekovář since we can establish the following group isomorphism.

Corollary 2.1. *The equivalence of categories induces a canonical group isomorphism*

$$\mathrm{Ext}^1_{\underline{\mathrm{Rep}}_{\mathrm{st}}(G_F)}(V, W) \cong \mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(D_{\mathrm{st}}(V), D_{\mathrm{st}}(W)) \tag{2.9}$$

for every V, W semistable p-adic representations of G_F.

2.2. **Extension of p-adic Galois representations.** One of the advantage of working with the linear algebraic category of filtered (φ, N)-modules is given by the characterization and explicit description that we can give of the extension group $\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}$.

Lemma 2.1. *Let D be an admissible filtered (φ, N)-module and assume N induces a bijection $N(D_0) = D_{-1}$. Then there is a canonical isomorphism*

$$\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(F_0, D) \cong D_F/\mathrm{Fil}^0 D_F \tag{2.10}$$

where F_0 denotes the filtered (φ, N)-module $D_{\mathrm{st}}(\mathbb{Q}_p) = (F_0, \mathrm{Fil}^0, \mathrm{Id}, 0)$. The map is often referred to as the Bloch-Kato isomorphism.

Proof. See [IS03, Lemma 2.1], [Mas11, Lemma 4.6]. Note the difference by a Tate shift $[n+1]$. □

It will be fundamental for our later computations to have a precise and explicit description of the isomorphism. We here recall the construction contained in the proof. Consider

$$e : 0 \to D \xrightarrow{\iota} E \xrightarrow{p} F_0 \to 0 \tag{2.11}$$

an extension of filtered (φ, N)-modules whose class belong to $\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(F_0, D)$ and satisfies the hypothesis of the Lemma. We have a decomposition of $E_F \cong D_F \oplus F$ compatible with filtration

$$\mathrm{Fil}^i E_F \cong \begin{cases} \mathrm{Fil}^i D_F \oplus F & \text{for } i \leq 0, \\ \mathrm{Fil}^i D_F & \text{for } i > 0 \end{cases} \tag{2.12}$$

and there exists a splitting of p compatible with *Frobenius and monodromy*

$$s : F_0 \to E_0 \tag{2.13}$$

where E_0 is the slope-0 part. We then define $\pi_{D_F}(s(1)) + \mathrm{Fil}^0 D_F \in D_F/\mathrm{Fil}^0 D_F$ the element associated to the class $[e]$ of (2.11), where π_{D_F} denotes the projection to D_F under the decomposition $E_F \cong D_F \oplus F$.

Remark 2.1. *Lemma 2.1 fundamentally states that the class of a simple extension $[e]$ is characterized by the comparison between the Frobenius-monodromy data and the filtration of the vector spaces.*

We will also need to characterize the extension groups for other filtered (φ, N)-modules not necessarily satisfying the hypothesis of Lemma 2.1. Consider the following exact sequence of admissible (φ, N)-modules

$$0 \to F_0[-1] \to E \to V \to 0 \tag{2.14}$$

where V satisfies the hypothesis above and $F_0[-1]$ denotes the filtered (φ, N)-module $D_{\mathrm{st}}(\mathbb{Q}_p(1))$. Assume furthermore that V satisfies the following property

$$V^{\varphi=1, N=0} = (V^\vee[-1])^{\varphi=1, N=0} = 0. \tag{2.15}$$

Lemma 2.2. *Let E be an admissible filtered (φ, N)-module defined by* (2.14)*. Then there is a canonical isomorphism*

$$\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(F_0, E) \cong E_F/\mathrm{Fil}^0 E_F \oplus \mathbb{Q}_p \tag{2.16}$$

where F denotes the filtered (φ, N)-module $D_{\mathrm{st}}(\mathbb{Q}_p)$ as before. The map is often referred to as the Bloch-Kato isomorphism.

Proof. We consider the complex $C^\bullet_{\mathrm{st}}(E)$ defined by

$$C^\bullet_{\mathrm{st}}(E) = [E \xrightarrow{(-i, \varphi - \mathrm{Id}, N)} E_F/\mathrm{Fil}^0 E_F \oplus E \oplus E[1] \xrightarrow{(0, N, \mathrm{Id} - \varphi)} E[1]]. \tag{2.17}$$

In [Nek93, Sec. 1.19], we have an isomorphism

$$\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(F_0, E) \cong H^1(C^\bullet_{\mathrm{st}}(E)). \tag{2.18}$$

Explicitly, let

$$0 \to E \to D \to F_0 \to 0 \tag{2.19}$$

be an extension of admissible (φ, N)-modules whose class belongs to $\mathrm{Ext}^1_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi,N)}(F_0, E)$. Then we have a commutative diagram of the form

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E_F & \longrightarrow & D_F & \longrightarrow & F & \longrightarrow & 0 \\ & & \downarrow & & \downarrow \pi & & \downarrow & & \\ 0 & \longrightarrow & E_F/\mathrm{Fil}^0 E_F & \xrightarrow{\sim} & D_F/\mathrm{Fil}^0 D_F & \longrightarrow & 0 & \longrightarrow & 0. \end{array} \tag{2.20}$$

Pick a F_0-section $s : F_0 \to D$ and define a triple

$$[-\pi \circ i(s(1), (\varphi - \mathrm{Id})s(1), Ns(1))] \in E_F/\mathrm{Fil}^0 E_F \oplus E \oplus E[1] \tag{2.21}$$

where π is the vertical map from diagram (2.20) and the cohomology class $\rho(D) \in H^1(C^\bullet_{\mathrm{st}}(E))$ is independent of the choice of the section s. We now need to show that $H^1(C^\bullet_{\mathrm{st}}(E)) \cong F_0 \oplus E_F/\mathrm{Fil}^0 E_F$. Consider the dual shifted sequence of (2.14)

$$0 \to V^\vee[-1] \to E^\vee[-1] \to F_0 \to 0. \tag{2.22}$$

Under the assumption (2.15), there exists a unique (φ, N)-equivariant decomposition of $E^\vee[-1]$ as in the proof of Lemma 2.1 and then a decomposition

$$E = F_0[-1] \oplus V. \tag{2.23}$$

We then have that

$$\mathrm{Im}(E \xrightarrow{((\varphi - \mathrm{Id}), N)} E \oplus E[1]) = \mathrm{Im}(F_0[-1] \oplus V \to F_0[-1] \oplus V \oplus F_0 \oplus V[1]) = \tag{2.24}$$

$$= F_0[-1] \oplus \mathrm{Im}(V \xrightarrow{((\varphi - \mathrm{Id}), N)} V \oplus V[1]) \tag{2.25}$$

since φ and N act respectively by $p^{-1}\sigma$ and trivially on $F_0[-1]$. On the other hand,

$$\ker(E \oplus E[1] \xrightarrow{(N,(\mathrm{Id}-\varphi))} E \oplus E[1]) = F_0[-1] \oplus \mathbb{Q}_p \oplus \ker(V \oplus V[1] \xrightarrow{(N,(\mathrm{Id}-\varphi))} V \oplus V[1]). \tag{2.26}$$

We can then express the cohomology group $H^1(C^\bullet_{\mathrm{st}})$ as

$$H^1(C^\bullet_{\mathrm{st}}(E)) = E_F/\mathrm{Fil}^0 E_F \oplus \mathbb{Q}_p \oplus \frac{\ker(V \oplus V[1] \xrightarrow{(N,(\mathrm{Id}-\varphi))} V \oplus V[1])}{\mathrm{Im}(V \xrightarrow{((\varphi-\mathrm{Id}),N)} V \oplus V[1])}. \tag{2.27}$$

Recall that under the assumption (2.15), every simple extension of V factors through a (φ, N)-equivariant map as a direct sum $V \oplus F_0$ not compatible with the filtration. In particular, we have

$$\mathrm{Ext}^1_{\mathrm{MF}^{\mathrm{ad}}_F(\varphi,N)}(F_0, V) \cong H^1(C^\bullet_{\mathrm{st}}(V)) \cong V_F/\mathrm{Fil}^0 V_F \tag{2.28}$$

implying the last summand in (2.27) is trivial. From this we conclude $H^1(C^\bullet_{\mathrm{st}}(E)) = E_F/\mathrm{Fil}^0 E_F \oplus \mathbb{Q}_p$. □

The explicit description of the *Bloch-Kato isomorphism* (2.16) is given as follows. First of all fix the unique (φ, N)-equivariant decomposition $E = F_0[-1] \oplus V$. Then for every extension

$$0 \to E \to D \to F_0 \to 0 \tag{2.29}$$

consider the φ-equivariant splitting $s : F_0 \to D$ inducing a φ-equivariant decomposition

$$D \cong F_0[-1] \oplus V \oplus F_0. \tag{2.30}$$

Fix a decomposition $D_F \cong E_F \oplus F$ as in the previous case (2.12) then we define

$$(\pi_{E_F}(s(1)), \pi_{F_0[-1]}(N(s(1)))) \in E_F/\mathrm{Fil}^0 E_F \oplus \mathbb{Q}_p \tag{2.31}$$

where π_{E_F} denotes the projection under the decomposition $D_F \cong E_F \oplus F$, while $\pi_{F_0[-1]}$ denotes the projection on the first factor of the decomposition (2.30).

3. Motives, p-adic local systems and F-isocrystals

3.1. **Shimura curves and p-adic uniformization.** Let N be a positive integer that can be factored as $N = pN^-N^+$ with

- p prime;
- N^- positive squarefree integer with odd number of prime divisors;
- N^+ positive integer relatively prime to pN^-.

Let $\mathcal{B}$ be the indefinite rational quaternion algebra of discriminant pN^-. Fix $\mathcal{R}^{\max} \subset \mathcal{B}$ a maximal order and $\mathcal{R} \subset \mathcal{R}^{\max}$ an Eichler order of level N^+.

Definition 3.1. *Let S be a $\mathbb{Q}$-scheme. An abelian surface with* quaternionic multiplication (QM) *by $\mathcal{R}^{\max}$, and level N^+ structure over S is a triple $(\mathcal{A}, i, C)$ with*

(i) $\mathcal{A}$ is a principally polarized abelian scheme over S of relative dimension 2;
(ii) $i : \mathcal{R}^{\max} \hookrightarrow \mathrm{End}_S(\mathcal{A})$ is an injective ring homomorphism;
(iii) C is a subgroup scheme of $\mathcal{A}$ which is étale locally isomorphic to $(\mathbb{Z}/N^+\mathbb{Z})^2$ stable and locally cyclic under the action of $\mathcal{R}$.

Definition 3.2. *The Shimura curve $X := X_{N^+,pN^-}/\mathbb{Q}$ is the coarse moduli scheme for the moduli problem over $\mathbb{Q}$*

$$\underline{\mathrm{Sch}}/\mathbb{Q} \to \underline{\mathrm{Sets}}$$
$$S \mapsto \{\textit{Abelian surfaces with QM by } \mathcal{R}^{\max} \textit{ and level } N^+ \textit{ structure over } S\}/\sim .$$

X is a smooth, projective, and geometrically connected curve over $\mathbb{Q}$.

Let B be the definite rational quaternion algebra of discriminant N^-, let R be an Eichler $\mathbb{Z}[1/p]$-order of level N^+ in B. Define the group Γ to be

$$\Gamma := \{x \in R^\times : \mathrm{nrd}(x) = 1\}.$$

Since the quaternion algebra B splits at p, fix an isomorphism

$$\iota_p : B_p = B \otimes_{\mathbb{Q}} \mathbb{Q}_p \xrightarrow{\sim} M_2(\mathbb{Q}_p).$$

The isomorphism ι_p identifies Γ with a discrete cocompact subgroup of $SL_2(\mathbb{Q}_p)$.

Theorem 3.1 (Čerednik–Drinfeld)**.** *There is an isomorphism of rigid analytic varieties over $\mathbb{Q}_{p^2}$.*

$$(X_{\mathbb{Q}_{p^2}})^{\mathrm{an}} \cong X_\Gamma := \Gamma \setminus \mathcal{H}_p.$$

Proof. See [BC91, Section 5.3]. □

In order to proceed with our construction we need to work with a finely representable moduli problem and with the universal abelian scheme associated. To do this, we rigidify the problem adding full level structure.

Definition 3.3. *Let $M \geq 3$ be an integer relatively prime to N. An abelian surface with* quaternionic multiplication (QM) *by $\mathcal{R}^{\max}$, level N^+ structure and full level M over S is a quadruple $(\mathcal{A}, i, G, \nu)$ such that $(\mathcal{A}, i, G)$ satisfies the properties of definition 3.1 and ν is a $\mathcal{R}^{\max}$-equivariant isomorphism*

$$\nu : (\mathcal{R}^{\max}/M\mathcal{R}^{\max})_S \xrightarrow{\sim} A[M]$$

from the constant group scheme $(\mathcal{R}^{\max}/M\mathcal{R}^{\max})_S$ to the group scheme of M-division points.

The corresponding moduli problem admits a fine moduli scheme over $\mathbb{Q}$ which we denote by X_M. This is a smooth projective curve over $\mathbb{Q}$ but it is not geometrically connected. The map induced by forgetting the full level structure

$$q : X_M \to X$$

is a Galois covering with Galois group $G/\{\pm 1\}$ with

$$G := G_M := (\mathcal{R}^{\max}/M\mathcal{R}^{\max})^\times \cong (\mathcal{R}/M\mathcal{R})^\times \cong GL_2(\mathbb{Z}/M\mathbb{Z}) \cong (R/MR)^\times. \tag{3.1}$$

Let Γ_M be the group

$$\Gamma_M := \{\gamma \in \Gamma : \gamma \equiv 1 \mod MR\}$$

Theorem 3.2 (Čerednik–Drinfeld)**.** *There is an isomorphism of rigid analytic varieties over $\mathbb{Q}_{p^f}$ for a suitable finite unramified extension*

$$(X_{M,\mathbb{Q}_{p^f}})^{\mathrm{an}} \cong \Gamma \setminus \left(\mathcal{H}_p \times (R/MR)^\times\right) \cong \coprod_{(\mathbb{Z}/M\mathbb{Z})^\times} \Gamma_M \setminus \mathcal{H}_p.$$

In particular, $(X_{M,\mathbb{Q}_{p^f}})^{\mathrm{an}}$ is the disjoint union of Mumford curves $X_{\Gamma_M} := \Gamma_M \setminus \mathcal{H}_p$.

Proof. See [BC91, Section 5.3]. □

3.2. **Chow motives and abelian schemes.** In this section we recall some of the main constructions and properties of motives and Galois representations attached to p-adic sheaves on Shimura curves. In Appendix 10.1 [IS03], Iovita-Spieß generalized the construction of [Sch90] to the case of Shimura curves. We briefly recall some of their main results.

Consider $\pi : \mathcal{A} \to X$ the universal abelian scheme associated with the moduli problem of Definition 3.3. The category $\mathrm{Mot}(X_M)$ of *relative Chow motives* is an additive pseudo-abelian $\mathbb{Q}$-linear rigid tensor category. The objects of $\mathrm{Mot}(X_M)$ are triples (Y, p, e) where

- Y is a projective X_M-scheme,
- p is an idempotent element in the ring of relative correspondences $\mathrm{Corr}_{X_M}(Y \times Y) := \bigoplus_i CH^1(Y_i \times_{X_M} Y)$ where Y_i are the connected components of Y,
- e is an integer.

For a general motive $\mathcal{M}$, we can define its associated *Chow group* $\mathrm{CH}^i(\mathcal{M})$ to be

$$\mathrm{CH}^i(\mathcal{M}) := \mathrm{Hom}_{\mathrm{Mot}(X_M)}(\mathbb{1}, \mathcal{M}(i)) \tag{3.2}$$

where $\mathbb{1}$ is the unit motive associated to X_M. For a smooth projective X_M-scheme Y we denote by $h(Y) = (Y, \Delta_Y, 0)$ its motive. Thanks to an important Theorem of Deninger-Murre [DM91] and Künnemann [Kün94], the motive associated to the abelian surface $\mathcal{A}$ decomposes into its graded pieces

$$h(\mathcal{A}) = h^0(\mathcal{A}) \oplus h^1(\mathcal{A}) \oplus h^2(\mathcal{A}) \oplus h^3(\mathcal{A}) \oplus h^4(\mathcal{A}) \tag{3.3}$$

such that

$$h^i(\mathcal{A}) \cong \wedge^i h^1(\mathcal{A}), \quad h^i(\mathcal{A})^\vee \cong h^{4-i}(\mathcal{A})(2). \tag{3.4}$$

The $\mathcal{R}^{\max}$-action induces an embedding $\mathcal{B} \hookrightarrow \mathrm{End}(h^1(\mathcal{A}))$. Let e_2 be the unique nonzero idempotent in $\mathrm{End}(h^2(\mathcal{A})) = \mathrm{End}(\wedge^2 h^1(\mathcal{A}))$ such that

$$x^2 \cdot e_2 = N(x) e_2 \quad \text{for all } x \in \mathcal{B} \tag{3.5}$$

where N is the reduced norm. The projector $\epsilon_2 \in \mathrm{Corr}_{X_M}(\mathcal{A}, \mathcal{A})$ is defined by

$$(\mathcal{A}, \epsilon_2) = h^2(\mathcal{A})_- := \ker(e_2). \tag{3.6}$$

This motive should be thought as the *non-quaternionic* piece inside $h^2(\mathcal{A})$. Let

$$(-,-) : \mathrm{Sym}^2(h^2(\mathcal{A})_-) \to \mathbb{Q}(-2) \tag{3.7}$$

be the restriction of the symmetric pairing

$$h^2(\mathcal{A}) \otimes h^2(\mathcal{A}) \cong \wedge^2 h^1(\mathcal{A}) \otimes \wedge^2 h^1(\mathcal{A}) \to \wedge^4 h^1(\mathcal{A}) \cong \mathbb{Q}(-2). \tag{3.8}$$

This pairing induce the *Laplace operator*

$$\Delta_m : \mathrm{Sym}^m(h^2(\mathcal{A})_-) \to (\mathrm{Sym}^{m-2}(h^2(\mathcal{A})_-))(-2) \tag{3.9}$$

given by

$$\Delta_m(x_1 x_2 \cdots x_m) = \sum_{1 \le i < j \le m} (x_i, x_j) x_1 \cdots \hat{x_i} \cdots \hat{x_j} \cdots x_m. \tag{3.10}$$

Iovita-Spieß proved that $\ker(\Delta_m)$ exists as a motive and they denote it as

$$(\mathcal{A}^m, \epsilon_{2m}) := \ker(\Delta_m). \tag{3.11}$$

Remark 3.1. *In Scholl's original construction* [Sch90] *for elliptic modular curves X, the motive of weight k cusp forms sits directly inside $\mathrm{Sym}^{k-2}(h^1(\mathcal{E}))$ with $\mathcal{E} \to X$ the universal elliptic curve. In this case, the $h^1(\mathcal{E})$ is already the standard representation of rank 2. On our Shimura curve, parametrising QM abelian surfaces, $h^1(\mathcal{A})$ is of rank 4 and decomposes as $V \otimes V'$ where V is the piece we are interested in. Taking the wedge product gives*

$$\wedge^2 h^1(\mathcal{A}) = h^2(\mathcal{A}) = (\mathrm{Sym}^2(V) \otimes \wedge^2 V') \oplus (\wedge^2 V \otimes \mathrm{Sym}^2 V')$$

and consequently the idempotent e_2 isolates the piece $h^2(\mathcal{A})_- = \mathrm{Sym}^2 V$. For this reason, the definition of $(\mathcal{A}^m, \epsilon_n) = \mathrm{Sym}^n V$ is extracted from $\mathrm{Sym}^m(h^2(\mathcal{A})_-) = \mathrm{Sym}^m(\mathrm{Sym}^2 V)$ through the Laplace operator Δ_m.

The Galois group G defined by (3.1) acs canonically as X-automorphisms on X_M and $\mathcal{A}^m$. In order to pass from the Shimura curve X_M to X, we consider the projector

$$p_G := \frac{1}{|G|} \sum_{g \in G} g \in \mathbb{Q}[G] \tag{3.12}$$

as an element in $\mathrm{Corr}_X(\mathcal{A}^m, \mathcal{A}^m)$. The Chow motive $\mathcal{M}_n$ on $\mathrm{Mot}(X)$ is then defined to be

$$\mathcal{M}_n := (\mathcal{A}^m, p_G \circ \epsilon_n) \tag{3.13}$$

for $m = n/2$.

3.3. **p-adic local systems.** We follow Sec. 5 [IS03]. In order to determine the p-adic realization of the motive $\mathcal{M}_n$, we consider the p-adic local system $R^2\pi_*\mathbb{Q}_p$ and mimic the construction in the light of Remark 3.1. The action of $\mathcal{R}^{\max}$ on $\mathcal{A}$ induces an action of $\mathcal{B}^*$ on $R^2\pi_*\mathbb{Q}_p$ allowing us to define

$$\mathbb{L}_2 := \bigcap_{b\in\mathcal{B}} \ker(b - N(b) : R^2\pi_*\mathbb{Q}_p \to R^2\pi_*\mathbb{Q}_p) \tag{3.14}$$

where $N(b)$ denotes the reduced norm of b. Consider the Laplace operator

$$\Delta_m : \mathrm{Sym}^m\mathbb{L}_2 \to (\mathrm{Sym}^{m-2}\mathbb{L}_2)(-2) \tag{3.15}$$

induced by the non-degenerated pairing

$$(-,-) : \mathbb{L}_2 \otimes \mathbb{L}_2 \hookrightarrow R^2\pi_*\mathbb{Q}_p \otimes R^2\pi_*\mathbb{Q}_p \to R^4\pi_*\mathbb{Q}_p \to \mathbb{Q}_p(-2). \tag{3.16}$$

For $n > 2$, we define $\mathbb{L}_n$ to be the kernel of Δ_m where $m = n/2$.

Lemma 3.1 (Lemma 10.1 [IS03])**.** *Let $R_p : \mathrm{Mot}(X_M) \to D^b(X_M, \mathbb{Q}_p)$ be the p-adic realization functor from $\mathrm{Mot}(X_M)$ to the bounded derived category of $\mathbb{Q}_p$-sheaves. Let $m = n/2$, then we have*

$$R_p(\mathcal{A}^m, \epsilon_n) = \mathbb{L}_n[-n].$$

If we consider $\mathcal{M}_n$ as a Chow motive over $\mathbb{Q}$ under the canonical functor $\mathrm{Mot}(X) \to \mathrm{Mot}(\mathbb{Q})$ then its p-adic realisation is given by

$$R\Gamma(X, R_p(\mathcal{M}_n)) = H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G[-n-1]. \tag{3.17}$$

The étale cohomology of the p-adic sheaf $\mathbb{L}_n$ will be correspond under the D_{st} functor to the de Rham cohomology of the F-isocrystal obtained from the relative de Rham cohomology. In particular, we want to closely study the p-adic Galois representation

$$V := H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G. \tag{3.18}$$

3.4. **Cycle map and p-adic Abel-Jacobi.** In order to compute the p-adic height pairing of Heegner cycles, we will need to construct the associated extensions of p-adic Galois representations. To do this, recall that to every element in the Chow group of a Chow motive we can associate its cycle class

$$cl : \mathrm{CH}^i(\mathcal{M}) \to R^{2i}\Gamma(X, R_p(\mathcal{M})(i)) \tag{3.19}$$

given by the p-adic realization of $\mathrm{Hom}_{\mathrm{Mot}(X_M)}(\mathbb{1}, \mathcal{M}(i))$. For the motive $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$ we have

$$cl : \mathrm{CH}^i(\mathcal{M}_n) \to (p_G \circ \epsilon_n)_* H^{2i}(\mathcal{A}^m \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)). \tag{3.20}$$

We denote the kernel of the cycle class map by $\mathrm{CH}^i(\mathcal{M})_0$. By Lemma 3.1, the codomain of the cycle map for the motive $\mathcal{M}_n$ vanishes for every i and we then have $\mathrm{CH}^i(\mathcal{M}_n) = \mathrm{CH}^i(\mathcal{M}_n)_0$.

The *p-adic Abel-Jacobi map*

$$cl_0 : \mathrm{CH}^i(\mathcal{M}_n)_0 \to H^1_{\text{cont}}(K, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i))) = \mathrm{Ext}^1_{\underline{\mathrm{Rep}}(G_K)}(\mathbb{Q}_p, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i))) \tag{3.21}$$

associates to every class $z \in CH^i(\mathcal{M})_0$ an extension

$$0 \to R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) \to E_z \to \mathbb{Q}_p \to 0. \tag{3.22}$$

The p-adic representation E_z is defined as follows. Consider $\mathcal{M} = (Y, p)$ a relative motive over X_M, let $z \in \mathrm{CH}^i(\mathcal{M})_0 = p_*\mathrm{CH}^i(Y)_0$ and Z be a cycle in the Chow group $\mathrm{CH}^i(Y)_0$ such that $z = p_*[Z]$. Then we can first consider the Gysin sequence

$$0 \to H^{2i-1}_{\text{ét}}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \to H^{2i-1}_{\text{ét}}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \to H^{2i}_{\text{ét},|Z|}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 \to 0 \tag{3.23}$$

where

$$H^{2i}_{\text{ét},|Z|}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 := \ker(H^{2i}_{\text{ét},|Z|}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \to H^{2i}_{\text{ét}}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))) \tag{3.24}$$

and then apply the projector p and take the pullback of the exact sequence by the map $\mathbb{Q}_p \to p_* H^{2i}_{\text{ét},|Z|}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))$ sending 1 to $cl(z)$

(3.25)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) & \longrightarrow & E_z & \longrightarrow & \mathbb{Q}_p & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow{\scriptstyle cl} & & \\ 0 & \longrightarrow & p_* H^{2i-1}_{\text{ét}}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H^{2i-1}_{\text{ét}}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H^{2i}_{\text{ét},|Z|}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 & \longrightarrow & 0. \end{array}$$

In the case of $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$, using once again the results of Lemma 3.1, we can associate to every $z \in \mathrm{CH}^{m+1}(\mathcal{M})$ an extension

$$0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to E_z \to \mathbb{Q}_p \to 0. \tag{3.26}$$

As discussed in Sec. 7 of [IS03], the extensions obtained for the motive $\mathcal{M}_n$ are *semistable*, implying that the p-adic Abel-Jacobi map factors through

$$cl_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n)_0 \to H^1_{\text{st}}(K_v, H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G) \tag{3.27}$$

where v is a place above p and H^1_{st} denotes the Selmer group introduced by Nekovář [Nek93] given by

$$H^1_{\text{st}}(K_v, V) := \ker(H^1(K_v, V) \to H^1(K_v, V \otimes_{\mathbb{Q}_p} B_{\text{st}})) \tag{3.28}$$

for any p-adic representation V of G_{K_v}. If V is semistable, then $H^1_{\text{st}}(K, V)$ can be identified with the group of extension classes of $\mathbb{Q}_p$ by V. In our case we then have that the p-adic Abel-Jacobi map factors through

$$cl_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n) \to \mathrm{Ext}^1_{\underline{\mathrm{Rep}}_{\text{st}}(G_K)}(\mathbb{Q}_p, H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G). \tag{3.29}$$

3.5. **Biextensions.** Let D be a reduced effective divisor on X_M defined over K such that all closed points reduce to distinct points on X under the map $q : X_M \to X$. There is then an exact sequence of G_K-modules given by the Gysin sequence

(3.30)

$$0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to H^1_{\text{ét}}((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x \in D} (\mathbb{L}_n)^{G_x}_x(m) \to 0$$

where G_x is the stabilizer of the point x. It's important to remark that the direct sum of the stalks of the p-adic sheaf $\mathbb{L}_n(m)$ is the p-adic realisation of the motive

$$R\Gamma^{2m}(X_{M,K}, R_p((\mathcal{A}^m_D, \epsilon_n)_K)(m)) = \bigoplus_{x \in D} (\mathbb{L}_n)_x(m) \tag{3.31}$$

where $\mathcal{A}^m_D = \coprod_{x \in D} \mathcal{A}^m_x$ is the disjoint union of products of abelian surfaces associated with $x \in D$.

Its dual exact sequence of p-adic representations of G_K is given by cohomology with support sequence

$$0 \to \bigoplus_{x \in D} (\mathbb{L}_n)^{G_x}_x(m) \to H^1_{\text{ét},|D|}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \to 0. \tag{3.32}$$

The motives $\mathcal{M}_n$ come equipped with the symmetric pairing (3.7) and, at the level of p-adic realization $\mathbb{L}$, this descends to (3.16) inducing Poincaré duality. In particular, we have

$$(-,-) : \mathbb{L}_n(m) \otimes \mathbb{L}_n(m) \to \mathbb{Q}_p.$$

The pairing between the exact sequences is then given by

$$\begin{array}{ccccc} H^1_{\text{ét}}((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H^1_{\text{ét},|D|}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H^2_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G & \xrightarrow{\sim} & \mathbb{Q}_p \\ \uparrow \quad \downarrow & & \| & & \\ H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H^2_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G & \xrightarrow{\sim} & \mathbb{Q}_p. \end{array} \tag{3.33}$$

Consider now two effective reduced divisors D_1, D_2 on X_M defined over K_1, K_2 with disjoint support such that all their closed points reduce to distinct points on X under the map $q : X_M \to X$. Consider the two short exact Gysin sequences

(3.34)
$$e_{D_1} : 0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to H^1_{\text{ét}}(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x \in D_1} (\mathbb{L}_n)_x^{G_x}(m) \to 0,$$

(3.35)
$$e_{D_2} : 0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to H^1_{\text{ét}}(U_2 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x \in D_2} (\mathbb{L}_n)_x^{G_x}(m) \to 0$$

where U_i is the open subscheme of X_M complementary to D_i in X_M. The cohomology group

(3.36)
$$H^1_{\text{ét},|D_2|}(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G$$

represents a biextension of the exact sequence e_{D_1} and of the twisted dual $e^\vee_{D_2}(1)$. In particular, we obtain a diagram with exact rows and exact columns of p-adic representations of the absolute Galois group of the composite field $G_{K_1K_2}$

(3.37)
$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{x\in D_2}(\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H^1_{\text{ét},|D_2|}(X_M\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & H^1_{\text{ét}}(X_M\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{x\in D_2}(\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H^1_{\text{ét},|D_2|}(U_1\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & H^1_{\text{ét}}(U_1\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \oplus_{x\in D_1}(\mathbb{L}_n)_x^{G_x}(m) & = & \oplus_{x\in D_1}(\mathbb{L}_n)_x^{G_x}(m) & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}.$$

We will construct maps of p-adic Galois representations $\mathbb{Q}_p \to \bigoplus_{x\in D_1}(\mathbb{L}_n)_x^{\Gamma_x}(m)$ and $\bigoplus_{x\in D_1}(\mathbb{L}_n)_x^{\Gamma_x}(m+1) \to \mathbb{Q}_p(1)$ and we will take the pull-back of the first one and the push-out by the second one to obtain the diagram

(3.38)
$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2^\vee[-1] & \longrightarrow & V & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \mathbb{Q}_p & = & \mathbb{Q}_p & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}.$$

Remark 3.2. *In light of equation* (3.31)*, the problem of constructing trivial subobjects and quotients of* $\bigoplus_{x\in D_1}(\mathbb{L}_n)_x(m)$ *will be addressed from the motivic point of view in Section 3.8 where we will use the construction in* [IS03] *of suitable classes in* $\mathrm{CH}^m((\mathcal{A}^m_D, \epsilon_n))$ *and associated trivial p-adic representations under the cycle class map* (3.19)*.*

This biextension diagram is going to be the starting point of our computations on the p-adic height pairing of Section 5.

3.6. **F-Isocrystals on the $\mathcal{H}_p$.** To explicitly describe the de Rham realization of the motives, we will use the p-adic upper half-plane $\mathcal{H}_p$ and the explicit p-adic uniformization of the curve X_M.

Let $\hat{\mathcal{H}}_p$ denote the canonical formal model of $\mathcal{H}_p$ over $\mathbb{Z}_p$. We briefly recall the convergent filtered F-isocrystals associated to a $GL_2 \times GL_2$ representation $(V, \rho) \in \underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2 \times GL_2)$ see [IS03, Section 4] for more details. We denote by (V, ρ_1) and (V, ρ_2) the elements of $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$ obtained restricting ρ respectively to the first and second component. Conversely, for any $(V_1, \rho_1), (V_2, \rho_2)$ in $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$ we can form a $GL_2 \times GL_2$ representation

$$(V_1, \rho_1) \odot (V_2, \rho_2) = (V_1 \otimes V_2, \rho_1 \otimes 1_{V_2}, 1_{V_1} \otimes \rho_2).$$

For V in $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$ and $m \in \mathbb{Z}$ we define

$$V\{m\} := V \odot \det^{\otimes m}, \quad \{m\}V := \det^{\otimes m} \odot V.$$

For $n \geq 0$, let $\mathcal{P}_n$ denote the space of all polynomials in X, Y which are homogeneous of degree n over $\mathbb{Q}_p$ with GL_2-action

$$P(X, Y) \cdot \gamma = P(aX + bY, cX + dY), \quad \text{for } \gamma^{-1} = (\det \gamma)^{-1} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{3.39}$$

Denote the dual representation by $V_n := \mathcal{P}_n^\vee = \mathrm{Hom}_{\mathbb{Q}_p}(\mathcal{P}_n, \mathbb{Q}_p)$.

- As an *isocrystal*, $\mathcal{E}(V) := V \otimes_{\mathbb{Q}_p} \mathcal{O}_{\hat{\mathcal{H}}}$ is given to be the isotrivial vector bundle over the p-adic upper half plane $\mathcal{O}_{\mathcal{H}_p}$ with the induced connection given by usual derivation d;
- Consider the endomorphism

$$\varphi_V := \rho_2 \begin{pmatrix} 0 & p \\ 1 & 0 \end{pmatrix} : V \to V. \tag{3.40}$$

  The *Frobenius* is then given by

$$\Phi = \varphi_V \otimes \Phi_{\mathcal{O}_{\mathcal{H}_p}} \tag{3.41}$$

  in particular, it only depends on the GL_2 representation ρ_2.
- The *filtration* is only defined in terms of ρ_1. We describe it only for the representations V_n

$$F^k V_n = \sum_{j=0}^{n-k} (zx + y)^k x^j y^{n-j-k} \mathcal{O}_{\mathcal{H}_p} \tag{3.42}$$

  where $x, y \in V_n$ are the dual elements corresponding to the variables X, Y in $\mathcal{P}_n$. In particular, we have

$$F^0 V_n = V_n \otimes_{\mathbb{Q}_p} \mathcal{O}_{\hat{\mathcal{H}}}, \quad F^n V_n = (zx + y)^n \mathcal{O}_{\hat{\mathcal{H}}}, \quad F^{n+1} V_n = 0. \tag{3.43}$$

In the case where $\Gamma \subseteq \mathrm{GL}_2$ is a discrete, cocompact, and *torsion-free subgroup*, the filtered isocrystals $\mathcal{E}(V)$ descend to filtered isocrystals on the Mumford curve X_Γ. In particular, the action of GL_2 on $\mathcal{E}(V)$ through ρ_1 is compatible with the action on $\mathcal{H}_p$. We will denote these filtered isocrystals again by $\mathcal{E}(V)$ and denote their associated locally free $\mathcal{O}_{X_\Gamma}$-module with connection and filtration by $E(V)$.

Consider the $\mathrm{GL}_2 \times \mathrm{GL}_2$-representation of two-dimensional matrices (M_2, ρ_1, ρ_2) given by

$$\rho_1(A)(B) := A \cdot B, \quad \rho_2(A)(B) := B \cdot \mathrm{adj}(A) \tag{3.44}$$

for $A \in \mathrm{GL}_2$ and $B \in M_2$ where $\mathrm{adj}(A)$ is the adjugate matrix.

The following result explains why the F-isocrystals $\mathcal{E}(V_n\{n/2\})$ will be the correct object of study.

Theorem 3.3. *Consider the F-isocrystal obtained from $\pi : \mathcal{A} \to X_M$ by*

$$R_{\mathrm{F-isoc}}(h^1(\mathcal{A})) = R^1 \pi_* \Omega^\bullet_{\mathcal{A}/X_M}. \tag{3.45}$$

There exists an (up to scaling) canonical isomorphism of filtered isocrystals on X_M

$$R_{\mathrm{F-isoc}}(h^1(\mathcal{A})) \otimes \mathbb{Q}_{p^{\mathrm{unr}}} \cong \mathcal{E}(M_2, \rho_1, \rho_2) \otimes \mathbb{Q}_{p^{\mathrm{unr}}}. \tag{3.46}$$

This implies that the de Rham cohomology $\mathcal{H}^1_{\mathrm{dR}}(\mathcal{A}/X_M)$ is isomorphic to the locally free $\mathcal{O}_{X_M}$-module $E(M_2)$ of rank 4.

Proof. See [IS03, Lemma 5.10]. □

Remark 3.3. *It is important to note that the previous Theorem establishes an isomorphism of isocrystal after passing to the maximal unramified extension of $\mathbb{Q}_p$. In* [IS03] *this is needed because they explicitly use the moduli description of the formal model of the upper-half plane $\hat{\mathcal{H}}_p \otimes \mathbb{Z}_{p^{\mathrm{unr}}}$ in terms of special formal groups.*

Now we can observe that (M_2, ρ_1, ρ_2) is canonically isomorphic to $V_1 \odot V_1$, giving

$$(\wedge^2 M_2, \wedge^2\rho_1, \wedge^2\rho_2) = \wedge^2 V_1 \odot \mathrm{Sym}^2 V_1 \oplus \mathrm{Sym}^2 V_1 \odot \wedge^2 V_1 = \{1\}V_2 \oplus V_2\{1\}. \tag{3.47}$$

In particular, we obtain

$$\mathcal{E}(V_2\{1\}) \cong \bigcap_{b\in\mathcal{B}} \ker(b - N(b) : \mathcal{E}(\wedge^2 M_2) \to \mathcal{E}(\wedge^2 M_2)) \tag{3.48}$$

where the action of $\mathcal{B}$ is given by the isomorphism $\mathcal{B}\otimes\mathbb{Q}_{p^2} \cong M_2(\mathbb{Q}_{p^2})$. Observe that $\mathcal{V}_2$ inherits a non-degenerate symmetric bilinear pairing by restriction to

$$(-,-) : V_2\{1\} \otimes V_2\{1\} \hookrightarrow \wedge^2 M_2 \otimes \wedge^2 M_2 \to \wedge^4 M_2 \cong \det^{\otimes 2} \odot \det^{\otimes 2} = \{2\}\{2\}. \tag{3.49}$$

Compare once again this to the pairings (3.7) and (3.16). We will need an explicit and computable description of this pairing that will be later use to identify Heegner cycles. We recall from [BDIS02, Sec. 1.2] the construction using the following isomorphic description.

Lemma 3.2. *Let $\mathcal{U} = \{U \in M_2 : \mathrm{tr}(U) = 0\}$ be endowed with the right GL_2 action defined by*

$$U \cdot A := \overline{A} U A \tag{3.50}$$

for $U \in \mathcal{U}$ and $A \in \mathrm{GL}_2$ where $A\overline{A} = \det(A)$. There is an isomorphism of right-GL_2-modules defined by

$$\begin{aligned} &\mathcal{U} \to \mathcal{P}_2 \\ &U \mapsto P_U(X,Y) := \mathrm{tr}\left(U \begin{pmatrix} X \\ Y \end{pmatrix} \begin{pmatrix} Y & -X \end{pmatrix} \right). \end{aligned} \tag{3.51}$$

The pairing on $\mathcal{P}_2$ arises from the *Cartan-Killing form* on $\mathcal{U}$

$$(P_{U_1}, P_{U_2}) := -\mathrm{tr}(U_1\overline{U}_2) \tag{3.52}$$

for every $U_1, U_2 \in \mathcal{U}$. This induces an isomorphism

$$\begin{aligned} &\mathcal{P}_2 \to V_2 \\ &P(X,Y) \mapsto (P(X,Y), -) \end{aligned} \tag{3.53}$$

and induces a non-degenerate symmetric pairing on V_2 coinciding with (3.49). Consider the Laplace operator

$$\Delta_m : \mathrm{Sym}^m V_2 \to \mathrm{Sym}^{m-2} V_2 \otimes \det^{\otimes 2} \tag{3.54}$$

defined analogously to (3.9) and (3.15). The GL_2-subrepresentation V_{2m} of $\mathrm{Sym}^m V_2$ appears without surprise to be the kernel Δ_m.

3.7. ***F*-isocrystals on X_Γ.** We now want to be able to compare the Galois representation side provided by the étale cohomology and the (φ, N)-structure on the F-isocrystal. In [Fal89],[Fal92] and [Fal02] Faltings gives a proof of Fontaine's *crystalline and semistable conjectures* based on the work of Fontaine–Messing [FM87] using the theory of almost étale extensions while Tsuji [Tsu99] gave a proof generalizing the syntomic cohomology approach of Kato–Messing [KM92]. In our context we can summarize their results in the following theorem.

Theorem 3.4 (Fontaine's semistable conjecture)**.** *Let $\mathbb{L}_n$ the p-adic sheaf on X_M defined in Section 3.3. There exists an F-isocrystal $\mathcal{V}_n$ on X_M such that*

$$D_{\mathrm{st}}(H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G) = H^1_{\mathrm{dR}}(X_M \otimes \mathbb{Q}_p, \mathcal{V}_n)^G. \tag{3.55}$$

In particular, there is a functorial map associating to the p-adic lisse sheaf $R_p(h^1(\mathcal{A}))$ the F-isocrystal given by the relative de Rham cohomology $R^1\pi_*\Omega^\bullet_{\mathcal{A}/X_M}$. This then gives us an explicit description of the F-isocrystal $\mathcal{V}_n$ when we pass to the maximal unramified extension.

Corollary 3.1. *Under the analytic isomorphism of Theorem 3.2, the pullback of $\mathcal{V}_n \otimes \mathbb{Q}_{p^{\mathrm{unr}}}$ on $\mathcal{H}_p$ is given by the F-isocrystal $\mathcal{E}(V_n\{n/2\})$.*

Proof. Recall that Cerednik-Drinfeld Theorem gives an isomorphism

$$X_{M,\mathbb{Q}_{p^{\mathrm{unr}}}} \cong \Gamma \setminus \left(\mathcal{H}_p \times (R/MR)^\times\right) \cong \coprod_{(\mathbb{Z}/M\mathbb{Z})^\times} \Gamma_M \setminus \mathcal{H}_p. \tag{3.56}$$

Let $\pi_{\Gamma_M} : \mathcal{H}_p \to \Gamma_M \setminus \mathcal{H}_p$ the projection map. By the previous Theorem, we associate $(R_p(h^1(\mathcal{A}))$ to $R^1\pi_*\Omega^\bullet_{\mathcal{A}/X_M}$ and by Theorem 3.3 we have that the pullback to the upper-half plane is given by

$$\pi^*_{\Gamma_M}(R^1\pi_*\Omega^\bullet_{\mathcal{A}/X_M}) \cong \mathcal{E}(M_2). \tag{3.57}$$

Under the non-degenerate bilinear symmetric pairing on the F-isocrystal (3.49) descending from the motivic pairing (3.7) we have that the F-isocrystal $\mathcal{E}(V_n\{n/2\})$ is isomorphic to the one cut from $R^1\pi_*\Omega^\bullet_{\mathcal{A}/X_M}$ and isomorphic to the pullback $\pi^*_{\Gamma_M}(\mathcal{V}_n)$. □

Remark 3.4. *It is important to note that the F-isocrystal on the upper half plane $\mathcal{H}_p$ associated to $\mathcal{V}_n$ can be constructed passing to a finite extension of $\mathbb{Q}_p$. This isocrystal becomes isomorphic to $\mathcal{E}(V_n\{n/2\})$ only after a base change to the maximal unramified extension. From the fact that $\mathcal{H}_p$ is a Stein space, we obtain that the isocrystal $\mathcal{V}_n$ is given by a free vector bundle. The unramified extension is then need to obtain an explicit horizontal basis for the Frobenius action.*

We will from now on work on a finite unramified extension of $\mathbb{Q}_p$ where the p-adic uniformization of Cerednik–Drinfel holds. When no confusion will arise, we will then denote the isocrystal on X_M and on the upper-half plane $\mathcal{H}_p$ in the same way.

Given such an object, we can define the de Rham cohomology of general quotients of the upper-half plane.

Definition 3.4. *Let $\Gamma \subseteq \mathrm{GL}_2$ be a discrete and cocompact subgroup. If Γ is torsion-free we define*

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) := \mathbb{H}^1(X_\Gamma, \mathcal{V}_n \otimes \Omega^\bullet_1) \tag{3.58}$$

where the complex $(\mathcal{V}_n \otimes \Omega^\bullet_1)$ is induced by the integrable connection. If Γ is not torsion-free, we consider a torsion-free subgroup $\Gamma' \subseteq \Gamma$ and we define

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) := H^1_{\mathrm{dR}}(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'} \tag{3.59}$$

In Section 4, we will study the properties of these cohomology groups. In particular, the F-isocrystal structure of $\mathcal{V}_n$ will induce the structure of a filtered (φ, N)-module on $H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)$. We then have a comparison isomorphism between the p-adic representations attached to $\mathbb{L}_n$ and the hypercohomology of the vector bundle with connection $\mathcal{V}_n$.

Theorem 3.5. *Let Γ the discrete cocompact subgroup of GL_2 associated to X as in Section 3.1. The p-adic representation $H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G$ is semistable and we have a canonical (up to scaling) isomorphisms of filtered (φ, N)-modules*

$$D_{\mathrm{st},\mathbb{Q}_{p^f}}(H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G) = H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n). \tag{3.60}$$

Proof. It follows from Theorem 3.4 and [IS03, Theorem 5.9] □

One of the important properties of this theorem and key element inside its proof is the compatibility between the symmetric products defined over the p-adic local system $\mathbb{L}_n$ (3.16) and the F -isocrystal $\mathcal{V}_n$ coming from the relative de Rham cohomology. This has indeed a motivic nature and comes from the pairing (3.7). One intance of this compatibility is the fact that the pairing

$$H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n) \otimes H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n) \to \mathbb{Q}_{p^2}[n+1] \tag{3.61}$$

is mapped under Fontaine's functor to the pairing

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \otimes H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \to H^2_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n \otimes \mathcal{V}_n) \to H^2_{\mathrm{dR}}(X_\Gamma, \mathcal{O}_{X_\Gamma})[n] \cong \mathbb{Q}_{p^2}[n+1]. \tag{3.62}$$

This property will be used later to pass from dual exact sequences in the category of p-adic representations to the dual of exact sequences in the de Rham realizations.

Consider now D to be an effective reduced divisor of X defined over K. Assume that $K/\mathbb{Q}$ is nontrivial unramified at p and all the closed points $x_1, x_2, \dots, x_r \in |D|$ are defined over K and such that all x_i reduce to distinct points on X under the map $q : X_M \to X$. We then have a Gysin exact sequence of étale cohomology groups

$$e_{\acute{e}t} : 0 \to H^1_{\acute{\mathrm{e}}\mathrm{t}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \to H^1_{\acute{\mathrm{e}}\mathrm{t}}(U \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \to \bigoplus_{x \in D} (\mathbb{L}_n)^{G_x}_x(-1) \to 0. \tag{3.63}$$

Let $z_1, z_2, \dots, z_r \in \mathcal{H}_p(K_p)$ points on the p-adic upper half plane such that $\Gamma z_i = q(x_i)$ under the p-adic uniformization of X Theorem 3.1.

Theorem 3.6. *Assume that $x_1, x_2, \dots, x_r$ specialize to smooth distinct points of the special fiber $X_{\mathbb{F}_{p^2}}$. The exact sequence obtained from* (3.63) *by applying the functor $D_{\mathrm{st},\mathbb{Q}_{p^f}}$ is isomorphic to the sequence of filtered (φ, N)-modules*

$$e_{\mathrm{dR}} : 0 \to H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \to H^1_{\mathrm{dR}}(X_\Gamma - D_\Gamma, \mathcal{V}_n) \to \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)^{\Gamma_z}_z[1] \to 0 \tag{3.64}$$

where Γ_z is the stabilizer of the point z.

Proof. See [CI10, Theorem 7.6] and [IS03, Theorem 5.13]. □

3.7.1. $\mathbb{Q}_{p^2}$*-points on* $\mathcal{H}_p$. It will be useful to identify the set of $\mathbb{Q}_{p^2}$-points on $\mathcal{H}_p$ with the set of $\mathbb{Q}_p$-algebra homomorphisms $\mathrm{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$. This will turn out useful when we will describe Heegner points geometrically on X and p-adic analytically on X_Γ.

Lemma 3.3. *There is a bijection between* $\mathrm{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ *and* $\mathcal{H}_p(\mathbb{Q}_{p^2})$.

Proof. Every $\Psi \in \mathrm{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ defines an action of $\mathbb{Q}^\times_{p^2}$ on $\mathcal{H}_p(\mathbb{Q}_{p^2})$. The corresponding point $z_\Psi \in \mathcal{H}_p(\mathbb{Q}_{p^2})$ is defined to be the unique point such that

$$\Psi(a) \cdot \begin{pmatrix} z \\ 1 \end{pmatrix} = a \begin{pmatrix} z \\ 1 \end{pmatrix} \tag{3.65}$$

for every $a \in \mathbb{Q}_{p^2}$. Conversely, for every point $z \in \mathcal{H}(\mathbb{Q}_{p^2})$ we can define the associated homomorphims $\Psi_z \in \mathrm{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ to be the linear transformation associated to multiplication by $\mathbb{Q}^\times_{p^2}$ on the projective coordinates of z. These two maps can be easily checked to be inverse of each other. □

Consider now a point $z \in \mathcal{H}_p(\mathbb{Q}_{p^2})$ and let $\Psi_z \in \mathrm{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ we want to compute the stalk of the isocrystal $\mathcal{E}(V_n\{n/2\})_{\mathbb{Q}_{p^2}}$ at the point z. This in particular will have the structure of a *filtered Frobenius module*. As a vector space this coincide with $V_{n,\mathbb{Q}_{p^2}} = V_n \otimes \mathbb{Q}_{p^2}$ and it is generated by $\mathrm{ev}_z \in V_{n,\mathbb{Q}_{p^2}} = \mathrm{Hom}(\mathcal{P}_n, \mathbb{Q}_{p^2})$ given by

$$\mathrm{ev}_z(P(X,Y)) := P(z,1) \tag{3.66}$$

that has the property

$$(\Psi_z(a) \cdot \mathrm{ev}_z)(P(X,Y)) = a^n P(z,1). \tag{3.67}$$

To define the filtration, consider first the subspaces

$$(V_{n,\mathbb{Q}_{p^2}})_i := \{v \in V_{n,\mathbb{Q}_{p^2}} : \rho_1(\Psi(a))(v) = a^i \sigma(a)^{n-i} v \quad \forall a \in \mathbb{Q}_{p^2}\} \tag{3.68}$$

and then define

$$F^i(V_{n,\mathbb{Q}_{p^2}})_z := \bigoplus_{j \geq i} (V_{n,\mathbb{Q}_{p^2}})_j. \tag{3.69}$$

The *Frobenius endomorphism* is given by $\phi_V \otimes \sigma$.

3.8. **Heegner points and special cycles.** In this section we will discuss special cycles in $\mathrm{CH}^{m+1}(\mathcal{M}_n)$ arising from abelian surfaces with complex multiplication. Under the Čerednik-Drinfelf p-adic uniformization, we will also study their characterization on the p-adic upper half plane. We will follow the construction of [IS03] Sec.8.

Recall that we want to construct interesting classes in $\mathrm{CM}^{m+1}(\mathcal{M}_n)$ or, as explained at the end of Section 3.2, construct trivial p-adic representations in the p-adic realization of motives $(\mathcal{A}_D^m, \varepsilon_n)$ for certain reduced effective divisors D on X_M. We will define these classes using special CM abelian surfaces.

Definition 3.5. *Consider A an abelian surface with QM by $\mathcal{R}^{\max}$. The abelian surface is said to have* complex multiplication *(CM) if $\mathrm{End}_{\mathcal{R}^{max}}(A) \neq \mathbb{Z}$. In this case it is an order $\mathcal{O}$ in an imaginary quadratic number field. A closed point on X is called* CM *if it can be represented by a triple (A, ι, C) where A has complex multiplication. It is furthermore called* Heegner point *if (A, ι, C) is CM by an order $\mathcal{O}$ and C is $\mathcal{O}$-stable.*

We will now fix a number filed K satisfying certain prescribed properties that will be key in the construction of the Chow classes.

Hypothesis 3.1 (Heegner)**.** *Let K be an imaginary quadratic field of discriminant D_K satisfying the following properties:*

- $(D_K, N) = 1$,
- *all prime divisors of $N^- p$ are* inert *in K,*
- *all prime divisors of $N+$ are* split *in K.*

Consider now an abelian surface A with CM by the maximal order $\mathcal{O}_K$. Thanks to the splitting behaviour of the primes in K we obtain

$$\mathrm{End}(A) \cong \mathcal{O}_K \otimes \mathcal{R}^{\max} \cong M_2(\mathcal{O}_K).$$

In particular, fixing an isomorphism $\mathrm{End}(A) \cong M_2(\mathcal{O}_K)$, we get an isomorphism

$$A \cong E \times E \tag{3.70}$$

where E is an elliptic curve with CM over H Hilbert class field of K and $\mathrm{End}(E) \cong \mathcal{O}_K$. As usual, we fix an isomorphism between the endomorphism ring and the maximal order. The special rigid structure of A is reflected in its associated motive. From the identifications of $\mathrm{End}(A)_{\mathbb{Q}} = K \otimes \mathcal{R}_{\max} = M_2(K) = K \otimes M_2(\mathbb{Q})$ and $A = E \times E$ we can explicitly describe the action of the endomorphisms on the motives. By the *Kunneth decomposition*

$$h^1(A) = h^1(E \times E) = h^1(E) \otimes_{\mathbb{Q}} \mathbb{Q}^2, \tag{3.71}$$

where K acts on $h^1(E)$ and $M_2(\mathbb{Q})$ acts on $\mathbb{Q}^2$. Considering the second piece of the motive of A and applying the decomposition once again we obtain

$$h^2(A) = \wedge^2 h^1(A) = \wedge^2(h^1(E) \otimes_{\mathbb{Q}} \mathbb{Q}^2) = (h^2(E) \otimes \mathrm{Sym}^2(\mathbb{Q}^2) \oplus (\mathrm{Sym}^2 h^1(E) \otimes \wedge^2 \mathbb{Q}^2). \tag{3.72}$$

Where now the matrix group M_2 acts by the determinant on the last summand showing how the action of $\mathcal{B}^*$ cuts, in the notation of (3.6), the part

$$h^2(A)_- = \mathrm{Sym}^2(h^1(E)). \tag{3.73}$$

Furthermore, since E has CM by $\mathcal{O}_K$, then $h^1(E)$ acquires an action of K and in particular the base change to K will decompose into two summands

$$h^1(E) \otimes_{\mathbb{Q}} K = M(\psi) \oplus M(\bar{\psi}) \tag{3.74}$$

where $M(\psi), M(\bar{\psi})$ are the motives associated to the Hecke character ψ of H attached to E and its conjugate as in Ch. 1 [Sch88]. From the fact that $\psi\bar{\psi} = N_{K/\mathbb{Q}}$, the symmetric square is then given by

$$\mathrm{Sym}^2(h^1(E)) \otimes_{\mathbb{Q}} K = M(\psi^2) \oplus \mathbb{1}(-1) \oplus M(\bar{\psi}^2). \tag{3.75}$$

Consider the cycle defined by

$$y := [\Gamma_{\sqrt{D_K}}] + D_K[\{O\} \otimes E] - [E \times \{O\}] \in CH^1(E \times E) \tag{3.76}$$

where $\Gamma_{\sqrt{D_K}}$ is the graph of the endomorphism $\sqrt{D_K}$ in $\mathrm{End}(E)$. Sec. 8 [IS03] proves that $y \in \mathrm{CH}^1(A, \epsilon_2)$ and will land in the $\mathbb{1}$-part of (3.75) under the cycle class map.

Let t be the number of prime factors of N and h the class number of K. Consider the following subgroups of $\mathrm{Aut}(X_H)$:

- $\mathcal{W} \cong (\mathbb{Z}/2\mathbb{Z})^t$ subgroup generated by *Atkin-Lehner involutions* ω_l for every l prime divisor of N,
- $\Delta = \mathrm{Pic}(\mathcal{O}_K)$ subgroup identified with $\mathrm{Gal}(H/K)$ via the Artin map.

By Lemma 2.5 [BD96] there are precisely $2^t h$ Heegner points with CM by $\mathcal{O}_K$ which are defined over the Hilber class field H of K and the group action of $\mathcal{W} \times \Delta$ on this set is free and transitive.

In order to construct the desired cycle classes in $\mathrm{CH}^{m+1}(\mathcal{M}_n)$ we need once again to pass through the fine moduli scheme X_M. Consider P closed Heegner point on X_H and let $\tilde{P}$ a closed point in the fiber $q^{-1}(P)$ in $(X_M)_H$. The fiber $\mathcal{A}_{\tilde{P}}$ above $\tilde{P}$ is an abelian surface with CM by $\mathcal{O}_K$. We can then consider $y_{\tilde{P}} \in \mathrm{CH}^1(\mathcal{A}_{\tilde{P}}, \epsilon_2)$ as defined above. For an abelian surface with CM, the cycle class map factors through algebraic equivalence, in particular we could modify the cycle $y_{\tilde{P}}$ by any element in $\mathrm{Pic}^0(\mathcal{A}_{\tilde{P}})$. We choose the representatives

$$z_{\tilde{P}} \in [y_{\tilde{P}}] \in \mathrm{NS}(\mathcal{A}_{\tilde{P}}) = \mathrm{Pic}(\mathcal{A}_{\tilde{P}})/\mathrm{Pic}^0(\mathcal{A}_{\tilde{P}}) \tag{3.77}$$

for every $\tilde{P}$ in such a way that they are compatible with the G-action, i.e.

$$g_*(z_{\tilde{P}}) = z_{g(\tilde{P})}. \tag{3.78}$$

Definition 3.6. *Define the cycle class* $y_P = y_P^{(2)} \in \mathrm{CH}^2((\mathcal{M}_n)_H)$ *to be the image of* $z_{\tilde{P}}$ *under*

$$\mathrm{CH}^1(\mathcal{A}_{\tilde{P}}, \epsilon_2) \xrightarrow{(p_G \circ \iota_{\tilde{P}})_*} \mathrm{CH}^2((\mathcal{M}_2)_H) \tag{3.79}$$

where $\iota_{\tilde{P}} : \mathcal{A}_{\tilde{P}} \hookrightarrow \mathcal{A}_H$. *For arbitrary even* $n \geq 2$, *first consider the m-th product of cycles under the projector* ϵ_n *as defined in* (3.11)

$$z_{\tilde{P}}^{(n)} = (\epsilon_n)_*(z_{\tilde{P}} \times \cdots \times z_{\tilde{P}}) \tag{3.80}$$

and then we define the cycle class $y_P^{(n)} \in \mathrm{CH}^{m+1}((\mathcal{M}_n)_H)$ *to be the image of* $z_{\tilde{P}}^{(n)}$ *under*

$$\mathrm{CH}^m(\mathcal{A}_{\tilde{P}}^m, \epsilon_n) \xrightarrow{(p_G \circ \iota_{\tilde{P}})_*} \mathrm{CH}^{m+1}((\mathcal{M}_n)_H). \tag{3.81}$$

Thanks to the compatibility under the G-action (3.78), the definition of y_P and $y_p^{(n)}$ is independent of the point $\tilde{P}$, but still depends on the set of compatible classes $z_{\tilde{P}}$.

3.8.1. *Heegner points on* $\mathcal{H}_p$. We briefly recall the description of p-adic analytic Heegner points on $\mathcal{H}_p$ following [BD98] and [IS03].

Definition 3.7. *An embedding* $\Psi : K \to B$ *is called* optimal *if* $\Psi(K) \cap R = \Psi(\mathcal{O}_K[1/p])$. *The group* Γ *acts by conjugation of the set of optimal embeddings and we denote* $\mathrm{emb}(\mathcal{O}_K, R)$ *the set of conjugacy classes* $[\Psi]$.

The set $\mathrm{emb}(\mathcal{O}_K, R)$ is endowed with an action of Δ and [BD98, Theorem 5.3] gives a bijection between the set of Heegner points on X_H and $\mathrm{emb}(\mathcal{O}_K, R)$ compatible with the Δ-action. Fix an embedding $H \hookrightarrow \mathbb{C}_p$ and let $v|p$ the corresponding place of H. Since p is inert in K and totally split in H, we then identify K_p and H_v with $\mathbb{Q}_{p^2}$. By Lemma 3.3, we can identify the classes in $\mathrm{emb}(\mathcal{O}_K, R)$ with K_p-points of X via

$$\mathrm{emb}(\mathcal{O}_K, R) \hookrightarrow \Gamma \backslash \mathrm{Hom}(K_p, M_2(\mathbb{Q}_p)) = X_\Gamma(K_p). \tag{3.82}$$

An Heegner point $P \in X_\Gamma(K_p)$ lies in the image of $\mathrm{emb}(\mathcal{O}_K, R)$ since the corresponding embedding Ψ_P can be defined by looking at the endomorphism ring of the associate CM abelian surface and at its reduction modulo p

$$\Psi_P : \mathcal{O}_K = \mathrm{End}(\mathcal{A}_P) \hookrightarrow \mathrm{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2}). \tag{3.83}$$

By the work of Čerednik–Drinfeld [BC91], we have $\mathrm{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2})[1/p] = R[1/p]$. This map is an optimal embedding by [GZ86, Prop. 7.3]. See [BD98] for the complete details.

4. de Rham cohomology and Bruhat-Tits cocycles

In order to compute the local p-adic component of Nekovar height, we need to understand the filtered (ϕ, N)-structure of the following cohomology groups

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n), \quad H^1_{\mathrm{dR}}(U, \mathcal{V}_n), \quad H^1_{\mathrm{dR}}(X_\Gamma, Z, \mathcal{V}_n), \quad H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$$

where $U \subseteq X_\Gamma$ is an open set and $Z \hookrightarrow X_\Gamma$ is a closed subscheme belonging to U. When Γ contains torsion, these groups are defined as follows: Pick a normal subgroup $\Gamma' \subset \Gamma$ which is torsion-free. Let $\pi : X_{\Gamma'} \longrightarrow X_\Gamma$ be the projection, $U' = \pi^{-1}(U)$ and $Z' = \pi^{-1}(Z)$, then define

$$\begin{aligned} H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) &:= H^1_{\mathrm{dR}}(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H^1_{\mathrm{dR}}(U, \mathcal{V}_n) &:= H^1_{\mathrm{dR}}(U', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H^1_{\mathrm{dR}}(X_\Gamma, Z, \mathcal{V}_n) &:= H^1_{\mathrm{dR}}(X_{\Gamma'}, Z', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) &:= H^1_{\mathrm{dR}}(U', Z', \mathcal{V}_n)^{\Gamma/\Gamma'}. \end{aligned}$$

These groups do not depend on the choice of Γ'. For the remainder of the paper, fix a choice of branch of p-adic logarithm $\log : \mathbb{C}_p^\times \longrightarrow \mathbb{C}_p$.

4.1. **The (ϕ, N)-module structure of H^1_{dR}.** In this section we recall the structure of $H^1_{\mathrm{dR}}(X, \mathcal{V}_n)$ as a (ϕ, N)-module following [IS03] and [Mas11]. Assume for now that Γ is torsion-free. Then the Mumford curve X_Γ over $\mathbb{Q}_p^{\mathrm{ur}}$ is a smooth projective curve with a semistable model $\mathfrak{X}_\Gamma$. Let $C_1, \dots C_r$ be its irreducible components of the special fiber, C, of $\mathfrak{X}_\Gamma$. Let $\mathfrak{G} = (\mathfrak{V}(\mathfrak{G}), \mathfrak{E}(\mathfrak{G}))$ the oriented intersection graph of C. Consider the reduction map

$$\mathrm{red} : X_\Gamma^{\mathrm{an}} \to C(\overline{\mathbb{F}}_p)$$

and define the admissible cover $\{U_v\}_{v \in \mathfrak{V}}$ where $U_v = \mathrm{red}^{-1}(C_v)$. For every edge $e \in \mathfrak{E}$ denote $s(e)$ the start vertex and $t(e)$ the target vertex. We can associate to each edge e a wide open annulus A_e given by

$$A_e = U_{s(e)} \cap U_{t(e)} = \mathrm{red}^{-1}(x)$$

where $x = C_{s(e)} \cap C_{t(e)}$.

4.2. **The general case.** Let $D = (\tilde{z}_1) + \cdots (\tilde{z}_r)$ and $Z = (\tilde{w}_1) + \cdots + (\tilde{w}_s)$ be disjoint divisors in X_Γ such that each of the points $\tilde{z}_i$ and $\tilde{w}_j$ reduces to a smooth point in $\mathfrak{X}_\Gamma$. We will denote by U the open subscheme $X_\Gamma - D$, and by

$$i : Z \hookrightarrow X_\Gamma$$

the closed subscheme associated with the same divisor. Our aim is to describe the (φ, N)-module structure on the relative de Rham cohomology group $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$.

Recall that $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ is defined as follows: Let $\Omega^\bullet_{(U,Z)}(\mathcal{V}_n)$ denote the cone associated to the morphism of complexes

$$i^* : \Omega^\bullet_U \otimes \mathcal{V}_n \to i_*(\Omega^\bullet_Z \otimes i^* \mathcal{V}_n).$$

Then $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ is the hypercohomology of the complex of sheaves $\Omega^\bullet_{(U,Z)}(\mathcal{V}_n)$.

For each $v \in \mathfrak{V}$, let $U_{v,D}$ denote the wide-open affinoid $U_v \backslash |D|$. Then the collection $(U_{v,D})_{v \in \mathfrak{V}}$ is an admissible covering of U. Computing $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ by Cech hypercohomology with this choice of covering, each cohomology class is represented by a cocycle

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \quad \text{for } \omega_v \in H^0(U_{v,D}, \Omega^1_{U_{v,D}} \otimes \mathcal{V}_n), \alpha \in H^0(Z, i^* \mathcal{V}_n), f_e \in H^0(A_e, \mathcal{V}_n),$$

which satisfy

$$\omega_{v_1}|_{A_e} - \omega_{v_2}|_{A_e} = df_e \quad \text{for every } e = (v_1, v_2) \in \mathfrak{E}.$$

A representative cocycle for a cohomology class is well-defined up to

$$\left((dg_v)_{v\in\mathfrak{V}}, (g_v)|_Z, (g_{s(e)} - g_{t(e)})_{e\in\mathfrak{E}}\right) \quad \text{for } g_v \in H^0(U_{v,D}, \mathcal{V}_n). \tag{4.1}$$

This implies that the map

$$\gamma_{U,Z} : H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \longrightarrow \ker(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E}})$$

given by

$$((\omega_v)_{v\in\mathfrak{V}}, \alpha, (f_e)_{e\in\mathfrak{E}}) \mapsto (\omega_v)_{v\in\mathfrak{V}}$$

is well-defined and surjective, where

$$H^i_{\mathfrak{V},D} = \bigoplus_{v\in\mathfrak{V}} H^i_{dR}(U_{v,D}, \mathcal{V}_n), \quad H^i_{\mathfrak{E}} = \bigoplus_{e\in\mathcal{E}} H^i_{dR}(A_e, \mathcal{V}_n).$$

The group $\ker(\gamma_{U,Z})$ is generated by elements of the form

$$((0)_{v\in\mathfrak{V}}, \alpha, (f_e)_{e\in\mathfrak{E}}) \quad \text{such that } df_e = 0,$$

modulo elements of the form in (4.1), under the condition that $dg_v = 0$ for all $v \in \mathfrak{V}$. Hence the kernel of $\gamma_{U,Z}$ is naturally identified with the quotient

$$\left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right) / H^0_{\mathfrak{V},D}.$$

Here $(H^0_{\mathfrak{e}})^-$ denotes the edge cochains which change sign when the orientation of an edge is reversed. We will denote this identification by $\iota_{U,Z}$ and note that we have an exact sequence

$$0 \to \left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right) / H^0_{\mathfrak{V},D} \xrightarrow{\iota_{U,Z}} H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma_{U,Z}} \ker(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E}}) \to 0. \tag{4.2}$$

Consider the retraction map

$$P_{U,Z} : H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \longrightarrow \left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right) / H^0_{\mathfrak{V},D}$$

defined on representatives by

$$((\omega_v)_{v\in\mathfrak{V}}, \alpha, (f_e)_{e\in\mathfrak{E}}) \mapsto \left((f_e - (F_{s(e)} - F_{t(e)}))_{e\in\mathfrak{E}}, \alpha - (F_v)|_Z\right),$$

where F_v is a choice of Coleman primitive of ω_v on $U_{v,D}$ with respect to the fixed choice of p-adic logarithm, which is well-defined up to constant sections of $\mathcal{V}_n|_{U_{v,D}}$. This defines a section

$$T_{U,Z} : \ker\left(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E}}\right) \longrightarrow H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$$

by

$$\alpha \mapsto \beta - \iota_{U,Z}(P_{U,Z}(\beta)),$$

where $\beta \in H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ is any element satisfying $\gamma_{U,Z}(\beta) = \alpha$.
Denote by

$$\varphi_1 : \ker(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E}}) \longrightarrow \ker(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E}})$$

and

$$\varphi_2 : \left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right) / H^0_{\mathfrak{V},D} \longrightarrow \left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right) / H^0_{\mathfrak{V},D}$$

the respective natural Frobenius action. The Frobenius action on $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ is defined by

$$\varphi(\omega) = \iota_{U,Z}(\varphi_2(P_{U,Z}(\omega))) + T_{U,Z}(\varphi_1(\gamma_{U,Z}(\omega))) \quad \text{for } w \in H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n).$$

Given $e \in \mathfrak{E}$, there is a natural residue map

$$\mathrm{Res}_e : H^1_{\mathrm{dR}}(A_e, \mathcal{V}_n^{an}) \longrightarrow H^0_{dR}(A_e, \mathcal{V}_n^{an}).$$

The monodromy operator is defined as

$$N(\omega) := \iota_{U,Z}\left(((\mathrm{Res}_e(\omega|_{A_e}))_{e\in\mathfrak{E}}, 0)\right) \quad \text{for } \omega \in H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n). \tag{4.3}$$

Proposition 4.1. *The admissible filtered (φ, N)-module $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ admits a slope decomposition*

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) = H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m \oplus H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_{m+1} \tag{4.4}$$

and $\mathrm{im}(\iota_{U,Z}) = H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m$, $\mathrm{im}(T_{U,Z}) = H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_{m+1}$. *Furthermore, there is a decomposition*

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) = F^{m+1}_{U,Z} \oplus H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m. \tag{4.5}$$

Here $F^{m+1}_{U,Z} = H^0(X_\Gamma, \Omega^1_{X_\Gamma}(\log(D)) \otimes \mathcal{V}_n)$, which is also identified with the vector space of $\mathcal{V}_n$-valued meromorphic differentials on U modulo exact differentials.

4.3. **Pulling back to $\mathcal{H}_p$.** By pulling back to $\mathcal{H}_p$, the decomposition of $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ obtains a more explicit combinatorial form, which allows us to define a (φ, N)-module for Γ containing torsion. For now, assume as before that Γ is torsion free.

Let $\mathcal{H}_p := \mathbb{P}^1(\mathbb{C}_p)\backslash\mathbb{P}^1(\mathbb{Q}_p)$ denote the p-adic upper half-plane. It is a rigid analytic space with an action of $\mathrm{GL}_2(\mathbb{Q}_p)$ by Mobius transformation.

Denote by $\mathcal{T}_p = (\mathcal{T}_p^0, \mathcal{T}_p^1)$ the Bruhat-Tits tree with vertex and oriented edge set $\mathcal{T}_p^0$ and $\mathcal{T}_p^1$, respectively. It is equipped with the usual action of $\mathrm{GL}_2(\mathbb{Q}_p)$, and there is a $\mathrm{GL}_2(\mathbb{Q}_p)$-equivariant reduction map

$$\mathrm{red}_p : \mathcal{H}_p \longrightarrow \mathcal{T}_p.$$

Let $C^0(V_n)$ (resp. $C^1(V_n)$) denote the $\mathrm{GL}_2(\mathbb{Q}_p)$ module of maps $f : \mathcal{T}_p^0 \to V_n$ (resp. $f : \mathcal{T}_p^1 \to V_n$ such that $f(e) = -f(\bar{e})$ for every oriented edge $e \in \mathcal{T}_p^1$)). Define the map

$$\Delta : C^0(V_n) \longrightarrow C^1(V_n)$$

by

$$\Delta(f)(e) = f(s(e)) - f(t(e)) \quad \text{for } e \in \mathcal{T}_p^1,$$

where $s(e)$ and $t(v)$, respectively, are the start and end vertices of e. This map sits in a short exact sequence

$$0 \longrightarrow V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0. \tag{4.6}$$

Let $\nabla : C^1(V_n) \longrightarrow C^0(V_n)$ be the $\mathrm{GL}_2(\mathbb{Q}_p)$-equivariant map obtained by

$$\nabla(f)(v) = \sum_{s(e)=v} f(e) \quad \text{for } f \in C^1(V_n),\ v \in \mathcal{T}_p^0,$$

where the sum runs over all edges starting at the vertex v. Let C^1_{har} denote the kernel of ∇. It sits in the exact sequence

$$0 \longrightarrow C^1_{har}(V_n) \longrightarrow C^1(V_n) \longrightarrow C^0(V_n) \longrightarrow 0.$$

The oriented intersection graph, $\mathfrak{G}$, is canonically identified with $\Gamma\backslash\mathcal{T}_p$. For each vertex $v \in \mathcal{T}_p^0$, let $N(v)$ be the neighborhood of v in $\mathcal{T}_p$. The reduction maps and the quotient by the action of Γ fit into the commutative diagram

$$\begin{array}{ccc} \mathcal{H}_p & \xrightarrow{\mathrm{red}_p} & \mathcal{T}_p \\ \downarrow & & \downarrow \\ X_\Gamma^{\mathrm{an}} & \xrightarrow{\mathrm{red}} & \mathfrak{G}. \end{array}$$

It follows that for each vertex $v \in \mathfrak{V}(\mathfrak{G})$, the wide open affinoid domain U_v is isomorphic to $\mathbb{P}^1$ minus a set of disjoint disks, and for each edge $e \in \mathfrak{E}(\mathfrak{G})$, the preimage A_e is indeed a wide open annulus. Therefore, there are compatible isomorphisms

$$C^0(V_n)^\Gamma \cong H^0_{\mathfrak{V}}$$

and

$$C^1(V_n)^\Gamma \cong \left(H^0_{\mathfrak{E}}\right)^-.$$

Let $w_i \in \mathcal{H}_p$, be any choice of lift of $\tilde{w}_i \in X_\Gamma(\mathbb{C}_p) \cong \mathcal{H}_p/\Gamma$. It follows that there is an isomorphism

$$\mathrm{Func}\left(\bigcup_{i=1}^{s} \Gamma \cdot w_i, V_n\right)^\Gamma \cong H^0(Z, i^*\mathcal{V}_n).$$

Note then that $\left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right)/H^0_{\mathfrak{V},D}$ is identified with the first cohomology of the total complex of the bi-complex

$$\begin{array}{ccc} C^1(V_n)^\Gamma & \longrightarrow & 0 \\ \uparrow & & \uparrow \\ C^0(V_n)^\Gamma & \longrightarrow & \mathrm{Func}(\bigcup_i \Gamma \cdot w_i, V_n)^\Gamma. \end{array} \tag{4.7}$$

Since the complex

$$V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0$$

is a free resolution of V_n, the cohomology of (4.7) computes the Γ-hypercohomology of the complex

$$\mathcal{K}_Z(V_n) = \left(V_n \longrightarrow \mathrm{Func}\left(\bigcup_i \Gamma \cdot w_i, V_n\right)\right),$$

where the arrow indicates the map

$$\alpha \mapsto (w \mapsto \alpha).$$

There is an isomorphism

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong \left((H^0_{\mathfrak{e}})^- \oplus H^0(Z, i^*\mathcal{V}_n)\right)/H^0_{\mathfrak{V},D}.$$

Definition 4.1. *Let $\{z_1, \dots, z_r\} \subset \mathcal{H}_p$ be a fixed choice of lifts of the points $\tilde{z}_i$ to $\mathcal{H}_p$, and write*

$$\mathcal{H}_U := \mathcal{H}_p - \bigcup_{i=1}^{r} \Gamma \cdot z_i.$$

Let $\mathcal{T}_U$ be the tree whose set of vertices is $\mathcal{T}_p^0 \cup \bigcup_{i=1}^r \Gamma \cdot z_i$, and whose set of edges is the set of edges of $\mathcal{T}_p$ along with an edge e_w for each $w \in \bigcup_{i=1}^r \Gamma \cdot z_i$, where the edge e_w connects the vertex associated with w and the vertex $\mathrm{red}_p(w)$.
Let $\mathcal{T}_U^1$ denote the set of oriented edges of $\mathcal{T}_U$, and $C^1_U(V_n)$ be the set of functions

$$c : \mathcal{T}_U^1 \longrightarrow V_n$$

satisfying

$$c(\bar{e}) = -c(e) \quad \textit{for all } e \in \mathcal{T}_U^1,$$

and

$$\sum_{s(e)=v} c(e) = 0 \quad \textit{for all } v \in \mathcal{T}_p^0, \tag{4.8}$$

where the sum in (4.8) runs over all edges in $\mathcal{T}_U^1$ whose starting vertex is v.

Proposition 4.2. *There is a natural identification*

$$C^1_U(V_n)^\Gamma \cong \ker(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{E},D}).$$

Proof. The group $\ker\left(H^1_{\mathfrak{V},D} \to H^1_{\mathfrak{e}}\right)$ is isomorphic to

$$\ker\left(\left(\bigoplus_{v \in \mathcal{T}_p^0} H^1_{\mathrm{dR}}((\mathrm{red}_p|_{\mathcal{H}_U})^{-1}(N(v)), \mathcal{V}_n^{an})\right)^\Gamma \to \left(\bigoplus_{e \in \mathcal{T}_p^1} H^1_{\mathrm{dR}}(\mathrm{red}_p^{-1}(e), \mathcal{V}_n^{an})\right)^\Gamma\right). \tag{4.9}$$

For every $v \in \mathcal{T}_p^0$, the pre-image

$$\left(\mathrm{red}_p|_{\mathcal{H}_U}\right)^{-1}(N(v))$$

is $\mathbb{P}^1(\mathbb{C}_p)$ minus a set of $(p+1)$ disjoint disks and minus a finite set of points. Each disk corresponds to an edge in $N(v)$ and each removed point corresponds to some $w \in \bigcup_{i=1}^{r} \Gamma \cdot z_i$ with $\mathrm{red}_p(w) = v$. It follows that an element of the de Rham cohomology group

$$H^1_{\mathrm{dR}}\left(\left(\mathrm{red}_p|_{\mathcal{H}_U}\right)^{-1}(N(v)), \mathcal{V}_n^{an}\right)$$

is determined by the residues on the removed disks and on the removed points, with the restriction that the sum of residues is 0. □

It follows that we have an exact sequence

$$0 \to H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \to 0 \tag{4.10}$$

The natural restriction map

$$C^1_U(V_n)^\Gamma \longrightarrow C^1(V_n)^\Gamma$$

defines a map

$$\mathrm{res} : C^1_U(V_n) \longrightarrow H^1(\Gamma, \mathcal{K}_Z(V_n))$$

and the monodromy operator can be described as the composition

$$N : H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \xrightarrow{\mathrm{res}} H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n).$$

4.4. **Removing the restriction on Γ.** If Γ is not necessarily torsion free, choose some normal subgroup $\Gamma' \subset \Gamma$, which is torsion free and let $G = \Gamma/\Gamma'$. Let

$$r : X_{\Gamma'} \longrightarrow X_\Gamma$$

be the natural reduction map. Let $D' = r^*D, Z' = r^*Z$, and $U' = r^{-1}(U)$. We define

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) := H^1_{\mathrm{dR}}(U', Z', \mathcal{V}_n)^G.$$

The group G also acts on $H^1(\Gamma, \mathcal{K}_{Z'}(V_n))$ and $C^1_{U'}(V_n)$. It is clear that the action of G commutes with the maps Frobenius action as well as the maps $\iota'_{U',Z'}, \gamma'_{U,Z}, P'_{U,Z}.T'_{U,Z}$ and therefore also the monodromy operator N on $= H^1_{\mathrm{dR}}(U', Z', \mathcal{V}_n)$.

Proposition 4.3. *There is an exact sequence*

$$0 \to H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C^1_U(V_n)^\Gamma \to 0.$$

Proof. We begin with the analogous exact sequence for Γ' and take G-invariance to obtain

$$0 \to H^1(\Gamma, \mathcal{K}_{Z'}(V_n))^G \xrightarrow{\iota'_{U',Z'}} H^1_{\mathrm{dR}}(U', Z', \mathcal{V}_n)^G \xrightarrow{\gamma'_{U',Z'}} (C^1_{U'}(V_n)^\Gamma)^G \to H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))).$$

Since G is a finite group, and $H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n)))$ is a vector space over a field of characteristic zero, and hence is divislbe, it follows that

$$H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))) = 0.$$

We remark that there $\mathcal{K}_{Z'} = \mathcal{K}_Z$ and $C^1_U(V_n) = C^1_{U'}(V_n)$. It follows by definition that

$$H^1_{\mathrm{dR}}(U', Z', \mathcal{V}_n)^G = H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$$

and it is immediate that

$$(C^1_{U'}(V_n)^\Gamma)^G = C^1_U(V_n)^{\Gamma'}.$$

By the inflcation-restriction sequence for group hypercohomology we obtain an exact sequence

$$0 \to H^1(G, H^0(\Gamma', \mathcal{K}_{Z'})) \to H^1(\Gamma, \mathcal{K}_Z(V_n)) \to H^1(\Gamma', \mathcal{K}'_Z(V_n))^G \to H^2(G, H^0(\Gamma', \mathcal{K}_{Z'}(V_n)).$$

Again, we note that G is a finite group and $H^0(\Gamma', \mathcal{K}_{Z'}(V_n)$ is a divisible group, and so it follows that

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong H^1(\Gamma', \mathcal{K}'_Z(V_n))^G,$$

and the result holds. □

Proposition 4.4. *For Γ not necessarily torsion-free, $H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)$ admits a (φ, N)-module structure that is independent of the choice of Γ', and that there is a slope decomposition*

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) = H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m \oplus H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_{m+1},$$

where

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m \cong H^1(\Gamma, \mathcal{K}_Z(V_n)),$$

and

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_{m+1} \cong C^1_U(V_n)^\Gamma.$$

There is also a decomposition

$$H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n) = \mathrm{Fil}^{m+1}_{U,Z} \oplus H^1_{\mathrm{dR}}(U, Z, \mathcal{V}_n)_m,$$

where $\mathrm{Fil}^{m+1}_{U,Z}$ is the space of Γ-invariant meromorphic differentials on $\mathcal{H}_p$ with poles supported on the set $\bigcup_{i=1}^r \Gamma \cdot z_i$ modulo exact differentials.

Proof. See [IS03, Appendix 10]. □

4.5. **Poincaré Duality.** As before, let $D_1 = (\tilde{z}_1) + \cdots + (\tilde{z}_r)$ and $D_2 = (\tilde{w}_1) + \cdots + (\tilde{w}_s)$ be two divisors on X_Γ defined over a field K. Let U_j denote the open subscheme $X_\Gamma - |D_j|$ for $j \in \{1, 2\}$ and

$$i_j : Z_j \longrightarrow X_\Gamma$$

denote the closed subscheme associated to the divisor D_j for $j \in \{1, 2\}$. Let $j, l \in \{1, 2\}$ be two distinct elements. We remark that the complex $\Omega^\bullet_{(U_j, Z_l)}(\mathcal{V}_n)$, whose hypercohomology computes $H^1_{\mathrm{dR}}(U_j, Z_l, \mathcal{V}_n)$, is easily seen to be quasi-isomorphic to its subcomplex

$$[\mathcal{O}_{U_j}(-D_l) \otimes \mathcal{V}_n \to \Omega^1_{U_j} \otimes \mathcal{V}_n].$$

It follows that $H^1_{\mathrm{dR}}(U_j, Z_l, \mathcal{V}_n)$ is computed as the hypercohomology of the complex

$$[\mathcal{O}_{X_\Gamma}(-D_l) \otimes \mathcal{V}_n \to \Omega^1_{X_\Gamma}(\log(D_j)) \otimes \mathcal{V}_n].$$

Using this complex, we have the usual cup product in de Rham cohomology

$$\langle \cdot, \cdot \rangle_{D_1, D_2} : H^1_{\mathrm{dR}}(U_1, Z_2, \mathcal{V}_n) \times H^1_{\mathrm{dR}}(U_2, Z_1, \mathcal{V}_n) \xrightarrow{\cup} H^2_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n \otimes \mathcal{V}_n) \xrightarrow{tr} K[2m+1]. \tag{4.11}$$

compatible with the (φ, N)-structure. In particular, from the slope decomposition of Proposition 4.4 we have that (4.11) induces a perfect pairing

$$\langle \cdot, \cdot \rangle_{D_1, D_2} : H^1_{\mathrm{dR}}(U_1, Z_2, \mathcal{V}_n)_m \times H^1_{\mathrm{dR}}(U_2, Z_1, \mathcal{V}_n)_{m+1} \to K[2m+1]. \tag{4.12}$$

By the identifications of Proposition 4.4, this is the perfect pairing

$$(-, -)_\Gamma : C^1_{U_j}(V_n)^\Gamma \times H^1(\Gamma, \mathcal{K}_{Z_j}(V_n)) \to K \tag{4.13}$$

described in [IS03, Section 10.2]. We recall briefly its explicit description. Fix $\Gamma' \subseteq \Gamma$ a free subgroup with generators $\gamma_1, \ldots, \gamma_t$. Let $\mathcal{F}$ be a good fundamental domain for $\mathcal{T}_p/\Gamma$ and $b_1, \cdots, b_t, c_1, \cdots, c_t$ its free edges in the sense of de Shalit [Sha89]. The pairing of $\mathfrak{f} = (f, f_1, \cdots, f_r) \in C^1_{U_1}(V_n)^\Gamma$ and $\mathfrak{g} = [(g, g_1, \cdots, g_r)] \in H^1(\Gamma, \mathcal{K}^\bullet_{D_1}(V_n))$ is given by

$$(\mathfrak{f}, \mathfrak{g})_\Gamma = \frac{1}{[\Gamma : \Gamma']} \sum_{i=1}^t (f(\gamma_i), g(c_i))_\Gamma - \sum_{i=1}^t \frac{(f_i, g_i)_\Gamma}{\#\Gamma_{z_i}} \tag{4.14}$$

where Γ_{z_i} denotes the stabilizer of z_i.

Lemma 4.1. *For all $\omega \in H^1_{\mathrm{dR}}(U_1, Z_2, \mathcal{V}_n)$ and $\eta \in H^1_{\mathrm{dR}}(U_2, Z_1, \mathcal{V}_n)$ we have*

$$(\omega, \eta) = (P_{D_2}(\omega), I_{U_2}(\eta))_\Gamma - (I_{U_1}(\omega), P_{D_1}(\eta))_\Gamma. \tag{4.15}$$

Proof. This follows from the orthogonality of the slope decomposition and the identifications

$$H^1_{\mathrm{dR}}(U_i, Z_j, \mathcal{V}_n)_m \cong H^1(\mathcal{K}^\bullet_{Z_j}(V_n)), \quad H^1_{\mathrm{dR}}(U_i, Z_j, \mathcal{V}_n)_{m+1} \cong C^1_{U_i}(V_n)^\Gamma. \tag{4.16}$$

See [IS03, Theorem 10.6]. □

Proposition 4.5. *The pairing $\langle\cdot,\cdot\rangle_{D_1,D_2}$ is non-degenerate. Furthermore, the subspaces $\mathrm{Fil}^{m+1}_{U_j,Z_l}$ and $\mathrm{Fil}^{m+1}_{U_l,Z_j}$ are orthogonal complements. This induces a perfect pairing*

$$\mathrm{Fil}^{m+1}_{U_j,Z_l} \times H^1_{\mathrm{dR}}(U_l, Z_j, \mathcal{V}_n)/\mathrm{Fil}^{m+1}_{U_l,Z_j} \to K. \tag{4.17}$$

Proof. The key input is Serre duality which implies that the cup products

$$H^1(X_\Gamma, \mathcal{O}_{X_\Gamma}(-D_j)\otimes\mathcal{V}_n) \times H^0(X_\Gamma, \Omega^1_{X_\Gamma}(\log(D_j)\otimes\mathcal{V}_n) \xrightarrow{\cup} H^1(X_\Gamma, \Omega^1_{X_\Gamma}) \xrightarrow{tr} K$$

are non-degenerate for each $j \in \{1,2\}$. The proof follows *mutatis mutandis* the proof of Poincaré duality. See [The26] □

5. Nekovář's p-adic heights

In this section we recall the construction of Nekovář's p-adic height pairing in the semistable case [Nek93].

5.1. **Global p-adic height pairing.** Let K be a number field and let $G_K = \mathrm{Gal}(\overline{K}/K)$ be its absolute Galois group. Let L be a finite extension of $\mathbb{Q}_p$ and consider V a p-adic Galois representation of G_K over L. We will denote by V_v the local p-adic representation of $G_{K_v} = \mathrm{Gal}(\overline{K}_v/K_v)$ obtained by restriction from V. Suppose that

- V is unramified for all but a finite set of places of K,
- $H^i(K_v, V) = H^i(K_v, V^\vee(1)) = 0$ for $i = 0,1,2$ and for all places $v \nmid p$,
- V_v is a semistable representation of $\mathrm{Gal}(\overline{K}_v/K_v)$ for all $v|p$,
- $D_{\mathrm{st}}(V_v)^{\varphi=1,N=0} = D_{\mathrm{st}}(V_v^\vee(1))^{\varphi=1,N=0} = 0$.

The definition of the *local p-adic height* will depend on the choice of the following two objects:

- a continuous idele class character $l_K : \mathbb{A}_K^\times/K^\times \to \mathbb{Q}_p$ such that for each $v|p$ we have a factorization

$$\begin{array}{ccc} \mathcal{O}_{K_v}^\times & \xrightarrow{\quad l_v \quad} & \mathbb{Q}_p \\ & {\scriptstyle \log_v} \searrow \quad \nearrow {\scriptstyle \mathrm{tr}} & \\ & K_v & \end{array} \tag{5.1}$$

- for each $v|p$ a choice of a subspace $W_v \subset DR(V_v)$ complementary to the filtered piece Fil^0, i.e. a decomposition $DR(V_v) = W_v \oplus \mathrm{Fil}^0(\mathrm{DR}(V_v))$.

For the remaining of the paper we fix the continuous *cyclotomic character* l_K^{cyc} defined as follows:

$$l_K^{\mathrm{cyc}} : \mathbb{A}_K^\times/K^\times \xrightarrow{\mathrm{rec}_K} \mathrm{Gal}(\overline{K}/K) \xrightarrow{\chi_p^{\mathrm{cyc}}} \mathbb{Z}_p^\times \xrightarrow{\log_p} \mathbb{Q}_p \tag{5.2}$$

where rec_K is the usual reciprocity map and $\log_p$ is the Iwasawa branch of the p-adic logarithm, i.e. normalized such that $\log_p(p) = 0$. This character satisfies the property $l_K^{\mathrm{cyc}} = l_{\mathbb{Q}}^{\mathrm{cyc}} \circ N_{K/\mathbb{Q}}$ and in particular we have the following description place by place

$$l_{K,v}^{\mathrm{cyc}}(x) = \begin{cases} \log_p(N_{K_v/\mathbb{Q}_p}) & v|p, \\ -\mathrm{ord}_v(x)\log_p(\#k_v) & v \nmid p. \end{cases} \tag{5.3}$$

We will refer to l_K^{cyc} simply as l_K.

Nekovář [Nek93, Theorem 2.2] constructed a family of continuous bilinear pairings

$$h : H^1_f(K, V) \times H^1_f(K, V^\vee(1)) \to \mathbb{Q}_p. \tag{5.4}$$

Let $e_1 \in H^1_f(K, V)$ and $e_2 \in H^1_f(K, V^\vee(1))$ be two Selmer classes, these correspond to semistable extensions of Galois representations

$$\begin{aligned} e_1 : \qquad & 0 \longrightarrow V \longrightarrow E_1 \longrightarrow \mathbb{Q}_p \longrightarrow 0 \\ e_2 : \qquad & 0 \longrightarrow V^\vee(1) \longrightarrow E_2 \longrightarrow \mathbb{Q}_p \longrightarrow 0. \end{aligned} \tag{5.5}$$

There exist a set of Galois representations called *biextensions* which fit in to a diagram with exact row and columns

$$\begin{array}{ccccccccc} & & & & 0 & & 0 & & \\ & & & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2^\vee(1) & \longrightarrow & V & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 & \longrightarrow & 0 \\ & & & & \downarrow & & \downarrow & & \\ & & & & \mathbb{Q}_p & = & \mathbb{Q}_p & & \\ & & & & \downarrow & & \downarrow & & \\ & & & & 0 & & 0 & & \end{array} \tag{5.6}$$

Every choice of such biextension E determines a splitting of the height pairing in its *local components*

$$h(e_1, e_2) = \sum_v h(e_1, e_v)_{E,v}. \tag{5.7}$$

5.1.1. *p-adic heights at $v \nmid p$.* First of all observe that by continuity, every idele character restricted to v

$$l_v : K_v^\times \to \mathbb{Q}_p \tag{5.8}$$

is unramified, i.e. it has to factors through $\mathrm{ord}_v : K_v \to \mathbb{Z}$. Under the Kummer isomorphism, the map l_v induces an isomorphism

$$(\mathrm{ord}_v)_* : H^1(K_v, \mathbb{Q}_p(1)) \xrightarrow{\sim} K_v^\times \hat{\otimes} \mathbb{Q}_p \xrightarrow{\sim} \mathbb{Q}_p \tag{5.9}$$

Furthermore, the assumption $H^1(K_v, V) = H^1(K_v, V^\vee(1)) = 0$ simplifies the biextension diagram because it implies that the extensions e_1, e_2 are split and consequently the biextension E_v can be written as $E_v = V_v \oplus U_v$ where U_v is a simple extension of G_{K_v} p-adic representation

$$[0 \to \mathbb{Q}_p(1) \to U_v \to \mathbb{Q}_p \to 0] \in H^1(K_v, \mathbb{Q}_p(1)). \tag{5.10}$$

The *local height pairing* is then defined to be

$$h(e_1, e_2)_{E,v} = \log_p(\#k_v)\, \mathrm{ord}_v([U_v]). \tag{5.11}$$

5.1.2. *p-adic heights at $v|p$.* The case for local heights at places divinding p is more involved and uses the structure of (φ, N)-modules. The top short exact sequence of the local version of the biextension diagram (5.6) induces a commutative diagram (Sec. 3.5 [Nek93]) with exact rows and columns

(5.12)

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & D_{\mathrm{st}}(E_{2,v}^{\vee}(1))^{\varphi=1} & \xrightarrow{\sim} & D_{\mathrm{st}}(V_v)^{\varphi=1} & & \\
 & & & & \downarrow{\scriptstyle (0,N,-i)} & & \downarrow{\scriptstyle (N,-i)} & & \\
0 & \longrightarrow & \mathbb{Q}_p \oplus K_v & \longrightarrow & \mathbb{Q}_p \oplus D_{\mathrm{st}}(V_v(-1))^{\varphi=1} \oplus DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0 & \longrightarrow & D_{\mathrm{st}}(V_v(-1))^{\varphi=1} \oplus DR(V_v)/\mathrm{Fil}^0 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H^1_{\mathrm{st}}(K_v, E_{2,v}^{\vee}(1)) & \longrightarrow & H^1_{\mathrm{st}}(K_v, V_v) & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

where we have denoted by i the inclusion map $D_{\mathrm{st}}(V_v) \hookrightarrow DR(V_v)$ and the same for $E_{2,v}^{\vee}(1)$. Recall that this map depends on the choice of a branch of the p-adic logarithm.

Under the assumption

$$D_{\mathrm{st}}(V_v)^{\varphi=1,N=0} = D_{\mathrm{st}}(V_v^{\vee}(1))^{\varphi=1,N=0} = 0 \tag{5.13}$$

the embeddings of the DR components give isomorphisms

$$DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0 \cong \mathrm{coker}(D_{\mathrm{st}}(E_{2,v}^{\vee}(1))^{\varphi=1} \to D_{\mathrm{st}}(V_v(-1))^{\varphi=1} \oplus DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0) \tag{5.14}$$

$$DR(V_v)/\mathrm{Fil}^0 \cong \mathrm{coker}(D_{\mathrm{st}}(V_v)^{\varphi=1} \to D_{\mathrm{st}}(V_v(-1))^{\varphi=1} \oplus DR(V_v)/\mathrm{Fil}^0). \tag{5.15}$$

This implies that from the exact diagram (5.12), it can be extracted the following commutative diagram with exact rows

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathbb{Q}_p \oplus K_v & \longrightarrow & \mathbb{Q}_p \oplus DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0 & \longrightarrow & DR(V_v)/\mathrm{Fil}^0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \\
0 & \longrightarrow & H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H^1_{\mathrm{st}}(K_v, E_{2,v}^{\vee}(1)) & \longrightarrow & H^1_{\mathrm{st}}(K_v, V_v) & \longrightarrow & 0
\end{array} \tag{5.16}$$

where the vertical isomorphisms are often referred to as *Bloch-Kato exponential maps*. This isomorphisms are explicitly given by Lemma 2.1 and Lemma 2.2. It is important to recall that the $\mathbb{Q}_p$ summands are mapped isomorphically under the first horizontal map simplifying to commutative diagram of the form

$$\begin{array}{ccccccccc}
0 & \longrightarrow & K_v & \longrightarrow & DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0 & \longrightarrow & DR(V_v)/\mathrm{Fil}^0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \\
0 & \longrightarrow & H^1_{\mathrm{f}}(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H^1_{\mathrm{f}}(K_v, E_{2,v}^{\vee}(1)) & \longrightarrow & H^1_{\mathrm{f}}(K_v, V_v) & \longrightarrow & 0.
\end{array} \tag{5.17}$$

Recall that the definition of the *p-adic height* depends on the choice of the following two objects the continuous homomorphism $l_v : K_v^{\times} \to \mathbb{Q}_p$ and a choice of a subspace $W_v \subset DR(V_v)$ complementary to the filtered piece Fil^0. In particular, the choice of the subspace W_v induces a splitting of the top exact sequence as follows

$$w' : DR(V_v)/\mathrm{Fil}^0(V_v) \xrightarrow{W_v} DR(V_v) \xrightarrow{\alpha} DR(E_{2,v}^{\vee}(1)) \to DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0(E_{2,v}^{\vee}(1)) \tag{5.18}$$

where the leftmost map is induced by the decomposition $DR(V_v) = W_v \oplus \mathrm{Fil}^0$ and α denotes the unique (φ, N)-equivariant splitting

$$\begin{array}{ccccccccc}
 & & & & & \overset{w'}{\curvearrowleft} & & & \\
0 & \longrightarrow & K_v & \longrightarrow & DR(E_{2,v}^{\vee}(1))/\mathrm{Fil}^0 & \longrightarrow & DR(V_v)/\mathrm{Fil}^0 & \longrightarrow & 0.
\end{array} \tag{5.19}$$

The splitting of the exact sequence induces a retraction in the bottom exact sequence of (5.16)

$$w : H^1_{\mathrm{st}}(K_v, E_2) \to H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)). \tag{5.20}$$

Recall that the *Kummer map* determines a canonical isomorphism

$$H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)) = K_v^\times \hat{\otimes} \mathbb{Q}_p. \tag{5.21}$$

Under this identification, the inverse of the left most vertical map of the diagram (5.19) is given by the *Bloch-Kato logarithm map*

$$\begin{aligned} H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)) &\to \mathbb{Q}_p \oplus K_v \\ x &\mapsto (v(x), \log_p(x)) \end{aligned} \tag{5.22}$$

for the given fixed choice of the branch of the p-adic logarithm used to define $B_{\mathrm{st}} \subset B_{\mathrm{dR}}$ (See [Nek93, 1.35]). We are now ready to give the definition of the p-adic height.

Definition 5.1. *The* local p-adic height *of the biextension* $[E_v] \in H^1_{\mathrm{st}}(K, E_2)$ *as in* (5.6) *is defined to be*

$$h(e_1, e_2)_{E,v} = -l_v(w([E_v]). \tag{5.23}$$

In the next section we will specialize this to the case of biextensions arising from Heegner cycles and we will provide an explicit formula for the local p-adic height.

5.2. **p-adic heights on the Shimura curve.** We now want to study the extensions of filtered (φ, N)-modules obtained from the extensions of p-adic Galois representations considered in Section 3.2. Recall that by Theorem 3.4 we have the following isomorphism

$$D_{\mathrm{st}}(H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G) = H^1_{\mathrm{dR}}(X_M \otimes \mathbb{Q}_p, \mathcal{V}_n)^G. \tag{5.24}$$

This isomorphism is defined over the field of definition $\mathbb{Q}$ of X_M. From the discussion and results of Section 3.7, we have that we can explicitly characterize the F-isocrystal $\mathcal{V}_n = \mathbb{D}(\mathbb{L}_n)$ after base changing to $\mathbb{Q}_{p^{\mathrm{unr}}}$. In order to construct the p-adic height of a biextension we need, as explained in the previous Section, to describe the unique splitting of (5.17). We will later characterize this splitting after the base change of $\mathbb{Q}_{p^{\mathrm{unr}}}$ so we will not need to worry and carry on the field of definition of all the objects. Recall that under this assumption we have that we can translate the de Rham cohomology computation to the curve X_Γ.

Let K be a number field unramified at p and fix a place v above p and K_v its corrsponding completion. Let D be a reduced effective divisor on X_M defined over K such that all closed points reduce to distinct points on X under the map $q : X_M \to X$, we have an extension

$$e_{\text{ét}} : 0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to H^1_{\text{ét}}((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x \in D} (\mathbb{L}_n)^{G_x}_x(m) \to 0 \tag{5.25}$$

. Applying the functor D_{st} corresponding to the local representations at v, thanks to Theorem 3.6, we get an extension in $\mathrm{MF}_{K_v}(\varphi, N)$

$$e_{\mathrm{dR}} : 0 \to H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] \to H^1_{\mathrm{dR}}(X_\Gamma - |D_\Gamma|, \mathcal{V}_n)[-m-1] \to \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)^{\Gamma_z}_z[-m] \to 0. \tag{5.26}$$

Recall from (2.8), the relation between the effect of the Tate twist in the two categories. By Theorem 2.1, the equivalence of categories between semistable p-adic representations and admissible filtered (φ, N)-modules given by Fontaine's functor D_{st} is compatible with exact sequences, tensor product and duality. See in particular the discussion in Subsection 3.7 for the compatibility between the different pairings. Recall that from Subsection 3.5, the dual exact sequence $e^\vee_{\text{ét}}$ is given by

$$e^\vee_{\text{ét}} : 0 \to \bigoplus_{x \in D} (\mathbb{L}_n)^{G_x}_x(m) \to H^1_{\text{ét},|D|}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \to 0 \tag{5.27}$$

while the corresponding dual sequence $e^\vee_{\mathrm{dR}}$ is given by

$$e^\vee_{\mathrm{dR}} : 0 \to \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)^{\Gamma_z}_z[-m] \to H^1_{\mathrm{dR}}(X_\Gamma, D_\Gamma, \mathcal{V}_n)[-m] \to H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m] \to 0. \tag{5.28}$$

Considering two different reduced disjoint divisors D_1, D_2 satisfying previous assumptions we can consider the biextension diagram constructed in Subsection 3.5 in the category $\mathrm{MF}_{K_v}(\varphi, N)$

(5.29)

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{z\in D_2}(\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{z\in D_2}(\mathcal{V}_n)_z^{\Gamma_z}[-m-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma - |D_1|, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma - |D_1|, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \oplus_{z\in D_1}(\mathcal{V}_n)_z^{\Gamma_z}[-m] & = & \oplus_{z\in D_1}(\mathcal{V}_n)_z^{\Gamma_z}[-m] & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}.$$

We now want to consider only simple extensions of p-adic Galois representations by $\mathbb{Q}_p$ and $\mathbb{Q}_p(1)$ and the corresponding simple extensions of filtered (φ, N)-modules. In Section 3.8 we have recalled the construction of special Heegner classes $\mathrm{CH}^m(\mathcal{A}^m_{\tilde{P}}, \epsilon_n)$ attached to a Heegner point P. Under the p-adic Abel-Jacobi map, these classes can be used to produce extension classes

$$cl_0 : \mathrm{CH}^{m+1}(\mathcal{M}_n) \to \mathrm{Ext}^1_{\underline{\mathrm{Rep}}_{\mathrm{st}}(G_K)}(\mathbb{Q}_p, H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G) \tag{5.30}$$

Applying the functor D_{st} this produces a simple extension class in

$$\mathrm{Ext}^1_{\mathrm{MF}^{\mathrm{ad}}_{K_v}(\varphi,N)}(K_v, H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1]). \tag{5.31}$$

5.3. **Heegner biextension.** In [IS03], the authors explicitly characterized these latter extension classes arising from Heegner cycles.

Let P be a Heegner point on X defined over K quadratic imaginary number field of discriminant D_K. Let $\psi : K \to B$ be an optimal embedding corresponding to P as in Lemma 3.3. Let $z_{\tilde{P}} \in \mathrm{CH}^m(\mathcal{A}^m_{\tilde{P}}, \epsilon_n)$ be the Heegner class defined in Definition 3.6. Let $P_\psi(X, Y) \in \mathcal{P}_2$ the homogeneous polynomial in two variable defined by

$$P_\psi(X, Y) = P_{\psi(\sqrt{D_K})}(X, Y) = \mathrm{tr}\left(\psi\left(\sqrt{D_K}\right)\begin{pmatrix} X \\ Y \end{pmatrix}\begin{pmatrix} Y & -X \end{pmatrix}\right) \tag{5.32}$$

as in the isomorphism described in Lemma 3.2

Lemma 5.1. *Let $w_\psi, \overline{w_\psi}$ be the two points on $\mathcal{H}_p(\mathbb{Q}_{p^2})$ fixed by the action of $\psi(K)$, then*

$$P_\psi(X, Y) = -\frac{2\sqrt{D_K}}{w_\psi - \overline{w_\psi}}(X - w_\psi Y)(X - \overline{w_\psi} Y). \tag{5.33}$$

Proof. This follows from the standard theory of the quadratic forms. Let $\psi(\sqrt{D_K}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the matrix representation in $M_2(\mathbb{Q}_p)$. We then have

$$P_\psi(X, Y) = -cX^2 + (a - d)XY + bY^2. \tag{5.34}$$

The fixed points $w_\psi, w_{\overline{\psi}}$ are roots of P_ψ, they have eigenvalues $\sqrt{D_k}, -\sqrt{D_k}$ and therefore

$$cw_\psi + d = \sqrt{D_K}, \quad c\overline{w_\psi} + d = -\sqrt{D_K}. \tag{5.35}$$

The associated polynomial is equal to

$$P_\psi(X, Y) = -\frac{\sqrt{D_k} - (-\sqrt{D_k})}{w_\psi - \overline{w_\psi}}(X - w_\psi Y)(X - \overline{w_\psi} Y). \tag{5.36}$$

□

Lemma 5.2. *The composition of the p-adic Abel-Jacobi map*

$$cl : \mathrm{CH}^m(\mathcal{A}^m_{\bar{P}}, \epsilon_n) \to \mathbb{L}_{n,P}(m) \tag{5.37}$$

with the canonical isomorphisms of (φ, N)-modules

$$D_{\mathrm{st},\mathbb{Q}_p^{\mathrm{unr}}}(\mathbb{L}_n(m)_{\bar{P}}) = (\mathcal{V}_n)_\psi[-m] \tag{5.38}$$

sends $z^{(n)}_{\bar{P}}$ to the element $(-, P^m_\psi) \in V_n = \mathcal{P}_n^\vee$.

Proof. See [IS03, Lemma 8.4] □

We can now define the *Heegner biextension* we will study. Let K_1, K_2 two quadratic imaginary satisfying the Heegner hypothesis 3.1 and $\psi_i \in \mathrm{emb}(\mathcal{O}_{K_i}, R)$ for $i = 1, 2$ two optimal embeddings to which correspond P_{ψ_i} Heegner points. These points are defined over the Hilbert class fields H_i. Consider the Heegner classes $z_{\overline{P}_i} \in \mathrm{CH}^m(\mathcal{A}^m_{\bar{P}}, \epsilon_n)$. We consider the divisors

$$D_{\psi_i} = (P_{\psi_i}) + (P_{\overline{\psi}_i}) \quad \text{for } i = 1, 2 \tag{5.39}$$

and the cycle class maps

$$\begin{aligned} \mathrm{cl}_{\psi_i} : \mathbb{Q}_p &\to (\mathbb{L}_n)_{P_{\psi_i}}(m) \oplus (\mathbb{L}_n)_{P_{\overline{\psi}_i}}(m) \\ 1 &\mapsto \left(\mathrm{cl}(z^{(n)}_{P_{\psi_i}}), (-1)^{m+1} \mathrm{cl}(z^{(n)}_{P_{\overline{\psi}_i}}) \right) \qquad \text{for } i = 1, 2. \end{aligned} \tag{5.40}$$

It is important to remark that the point $P_{\overline{\psi}_i}$ corresponds to $w_P(\overline{P}_{\psi_i})$ where w_p is the Atkin-Lehner involution and $\overline{P}_{\psi_i}$ is the action of complex conjugation. See [BC91].

Lemma 5.3. *Let $z_i, \bar{z}_i$ the points on $\mathcal{H}_p$ corresponding to the optimal embeddings $\psi_i, \overline{\psi}_i$. Under the reduction map modulo p, the two points specialize to the same vertex and the cycle class map defines a morphism of filtered φ-modules*

$$\mathrm{cl}_{\psi_i} : \mathbb{Q}_p^{\mathrm{unr}} \to (V_n)_{z_i}[-m] \oplus (V_n)_{\bar{z}_i}[-m] \tag{5.41}$$

such that, denoting $Q_i = \mathrm{cl}_{\psi_i}(1)$ we obtain

$$(Q_i)_{z_i} + (Q_i)_{\bar{z}_i} = 0. \tag{5.42}$$

Proof. See [IS03, Theorem 9.2]. □

The *Heegner biextension* is then obtained from (5.29) taking the pullback and pushout along the cycle class maps cl_{ψ_i}.

5.4. ***p*-adic heights as Poincaré pairing on de Rham cohomology.** In this section we will study how the p-adic pairing described in Section 5.1.2 can be explicitly computed in terms of the Poincaré pairing of special classes in de Rham cohomology. As an important assumption, as explained at the beginning of the previous Section, we will work over the maximal unramified field extension $\mathbb{Q}_p^{\mathrm{unr}}$.

We start by fixing $D_1, D_2 \hookrightarrow X_\Gamma$ two disjoint effective reduced divisors on X_Γ. Assume that every closed point of D_1, D_2 is defined over $\mathbb{Q}_{p^2}$. A simple extension class of $H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1]$ by $\mathbb{Q}_p^{\mathrm{unr}}$ can be constructed as the pullback along a map of filtered φ-modules $\mathrm{cl}_i : \mathbb{Q}_p^{\mathrm{unr}} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m]$ of the Gysin exact sequence

(5.43)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & L_i & \longrightarrow & \mathbb{Q}_p^{\mathrm{unr}} & \longrightarrow & 0 \\ & & \| & & \downarrow{\scriptstyle \iota_{L_i}} & & \downarrow{\scriptstyle \mathrm{cl}_i} & & \\ 0 & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \xrightarrow{\iota} & H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n)[-m-1] & \xrightarrow{\mathrm{Res}} & \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m] & \longrightarrow & 0. \end{array}$$

The notation cl is reminiscent of the fact that we will later construct such extensions from the Heegner classes previously described.

Lemma 5.4. *The exact sequence of filtered (φ, N)-modules*

(5.44)
$$0 \longrightarrow H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] \xrightarrow{\iota} H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n)[-m-1] \xrightarrow{\mathrm{Res}} \oplus_{z\in D_i}(\mathcal{V}_n)_z^{\Gamma_z}[-m] \longrightarrow 0.$$

admits a unique Frobenius-monodromy equivariant splitting

$$s_i : \bigoplus_{z\in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[1] \to H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n) \tag{5.45}$$

where $s_i(Q) = \omega_Q \in H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n)$ is the unique element satisfying

(i) $\omega_Q \in H^1_{\mathrm{dR}}(X_\Gamma - D_i\mathcal{V}_n)_{m+1}$
(ii) $\mathrm{Res}(\omega_Q) = Q$,
(iii) $N(\omega_Q) = 0$.

Proof. First of all observe that $\bigoplus_{z\in D_i}(\mathcal{V}_n)_z^{\Gamma_z}[1]$ is isotrivial of slope $m+1$ since Frobenius acts by $\det^{\otimes m}$ on $V_n\{m\}$ and has trivial monodromy. In particular, Frobenius acts on every element Q as by $(-1)^{m+1}p^{m+1}\sigma$. See Section 3.6. By the description in Proposition 4.4 and the constructions of the Frobenius stable maps I, P in Section 4, we have a commutative diagram with exact rows and columns:

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) & \xrightarrow{I} & C_{\mathrm{har}}(V_n)^\Gamma & \longrightarrow & 0 \\
 & & \| & \overset{P}{\curvearrowleft} & \downarrow \iota & & \downarrow & & \\
0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma - D, \mathcal{V}_n) & \xrightarrow{I_U} & C_U(V_n)^\Gamma & \longrightarrow & 0 \\
 & & & \overset{P_U}{\curvearrowleft} & \downarrow \mathrm{Res} & \overset{T_U}{\curvearrowleft} & \downarrow & & \\
 & & & & \oplus_{z\in D}(\mathcal{V}_n)_z[1] & \longrightarrow & \oplus_{z\in D}\mathrm{Ind}^\Gamma(V_n)^\Gamma & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & H^1(\Gamma, C_{\mathrm{har}}(V_n)). & &
\end{array} \tag{5.46}$$

From the diagram we then see that the intersection of the trivial monodromy $N = 0$ and the slope $m+1$ cuts out a non trivial subspace. The trivial monodromy will indeed land in $T_U(C_U(V_n)^\Gamma)$ where T_U is the section of the middle row induced by P_U. Suppose ω, ω' are two elements satisfying the previous properties then their difference lie in

$$\omega - \omega' \in \iota(H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)_{m+1})^{N=0} = \{0\}$$

concluding the uniqueness by the fact that N defines a bijection on the global cohomology decomposition $N : H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)_{m+1} \to H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)_m$. □

Lemma 5.5. *Under the Bloch-Kato exponential map* (2.10) *we have an isomorphism*

$$\mathrm{Ext}^1_{\mathrm{MF}^{\mathrm{ad}}_{\mathbb{Q}_p^{\mathrm{unr}}}(\varphi,N)}(\mathbb{Q}_p^{\mathrm{unr}}, H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1]) \cong (\mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n))^\vee \cong M_k(X_\Gamma)^\vee \tag{5.47}$$

and the extension class $[L_i]$ of (5.43) *is identified with the element $(-, \eta_i) \in \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)^\vee$ where $\eta_i \in H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)$ is the unique element* $\mod \mathrm{Fil}^{m+1}$ *satisfying*

$$\iota(\eta_i) \equiv s_i(\mathrm{cl}_i(1)) \mod \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n) \tag{5.48}$$

where $s_i(\mathrm{cl}_i(1)) \in H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n)$ is the element defined by the unique (φ, N)-equivariant splitting of Lemma 5.4.

Proof. First of all, by the Bloch-Kato exponential the LHS of (5.47) is identified with $H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)/\mathrm{Fil}^{m+1}$ and by Proposition 4.5 it is isomorphic to $\mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)^\vee$. Furthermore we have the identification between filtered piece and modular forms thanks to [IS03, Proposition 6.1].

Consider the pullback diagram (5.43). The construction in Lemma 5.4 induces a Frobenius-monodromy splitting of the top row of the diagram

$$s : \mathbb{Q}_p^{\mathrm{unr}} \to L_i \qquad 1 \mapsto s_i(\mathrm{cl}_i(1)). \tag{5.49}$$

From the explicit construction of the Bloch-Kato discussed after Lemma 2.1, we have that ι induces an isomorphism

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)/\mathrm{Fil}^{m+1} \cong L_i/\mathrm{Fil}^{m+1}L_i \tag{5.50}$$

and η_i is defined to be the corresponding element in $H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)$ such that

$$\iota(\eta_i) \equiv s_i(\mathrm{cl}_i(1)) \mod \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n) \tag{5.51}$$

concluding the description of the extension class $[L_i]$ under the isomorphism (5.47). □

Similarly, we can describe the extension class of the middle column of (5.29) using Lemma 2.2. Fix two classes $\mathrm{cl}_i : \mathbb{Q}_p^{\mathrm{unr}} \to \bigoplus_{z\in D_1}(\mathcal{V}_n)_z^{\Gamma_z}[-m]$, let Q_1, Q_2 the respective images of the element 1. Consider the dual shifted extension of L_2 that can be obtained as the push-out along the cl map

(5.52)

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \bigoplus_{z\in D_2}(\mathcal{V}_n)_z^{\Gamma_z} & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]) & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \| & & \\
0 & \longrightarrow & \mathbb{Q}_p^{\mathrm{unr}}[-1] & \longrightarrow & L_2^\vee[-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0
\end{array}$$

In the previous top exact sequence we have used the fact that the relative de Rham cohomology is dual to the open cohomology under the Poincaré pairing described in Section 4.5. Construct the biextension obtained from the two classes cl_i pulling back and pushing out the diagram (5.29)

(5.53)

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p^{\mathrm{unr}}[-1] & \longrightarrow & L_2^\vee[-1] & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p^{\mathrm{unr}}[-1] & \longrightarrow & L & \longrightarrow & L_1 & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \mathbb{Q}_p^{\mathrm{unr}} & = & \mathbb{Q}_p^{\mathrm{unr}} & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

Lemma 2.2 explicitly identifies the extension class $[L]$ as an element in

$$\mathrm{Ext}^1_{\mathrm{MF}^{\mathrm{ad}}_{\mathbb{Q}_p^{\mathrm{unr}}}(\varphi,N)}(\mathbb{Q}_p^{\mathrm{unr}}, L_2^\vee[-1]) \cong \mathbb{Q}_p \oplus L_2^\vee[-1]/\mathrm{Fil}^0(L_2^\vee[-1]) \tag{5.54}$$

where $L_2[-1]$ satisfies all of the assumptions of Lemma 2.2. Observe that, by the Poincaré pairing and the interplay with filtration (Proposition 4.5), we can identify the group of extensions with

$$L_2^\vee[-1]/\mathrm{Fil}^0(L_2^\vee[-1]) \cong (\mathrm{Fil}^0 L_2)^\vee. \tag{5.55}$$

The extension class corresponding to L in (5.53) can be determined as follows.

Lemma 5.6. *Let ι_{L_2} be the injection of L_2 in $H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)$ (5.43) and ι be the injection of $H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)$ in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)$. The element $[L]$ in the extension group of $L_2^\vee[-1]$ by $\mathbb{Q}_p^{\mathrm{unr}}$ is given by*

$$(N_L, (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\mathrm{Fil}^0 L_2)^\vee \tag{5.56}$$

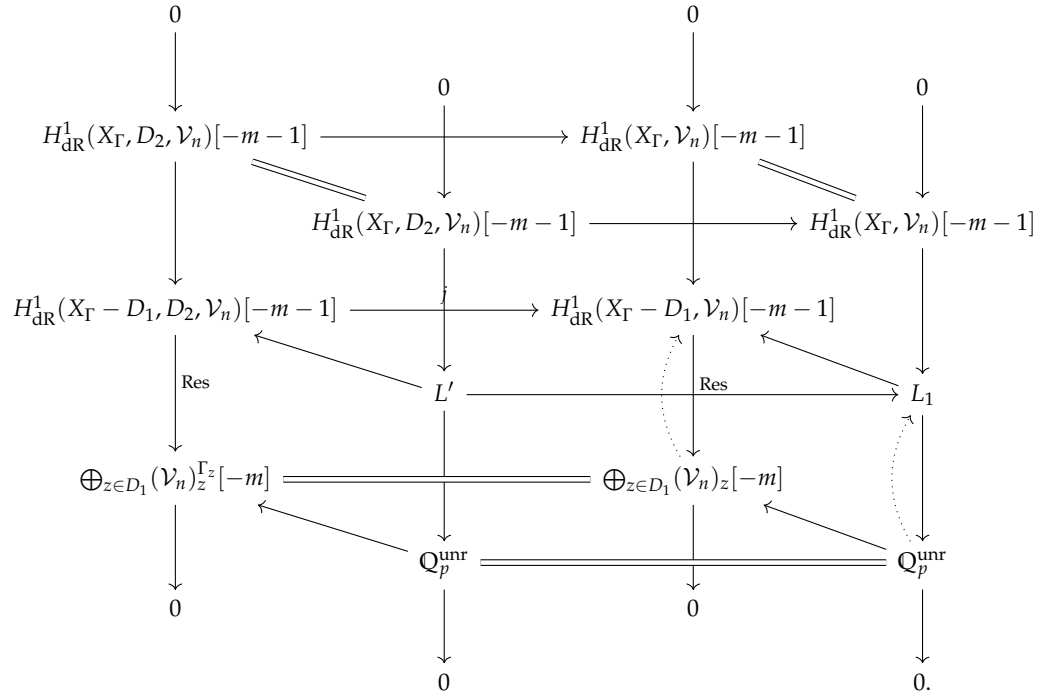
where $N_L \in \mathbb{Q}_p$, and $\eta_L \in H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)$ satisfies the following properties

(i) $\iota(\eta_L) \equiv \omega_{Q_1} \mod \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ *for* $\omega_{Q_1} \in H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)_{m+1}$,
(ii) $\mathrm{Res}(\omega_{Q_1}) = Q_1$,
(iii) $N(j(\omega_{Q_1})) = 0$ *in* $H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n)$.

The cohomology class ω_{Q_1} is uniquely characterized by these properties.

Proof. This time we need to apply Lemma 2.2, the general version of Lemma 2.1. First of all, we consider the pull-back of the last two columns of the biextension diagram (5.29) along the map $\mathrm{cl}_i : \mathbb{Q}_p^{\mathrm{unr}} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[-m]$ obtaining the following diagram.

(5.57)



As defined in Lemma 5.4, we have a Frobenius-monodromy equivariant splitting of the extension corresponding to L_1 given by (5.45) and denoted by the dottet arrows in diagram (5.57). We cannot hope to define such a splitting for the extension corresponding to L' since the (φ, N)-module $H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$ contains a non-trivial submodule with trivial monodromy $N = 0$. In fact, $H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$ is obtained as an extension of $H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1]$ by $\bigoplus_{z \in D_2} (\mathcal{V}_n)_z^{\Gamma_z}[-m-1]$. As in Lemma 2.2 and in the discussion that follows, we construct a φ-equivariant splitting of the surjective map Res

$$s : \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[-m] \to H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)[-m-1] \tag{5.58}$$

that we can describe as follows. First of all we use the commutative diagram described by decomposing the cohomology groups according to Proposition 4.4

(5.59)

$$\begin{array}{ccccc}
C_{\mathcal{U}_1}(V_n)^\Gamma & = & C_{\mathcal{U}_1}(V_n)^\Gamma & & \\
\uparrow{\scriptstyle I_{U_1}} & & \uparrow{\scriptstyle I_{U_1}} & & \\
H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) & \xrightarrow{j} & H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n) & \longrightarrow & 0 \\
\uparrow & & \uparrow & & \\
H^1(\Gamma, \mathcal{K}^\bullet_{D_2}(V_n)) & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & 0.
\end{array}$$

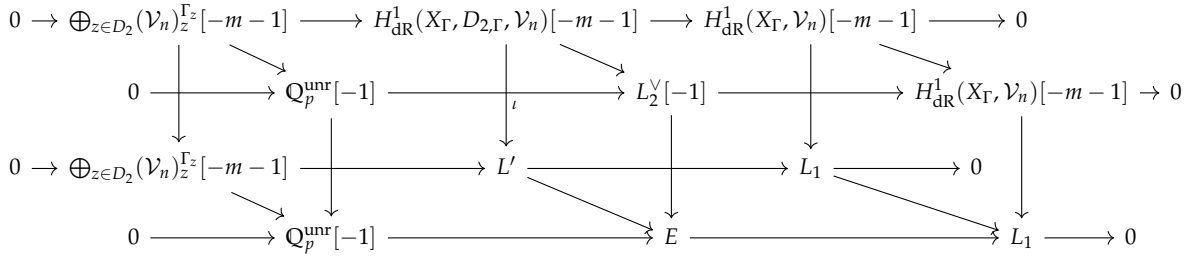
In this way, we can identify the following subspaces

$$H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)_{m+1} \cong H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n)_{m+1} \cong C_{U_1}(V_n)^\Gamma \tag{5.60}$$

where the isomorphisms are given by the surjection j and the Frobenius equivariant maps I_U. For every $Q \in \bigoplus_{z\in D_1}(\mathcal{V}_n)_z[1]$, define $s(Q) \in H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ to be the class defined in Lemma 5.4. Let now $\omega_{Q_1} = s(Q_1)$ be the class in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)_{m+1}$, this is characterize by the properties (ii), (iii) of the Lemma. It's important to recall that the $j(\omega_{Q_1})$ has trivial monodromy in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n)$ but it can acquire it in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ justifying once again the distinction between the Bloch-Kato exponential maps in Lemma 2.1 and Lemma 2.2.

Consider now the push-out of (5.57) along the map $\mathrm{cl}_2^\vee : \bigoplus_{z\in D_2}(\mathcal{V}_n)_z^{\Gamma_z}[-m-1] \to \mathbb{Q}_p^{\mathrm{unr}}[-1]$

(5.61)



Observe that we have a φ-stable decomposition of L induced by the Frobenius splitting s that is given by

$$L = \mathbb{Q}_p^{\mathrm{unr}}[-1] \oplus H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)[-m-1] \oplus \mathbb{Q}_p^{\mathrm{unr}} \tag{5.62}$$

and the element ω_{Q_1} lands in the $\mathbb{Q}_p^{\mathrm{unr}}$-part. We now observe that under the inclusion

$$\iota : H^1_{\mathrm{dR}}(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \to L' \tag{5.63}$$

we have an isomorphism of $\mathbb{Q}_{p^{\mathrm{unr}}}$-vector spaces given by

$$H^1_{\mathrm{dR}}(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)/\mathrm{Fil}^{m+1} \cong L'/\mathrm{Fil}^0(L') \tag{5.64}$$

that means that there exists a class $\eta_L \in H^1_{\mathrm{dR}}(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)$ satisfying the following property

$$\iota(\eta_L) \equiv \omega_{Q_1} \mod \mathrm{Fil}^0(L'). \tag{5.65}$$

We can then define $\tilde{\eta}_L$ to be the image of η_L under the surjection map

$$H^1_{\mathrm{dR}}(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)/\mathrm{Fil}^{m+1} \to L_2^\vee[-1]/\mathrm{Fil}^0(L_2^\vee[-1]). \tag{5.66}$$

The RHS of the last equation is isomorphic to $(\mathrm{Fil}^{m+1}L_2)^\vee$ under Poincaré duality. Recall that L_2 injects into

$$\iota_{L_2} : L_2 \hookrightarrow H^1_{\mathrm{dR}}(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \tag{5.67}$$

as filtered (φ, N)-modules and that we have a Poincaré pairing (4.11)

$$H^1_{\mathrm{dR}}(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \otimes H^1_{\mathrm{dR}}(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m] \to \mathbb{Q}_p^{\mathrm{unr}} \tag{5.68}$$

This implies that element $\tilde{\eta}_L$ can be identified with the functional

$$\tilde{\eta}_L = (\tilde{\eta}_L, -) = (\eta_L, \iota_{L_2}(-)) \in (\mathrm{Fil}^0 L_2)^\vee. \tag{5.69}$$

Under the Bloch-Kato exponential map we can then identify the class $[E]$ with

$$[L] = (N_L, (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\mathrm{Fil}^0 L_2)^\vee \tag{5.70}$$

where N_L is the projection onto the first component of $N(\omega_{Q_1})$. This concludes the proof. □

Recall from Proposition 4.1 that we have a decomposition of the de Rham cohomology by

$$H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) = \mathrm{Fil}^{m+1} \oplus H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n)_m. \tag{5.71}$$

This will be our choice of decomposition of the de Rham realization needed to define the p-adic height pairing.

Definition 5.2. *For every $Q \in \bigoplus_{z\in D_i}(\mathcal{V}_n)_z^{\Gamma_z}$ define the* Coleman-Gross differential *associated to Q to be the unique class*

$$\Psi_Q \in \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_i, \mathcal{V}_n) \tag{5.72}$$

such that $\mathrm{Res}(\Psi_Q) = Q$ *and* $N(\Psi_Q) = 0$.

Theorem 5.1. *The local p-adic height pairing of the biextension $[E]$ obtained from the class maps cl_i is given by*

$$h(e_1, e_2)_{E,v} = -\mathrm{Tr}_{K_v/\mathbb{Q}_p}(\eta_L, \Psi_{Q_2}) \tag{5.73}$$

where $\eta_L \in H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)$ is the class described in Lemma 5.6.

Proof. Recall from Section 5.1.2 that for a biextension E defined over a number field K where p is unramified and a place $v|p$, the local p-adic height of the class $[E] \in H^1_{\mathrm{st}}(K_v, E_2)$ is defined by

$$h(e_1, e_2)_{E,v} = -l_v(w([E_v]). \tag{5.74}$$

where w is the splitting of the bottom exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Q}_p \oplus K_v & \longrightarrow & \mathbb{Q}_p \oplus DR(E_2)/\mathrm{Fil}^0 & \longrightarrow & DR(V)/\mathrm{Fil}^0 & \longrightarrow & 0 \\ & & \downarrow\sim & & \downarrow\sim & \overset{w'}{\dashleftarrow} & \downarrow\sim & & \\ 0 & \longrightarrow & H^1_{\mathrm{st}}(K_v, \mathbb{Q}_p(1)) & \longrightarrow & H^1_{\mathrm{st}}(K_v, E_2) & \longrightarrow & H^1_{\mathrm{st}}(K_v, V) & \longrightarrow & 0 \\ & & & \overset{w}{\dashleftarrow} & & & & & \end{array} \tag{5.75}$$

where the p-adic Galois representations are given by

$$V = H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G(m+1) \tag{5.76}$$

and E_2 is the push-out of the dual of the extension

$$0 \to H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to H^1_{\text{ét}}((X_M - D_2) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x\in D}(\mathbb{L}_n)_x^{G_x}(m) \to 0 \tag{5.77}$$

under the class map cl_2. The map w is induced by an explicit splitting of the upper exact sequence. Recall that K_v is mapped isomorphically to itself through the first horizontal map (5.17). Furthermore, we can identify under Poincaré duality the quotients in the upper exact sequence as

$$0 \longrightarrow K_v \longrightarrow (\mathrm{Fil}^0\mathrm{DR}(E_2))^\vee \longrightarrow (\mathrm{Fil}^0\mathrm{DR}(V))^\vee \longrightarrow 0. \tag{5.78}$$

In order to study and explicitly describe the unique splitting of this exact sequence, we now pass to the maximal unramified extension $\mathbb{Q}_{p^{\mathrm{unr}}}$. In this way, we can apply the explicit description in terms of the analytic quotient X_Γ. In particular we obtain a sequence

$$0 \longrightarrow \mathbb{Q}_p^{\mathrm{unr}} \longrightarrow (\mathrm{Fil}^0 L_2)^\vee \longrightarrow (\mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n))^\vee \longrightarrow 0. \tag{5.79}$$

inducing a dual exact sequence contained in the usual Gysin extension

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \mathrm{Fil}^0 L_2 & \longrightarrow & \mathbb{Q}_p^{\mathrm{unr}} & \longrightarrow & 0 \\ & & \| & \overset{w'^\vee}{\dashleftarrow} & \downarrow{\scriptstyle \iota_{E_2}} & & \downarrow{\scriptstyle \mathrm{cl}_2} & & \\ 0 & \longrightarrow & \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) & \xrightarrow{\iota} & \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n) & \xrightarrow{\mathrm{Res}} & \bigoplus_{z\in D_2}\mathrm{Fil}^m(\mathcal{V}_n)_z^{\Gamma_z} & \longrightarrow & 0. \end{array} \tag{5.80}$$

By Lemma 5.6 we have that the class of $[L]$, under the Bloch-Kato isomorphism, is send to

$$(N_L, (\eta_L, \iota_{L_2}(-))) \in \mathbb{Q}_p \oplus (\mathrm{Fil}^0 L_2)^\vee \tag{5.81}$$

with $\eta_L \in H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)$ satisfying prescribed properties of the Lemma. In Subsection 5.1.2 we have described a section of the top exact sequence that Nekovář constructs in [Nek93]. Explicitly this is given by

$$w'^\vee : \mathrm{Fil}^0 L_2 \longrightarrow L_2 \xrightarrow{\alpha} H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \longrightarrow \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \tag{5.82}$$

where the first map is the canonical inclusion, α is the unique (φ, N)-equivariant splitting of the canonical inclusion and the last morphism it the projection according to the decomposition (5.71). This map exists in more generality as the composition of the equivalent maps

(5.83)
$$\begin{array}{ccccccc} \mathrm{Fil}^0 L_2 & \longrightarrow & L_2 & \xrightarrow{\alpha} & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) \\ \downarrow & & \downarrow & & \| & & \| \\ \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma, \mathcal{V}_n). \end{array}$$

The Frobenius-monodromy equivariant splitting is given by Lemma 5.4. Let $\omega_{Q_2} \in H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n)$ the (φ, N)-equivariant lift of Q_2. The splitting of the dual filtered exact sequence (5.80) can be then given as

$$\mathbb{Q}_p^{\mathrm{unr}} \to \mathrm{Fil}^0 L_2 \tag{5.84}$$
$$1 \mapsto \Psi_{Q_2} = \omega_{Q_2} - \eta_{Q_2} \tag{5.85}$$

for $\eta_Q \in H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n)_m \cong H^1(\Gamma, V_n)$ the unique slope m element such that

$$\omega_Q \equiv \eta_Q \mod \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n). \tag{5.86}$$

In particular, Ψ_{Q_2} is the Coleman-Gross differential associated to Q_2. We can then see that the splitting of the exact sequence (5.75) is given by

$$\begin{aligned} \mathbb{Q}_p \oplus DR(E_2)/\mathrm{Fil}^0 &\to \mathbb{Q}_p \oplus K_v \\ (N_L, (\eta_L, \iota_{L_2}(-))) &\mapsto (N_L, (\eta_L, \Psi_{Q_2})). \end{aligned} \tag{5.87}$$

Recall that we chose $l_v = \log_p \circ N_{K_v/\mathbb{Q}_p}$ to be the cyclotomic logarithm along the Iwasawa branch $\log_p(p) = 0$. Under the Kummer isomorphism (5.21), we conclude that the p-adic height is given by the projection of the second component of (5.87) composed with the trace map. This concludes the proof. □

Remark 5.1. *It is important to point out that a different choice of branch of the p-adic logarithm will introduce a correction by the monodromy factor N_L.*

We conclude this part giving an explicit description of the Poincaré pairing, and then of the height pairing, in some favorable conditions.

Theorem 5.2. *Consider the class maps $\mathrm{cl}_i : \mathbb{Q}_p^{\mathrm{unr}} \to \bigoplus_{z \in D_i} (\mathcal{V}_n)_z^{\Gamma_z}[1]$ with $\mathrm{cl}_i(1) = Q_i$ and suppose that D_1 reduces to a unique vertex and $\sum_{z \in D_1} Q_{1,z}/\#\Gamma_z = 0$. The local p-adic height of the biextension $[E]$ can be computed by*

$$h(e_1, e_2)_{E,v} = \mathrm{Tr}_{K_v/\mathbb{Q}_p}\left(\int_{D_1} (\Psi_{Q_2}, Q_1) dz\right) = \mathrm{Tr}_{K_v/\mathbb{Q}_p}\left(\sum_{z \in D_1} \frac{(F_{\Psi_{Q_2}}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}\right). \tag{5.88}$$

Proof. By Lemma 4.1 We have

$$(\eta_L, \Psi_{Q_2}) = (P_{D_2}(\eta_L), I_{U_2}(\Psi_{Q_2}))_\Gamma - (I(\eta_L), P(\Psi_{Q_2}))_\Gamma. \tag{5.89}$$

Since we have a decomposition $H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n) = \mathrm{Fil}^{m+1} \oplus H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)_m$ and modifying η_L up to Fil^{m+1} will not change the cup product. We can assume

$$\eta_L \in H^1_{\mathrm{dR}}(X_\Gamma, D_2, \mathcal{V}_n)_m = \ker(I) \tag{5.90}$$

simplifying the previous expression to

$$(\eta_L, \Psi_{Q_2}) = (P(\eta_L), I_{U_2}(\Psi_{Q_2}))_\Gamma. \tag{5.91}$$

Consider now $\Psi_{Q_1} \in \mathrm{Fil}^{m+1} H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ the element that, under condition (i) of Lemma 5.6, is given by

$$\Psi_{Q_1} = \omega_{Q_1} - \iota(\eta_L). \tag{5.92}$$

This element satisfies the following properties:

- $\mathrm{Res}(\Psi_{Q_1}) = \mathrm{Res}(\omega_{Q_1}) = Q_1$,
- $P(\Psi_{Q_1}) = -P(\eta_L)$,
- $I_{U_1}(\Psi_{Q_1}) = I_{U_1}(\omega_{Q_1})$,

and it is the Coleman-Gross differential associated to the differential Q_1 of Definition 5.2 under the identification

$$\mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) = \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n). \tag{5.93}$$

Since the Poincaré pairing is orthogonal with respect to the filtration Fil^{m+1} we obtain

$$0 = (\Psi_{Q_1}, \Psi_{Q_2}) = (P(\Psi_{Q_1}), I_{U_2}(\Psi_{Q_2}))_\Gamma - (I_{U_1}(\Psi_{Q_1}), P(\Psi_{Q_2}))_\Gamma = \tag{5.94}$$

$$= -(P(\eta_L), I_{U_2}(\Psi_{Q_2}))_\Gamma - (I_{U_1}(\omega_{Q_1}), P(\Psi_{Q_2}))_\Gamma \tag{5.95}$$

where we have used the identification

$$\mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_2, \mathcal{V}_n) = \mathrm{Fil}^{m+1}H^1_{\mathrm{dR}}(X_\Gamma - D_2, D_1, \mathcal{V}_n). \tag{5.96}$$

This simplifies the initial pairing (5.89) to the computation

$$(\eta_L, \Psi_{Q_2}) = -(I_{U_1}(\omega_{Q_1}), P_{D_1}(\Psi_{Q_2}))_\Gamma. \tag{5.97}$$

Explicitly we have

$$P_{D_1}(\Psi_{Q_2}) = ([F_{\Psi_{Q_2}}], (F_{\Psi_{Q_2}}(z))_{z\in D_1}), \tag{5.98}$$

$$I_{U_1}(\omega_{Q_1})(\gamma_z)_{z\in D_1} = ((\mathrm{Res}_e(\omega_{Q_1}))_{e\in\mathfrak{E}}, (\mathrm{Res}_{\gamma_z(z)}(\omega_{Q_1}))_{z\in D_1}). \tag{5.99}$$

It's important to note that from the construction of Lemma 5.6 we have that the element ω_{Q_1} maps under j to the unique element in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n)_{m+1} = \ker(P) \cong C_{U_1}(V_n)^\Gamma$ with trivial monodromy

$$\begin{array}{ccccc} C_{U_1}(V_n)^\Gamma & = & C_{U_1}(V_n)^\Gamma & & \\ {\scriptstyle I_{U_1}}\uparrow\downarrow & & {\scriptstyle I_{U_1}}\uparrow\downarrow & & \\ H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) & \xrightarrow{j} & H^1_{\mathrm{dR}}(X_\Gamma - D_1, \mathcal{V}_n) & \longrightarrow & 0 \\ \uparrow\downarrow{\scriptstyle P_{D_2}} & & \uparrow\downarrow{\scriptstyle P} & & \\ H^1(\Gamma, \mathcal{K}^\bullet_{D_2}(V_n)) & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & 0 \end{array} \tag{5.100}$$

where the dotted arrows are Frobenius equivariant.

Fix $\Gamma' \subseteq \Gamma$ a free subgroup with generators $\gamma_1, \ldots, \gamma_g$. Let $\mathcal{F}$ be a good fundamental domain for $\mathcal{T}_p/\mathcal{F}$ and $b_1, \cdots, b_g, c_1, \cdots, c_g$ its free edges in the sense of de Shalit [Sha89]. The pairing is then given by the explicit formula (4.14)

$$(\eta_L, \Psi_{Q_2}) = -\frac{1}{[\Gamma:\Gamma']}\sum_{i=1}^{g}([F_{\Psi_{Q_2}}](\gamma_1), (\mathrm{Res}_{c_i}(\omega_{Q_1})))_\Gamma - \sum_{z\in D_1}\frac{(F_{\Psi_{Q_2}}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}. \tag{5.101}$$

Observe now that under the condition that D_1 reduce to only one vertex and

$$\sum_{z\in D_1} Q_{1,z}/\#\Gamma_z = \sum_{z\in D_1}\mathrm{Res}_z(\omega_{Q_1})/\#\Gamma_z = 0, \tag{5.102}$$

the element $(\mathrm{Res}_e(\omega_{Q_1}))_{e\in\mathcal{E}} \in C^1(V_n)$ lies in $C^1_{\mathrm{har}}(V_n)^\Gamma$ by (4.8). Since $N(j(\omega_{Q_1})) = 0$ then the harmonic part has to be trivial, concluding $(\mathrm{Res}_e(\omega_{Q_1}))_{e\in\mathcal{E}} = 0$. The first term of the pairing (5.101) is then trivial giving the simplified expression

$$(\eta_L, \Psi_{Q_2}) = -\sum_{z\in D_1}\frac{(F_{\Psi_{Q_2}}(z), Q_{1,z})_\Gamma}{\#\Gamma_z}. \tag{5.103}$$

As a sanity check, conclude observing that the formula is invariant under rescaling of the Coleman integral of ΨQ_2 since the residues of Q_1 add up to 0. □

Remark 5.2. *The explicit computation in the previous proof shows that the height pairing is symmetric. In particular, we have used the fact that the pairing between the Coleman differentials Ψ_{Q_1}, Ψ_{Q_2} is 0 in $H^1_{\mathrm{dR}}(X_\Gamma - D_1, D_2, \mathcal{V}_n) \times H^1_{\mathrm{dR}}(X_\Gamma - D_2, D_1, \mathcal{V}_n)$ as in* (5.94) *leading to the equality* (5.95)

$$(P(\Psi_{Q_1}), I_{U_2}(\Psi_{Q_2}))_\Gamma = (I_{U_1}(\Psi_{Q_1}), P(\Psi_{Q_2}))_\Gamma. \tag{5.104}$$

5.5. **Local heights away from p.** Let $V = H^1_{\text{ét}}(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G(m+1)$ be the p-adic Galois representation. We recall from [Bry89], [Sch94] and [Nek95] the known results on the local heights away from p.

Lemma 5.7. *Let F be a finite extension of $\mathbb{Q}_l$ for $l \neq p$ then*

$$H^i(F, V) = 0 \quad \textit{for } i = 0, 1, 2. \tag{5.105}$$

Proof. Let $\mathbb{T}$ be the subalgebra of $\mathrm{End}(M_k(X, \mathbb{Q}))$ generated by the Hecke operators T_l for $l \nmid N$ and U_l for $l \mid N$. The p-adic representation V is a free $\mathbb{T}_{\mathbb{Q}_p}$-module of rank 2 that decomposes in eigenspaces $V_{\mathfrak{p}}(f)$ for $f \in M_k(X, \mathbb{Q})$ eigenforms. Under the Jacquet-Langlands correspondence, Iovita and Spiess [IS03, Lemma 5.8] proved that these eigenspaces are isomorphic to p-adic representations $V_{\mathfrak{p}}(f_\infty)$ coming the associate elliptic modular form f_∞. The proof of the Lemma follows then from [Nek95, Lemma 2.12] in the elliptic modular case. □

Let now $\mathcal{X}$ be a model for $X_M \otimes K_v^{\mathrm{unr}}$ i.e. a proper flat scheme over $\mathcal{O}_{K_v}^{\mathrm{unr}}$ such that

$$\mathcal{X} \times \mathrm{Spec}(K_v^{\mathrm{unr}}) \cong X_M \otimes K_v^{\mathrm{unr}}. \tag{5.106}$$

We distinguish two cases:

- if $v \nmid \mathrm{disc}(B)$ then we consider $\mathcal{X}$ the regular model constructed by Buzzard [Buz97];
- if $v \mid \mathrm{disc}(B)$ then we consider the algebrization of the formal model described by Boutot-Carayol [BC91] of the v-adic uniformization of Čerednik–Drinfeld.

In particular, if $v \nmid NM$ then the model is smooth over $\mathcal{O}_{K_v}^{\mathrm{unr}}$. Consider once again two reduced effective divisors D_1, D_2 defined over K_1, K_2 with disjoint support. Consider $\underline{D_1}$, $\underline{D_2}$ the Zariski closure of D_1, D_2 in the model. Assume that both divisors are contained in the regular locus $\mathcal{X}_M^{\mathrm{reg}} \subset \mathcal{X}_M$. Consider two maps of Galois representations

$$\mathrm{cl}_i : \mathbb{Q}_p \to \oplus_{x \in D_i} (\mathbb{L}_n)_x^{G_x}(m) \tag{5.107}$$

and let $Q_i = \mathrm{cl}_i(1)$. The maps cl_i induce two extensions e_1, e_2 and the natural biextension E splits as p-adic local Galois representation at v as $E_v = V_v \oplus U_v$ as explained in Section 5.1.1. Assume now that there exists cohomology classes on the model $\mathcal{Q}_i \in H^2_{\text{ét},\underline{D_i}}(\mathcal{X}, \mathbb{L}_n(m+1))^G$ restricting to

$$Q_i \in H^2_{\text{ét},D_i}(X \otimes K_v^{\mathrm{unr}}, \mathbb{L}_n(m+1))^G \cong H^0_{\text{ét}}(D_i, \iota_{D_i *}\mathbb{L}_n(m))^G = \oplus_{x \in D_i} (\mathbb{L}_n)_x^{G_x}(m) \tag{5.108}$$

under the pullback to the generic fiber. We can then express the height pairing in terms of these classes.

Lemma 5.8. *Assume that the class $\mathcal{Q}_i$ has image zero in $H^2_{\text{ét}}(\mathcal{X}, \mathbb{L}_n(m+1))^G$ under the connecting morphism of the long exact sequence of relative cohomology. Then*

$$h(e_1, e_2)_{E,v} = -l_v(\pi_v) \cdot \mathrm{Tr}(\mathcal{Q}_1 \cup \mathcal{Q}_2) \tag{5.109}$$

where the cup product is an element of $H^4_{\text{ét},c}(\mathcal{X}, \mathbb{Q}_p(2))^G$ and Tr *is the trace map*

$$\mathrm{Tr} : H^4_{\text{ét},c}(\mathcal{X}, \mathbb{Q}_p(2))^G \to \mathbb{Q}_p. \tag{5.110}$$

Proof. See [Bry89, Proposition 4.10] and [Sch94]. As pointed out by [Nek95, II.2.16], there is a minus sign correction that does not appear in the work of [Bry89]. □

We would like to explicitly characterize the cup product of the extended cohomology class in terms of intersection of divisors. To do this, assume that Q_i extend to sections

$$\underline{Q_i} \in H^0_{\text{ét}}(\underline{D_i}, \iota_{D_i,*}\mathbb{L}_n(m))^G. \tag{5.111}$$

If y, z are two points of $\underline{D_1}, \underline{D_2}$, following [Nek95], we define the *local intersection index* at a point x in $\mathcal{X}$ to be

$$(\underline{Q_1}, \underline{Q_2})_x := (\underline{y}, \underline{z})_x(\underline{Q_1}_x, \underline{Q_2}_x) \tag{5.112}$$

where $(\underline{y}, \underline{z})_x$ denotes the usual intersection number on $\mathcal{X}^{\mathrm{reg}}$.

Theorem 5.3. *Assume that the cohomology classes*

$$\mathcal{Q}_i = \underline{Q_i} \cup \mathrm{cl}(\underline{D_i}) \in H^2_{\text{ét},\underline{D_i}}(\mathcal{X}, \mathbb{L}_n(m+1))^G \tag{5.113}$$

satisfy the assumption of Lemma 5.8. The local height pairing is given by

$$h(e_1, e_2)_{E,v} = -l_v(\pi_v) \cdot \sum_{x \in \underline{D_1} \cap \underline{D_2}} (\underline{Q_1}, \underline{Q_2})_x \tag{5.114}$$

Proof. See [Nek95, II.2.16] and [Sch94], [Bry89, Theorem 4.3]. □

Remark 5.3. *The picture simplifies when the global cohomology group $H^2_{\text{ét}}(\mathcal{X}, \mathbb{L}_n(m+1)) = 0$. In this case, the assumption of Lemma 5.8 is satisfied implying that the local height computation simplifies to the expression* (5.114)*. Under smooth proper base change, this is always the case when v is a place of good reduction. The non-vanishing of the global H^2 is the same obstacle that the explicit local height with trivial coefficients has when dealing with bad reduction. In that case, a correction by a vertical divisor needs to be introduced.*

5.5.1. *Local intersection on v-adic uniformized curve.* To study the global cohomology we will adapt some of the arguments from [KM85, 14.5.5]. Consider $l | \mathrm{disc}(B)$ a prime different from p and let v a place above l. Thanks to Čerednik-Drinfeld Theorem 3.2 we can explictly characterize the bad reduction of the special fiber $\overline{C} = \mathcal{X} \times \mathrm{Spec}(\overline{\mathbb{F}_l})$

$$\begin{array}{ccc} \overline{C} & \xhookrightarrow{\quad j \quad} & \mathcal{X} \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} \\ \mathrm{Spec}(\overline{\mathbb{F}_l}) & \hookrightarrow & \mathrm{Spec}(\mathcal{O}_v^{\mathrm{unr}}). \end{array} \tag{5.115}$$

In particular, the irreducible components C_i of $\overline{C}$ are isomorphic to $\mathbb{P}^1_{\overline{\mathbb{F}}_l}$. By the proper base change theorem and the property of strictly henselian bases, we have

$$H^2_{\text{ét}}(\mathcal{X}, \mathbb{L}_n(m+1)) \cong (R^2 g_* \mathbb{L}_n(m+1))_v \cong H^2_{\text{ét}}(\overline{C}, j^* \mathbb{L}_n(m+1)). \tag{5.116}$$

Consider now the normalization of the special fiber

$$\nu : \coprod_j C_j \to \overline{C}, \tag{5.117}$$

this gives us a long exact sequence in cohomology that identifies the second étale cohomology

$$H^2_{\text{ét}}(\overline{C}, \mathbb{L}_n(m+1)) \cong \bigoplus_j H^2_{\text{ét}}(C_j, \mathbb{L}_n(m+1)). \tag{5.118}$$

Under Poincaré duality, the last term can be identified with

$$\bigoplus_j H^0_{\text{ét}}(C_j, \mathbb{L}_n(m)) \cong \bigoplus_j \mathbb{L}_n(m)_{z_j} \tag{5.119}$$

where the z_j are points on distinct components of C_j. The following Proposition will help us simplify the computation of the local height pairing in the case of Heegner cycles.

Proposition 5.1. *Let $\underline{Q_i} \in H^0_{\text{ét}}(\underline{D_i}, \iota_{D_i,*}\mathbb{L}_n(m))^G$ the extended cohomology class supported on the divisor $\underline{D_i}$. Supposed that the specialization on ever irreducible components C_j satisfies the property*

$$\sum_{\overline{x} \in C_j} \underline{Q_i}_x / \#\Gamma_{z_j} = 0. \tag{5.120}$$

Then the cohomology class $\mathcal{Q}_i$ defined by (5.113) *satisfies the assumption of Lemma 5.8.*

Proof. Under the previous characterization the connecting map in relative cohomology is given by (5.121)

$$H^1_{\text{ét}}(\mathcal{X}-\underline{D}_i,\mathbb{L}_n(m+1))^G\to H^2_{\text{ét},\underline{D}_i}(\mathcal{X},\mathbb{L}_n(m+1))^G\to H^2_{\text{ét}}(\mathcal{X},\mathbb{L}_n(m+1))^G\cong\bigoplus_j\mathbb{L}_n(m)_{z_j}^{\Gamma_{z_j}}$$

where the right most map sends

$$\mathcal{Q}_i\mapsto\left(\sum_{\overline{x}\in C_j}\underline{Q_i}_x/\#\Gamma_{z_j}\right)_j. \tag{5.122}$$

From this we deduce that the cohomology class is map to zero exactly when the weighted sum of the stalks is 0. □

5.5.2. *Intersection of Heegner cycles.* Recall the Heegner hypothesis described in Section 3.8. The imaginary quadratic field K of discriminant D_K satisfies the following properties:

- $(D_K,N)=1$,
- all prime divisors of N^-p are *inert* in K,
- all prime divisors of $N+$ are *split* in K.

This in particular implies that the fake elliptic curve A associated with an Heegner point is given by the product of two copies of an elliptic curve E with complex multiplication by $\mathcal{O}_K$. The fake elliptic curve A has then ordinary reduction for every $l\mid N^+$ and supersingular reduction for every $l|N^-p$. Recall that two CM elliptic curve associated to two different quadratic imaginary fields can reduce modulo v to the same elliptic curve only if their reduction is supersingular. This implies that if we want to compute the intersection of two Heegner cycles $\mathrm{cl}_{\psi_1},\mathrm{cl}_{\psi_2}$, we only need to consider the cases $l\nmid pN^-N^+$ and $l\mid N^-$.

- In the first case, for a suitable choice of M, $\mathcal{X}$ is smooth and by Remark 5.3 then the cycles $\mathcal{Q}_i$ described in Theorem 5.3 satisfy the hypothesis of Lemma 5.8 giving us the explicit formula of Theorem 5.3.
- In the case $l\mid N^-$ we can use once again the l-adic uniformization described by Boutot–Carayol [BC91] and the description of Section 5.5.1.

Consider the latter case. Fix $\mathrm{cl}_{\psi_1},\mathrm{cl}_{\psi_2}$ the two Heegner cycles defined by the cycle class map (5.40) and supported on the divisors D_1,D_2

$$D_i=(P_{\psi_i})+(P_{\overline{\psi}_i}). \tag{5.123}$$

Consider $\underline{D}_1,\underline{D}_2$ the Zariski closure of the divisors on the model $\mathcal{X}$. These divisors are supported on the regular locus $\mathcal{X}^{\text{reg}}$ since K_i are inert at l and then their points are $\mathbb{Q}_{l^2}$ points. Proposition 5.1 tells us that the classes $\mathcal{Q}_i\in H^2_{\text{ét},\underline{D}_i}(\mathcal{X},\mathbb{L}_n(m+1))^G$ defined in Theorem 5.3 reduce to

$$\left(\delta_j\cdot\underline{Q_i}_{z_j}\right)_j\in H^2_{\text{ét}}(\overline{C},\mathbb{L}_n(m+1))\cong\bigoplus_j\mathbb{L}_n(m)_{z_j} \tag{5.124}$$

where δ_j is equal to 0 if $\underline{D_i}$reduces to single component C_j or it does not intersect it at all and it is equal to 1 otherwise. The classes $\mathcal{Q}_1$, $\mathcal{Q}_2$ can be constructed by modifying the divisors $\underline{D}_1,\underline{D}_2$ by vertical divisors supported on the irreducible components containing the reduction modulo v of D_1,D_2 such that $\mathcal{Q}_i$ have trivial image in $H^2_{\text{ét}}(\mathcal{X},\mathbb{L}_n(m+1))^G$

$$\mathcal{Q}_i=\underline{Q_i}\cup\mathrm{cl}(\underline{D_i})+V_i,\quad V_i\in H^2_{\text{ét}}(\overline{C},\mathbb{L}_n(m+1))\cong\bigoplus_j\mathbb{L}_n(m)_{z_j} \tag{5.125}$$

From this description we then have the explicit computation for the Heegner cycles summarized by the following Theorem.

Theorem 5.4. *Let K_1,K_2 two quadratic imaginary field with p inert and $\mathcal{Q}_i$ the cycles on the model attached to cl_{ψ_i} defined above. The local height pairing is given by*

(i) if $l\mid N^+$ then $h(e_1,e_2)_{E,v}=0$

(ii) if $l \nmid N^- N^+$ then

$$h(e_1,e_2)_{E,v} = -l_v(\pi_v)\cdot \frac{2^m}{\binom{2m}{m}} \sum_{x\in \underline{D_1}\cap \underline{D_2}} (-1)^m P_m\left(\frac{n_x}{\sqrt{D_{K_1}D_{K_2}}}\right) \tag{5.126}$$

with $n_x = -\mathrm{tr}(\psi_{1,x}(\sqrt{D_1})\psi_{2,x}(\sqrt{D_2}))$.

(iii) if $l \mid \mathrm{disc}(B)$ the same sum hold plus the vertical term given by $l_v(\pi_v)\cdot \alpha_v$ with $\alpha_v \in \mathbb{Q}$.

Proof. The case $l|N^+$ follows trivially since the support of the Heegner cycles do not intersect. In the case (ii), the pairing formula follows from [Nek95, II.3.4] and [Zha97, Proposition 3.3.1]. In the case (iii), observe that the vertical divisor V_i is given by

$$V_i = \sum_j a_j \cup \mathrm{cl}(C_j) \in H^2_{\text{ét}}(\overline{C}, \mathbb{L}_n(m+1)) \cong \bigoplus_j \mathbb{L}_n(m)_{z_j} \tag{5.127}$$

where $a_j \in \mathbb{L}_n(m)_{z_j}$ and C_j irreducible component of the special fiber $\overline{C}$. Suppose that the classes $\mathcal{Q}_i' = \mathcal{Q}_i + V_i$ have image zero in the global second cohomology group, then by Lemma 5.8 we obtain

$$\begin{aligned} h(e_1,e_2)_{E,v} &= -l_v(\pi_v)\cdot(\mathrm{Tr}(\mathcal{Q}_1\cup\mathcal{Q}_2)+\mathrm{Tr}(V_1\cup\mathcal{Q}_2)+\mathrm{Tr}(\mathcal{Q}_1\cup V_2)+\mathrm{Tr}(V_1\cup V_2)) = \\ &= -l_v(\pi_v)\cdot(\mathrm{Tr}(\mathcal{Q}_1\cup\mathcal{Q}_2)+\mathrm{Tr}(V_1\cup\mathcal{Q}_2)) \end{aligned} \tag{5.128}$$

where the simplification is analogous to the classical case of intersection of divisors on a curve where the vertical correction is needed on one of the two classes. The first term $\mathrm{Tr}(\mathcal{Q}_1\cup\mathcal{Q}_2)$ is then computed by (ii). □

5.6. **Hecke action on extensions.** Let us briefly recall the Hecke action on the motive $h(\mathcal{A}^m)$. For an integer l prime to M and N, consider $X_M(l)$ the moduli space of abelian surfaces over $\mathbb{Q}$ parametrizing abelian surfaces (A,i,C,v,λ) where the first triple is the data parametrized by X_M by Definition 3.3 and $\lambda : A \to A'$ is an isogeny of degree l. There are two finite maps

$$\begin{array}{ccccc} & & X_M(l) & & \\ & \overset{\pi_1}{\swarrow} & & \overset{\pi_2}{\searrow} & \\ X_M & & & & X_M \end{array} \tag{5.129}$$

given my the moduli action

$$\pi_1(A,i,C,v,\lambda) = (A,i,C,v), \quad \pi_2(A,i,C,v,\lambda) = (A',\lambda_*(i),\lambda_*(C),\lambda_*(v)). \tag{5.130}$$

At the level of the universal abelian surface we have a commutative diagram of the form

$$\begin{array}{ccccccc} \mathcal{A}^m & \xleftarrow{\phi_1} & \mathcal{A}^m\times_{X_M,\pi_1} X_M(l) & \xrightarrow{\psi} & \mathcal{A}^m\times_{X_M,\pi_2} X_M(l) & \xrightarrow{\phi_2} & \mathcal{A}^m \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X_M & \xleftarrow{\pi_1} & X_M(l) & = & X_M(l) & \xrightarrow{\pi_2} & X_M \end{array} \tag{5.131}$$

where the map ψ is induced by the universal isogeny and the left and right diagram are cartesian. We can then define the *m-th Hecke correspondence* to be the graph $\Gamma_{\phi_1,\phi_2\psi}$ of

$$(\phi_1, \phi_2\circ\psi) : \mathcal{A}^m \times_{X_M,\pi_1} X_M(l) \to \mathcal{A}^m. \tag{5.132}$$

Fix now D a reduced divisor on X_M defined over a number field K and a choice of a map $\mathrm{cl} : \mathbb{Q}_p \to \oplus_{x\in D}\mathbb{L}_n(m)^{G_x}$. Consider now an extension of the p-adic Galois representation $V = H^1_{\text{ét}}(X_M\otimes\overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G$ of $G_K = \mathrm{Gal}(\overline{K}/K)$

$$0 \to V \to H^1_{\text{ét}}((X_M - |D|)\otimes\overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x\in D} (\mathbb{L}_n)^{G_x}_x(m) \to 0 \tag{5.133}$$

An Hecke operator T_l then sends this sequence to the exact sequence

$$0 \to V \to H^1_{\text{ét}}((X_M - |D'|)\otimes\overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \to \bigoplus_{x\in D'} (\mathbb{L}_n)^{G_x}_x(m) \to 0 \tag{5.134}$$

with $D' = \pi_1 \circ \pi_2^{-1}(D)$.

Lemma 5.9. *Consider x a closed point of X_M, let $Q \in (\mathbb{L}_n)_x^{G_x}(m)$. Let (A, i, C, v) be the data associated to x. Then the action of the Hecke operator T_l on Q is given by*

$$T_l Q = \sum_{\substack{\mu: A' \to A \\ \deg(\mu) = l}} \mu^*(Q) \in \bigoplus_{y \in \pi_1 \pi_2^{-1}\{x\}} (\mathbb{L}_n)_y^{G_y}(m). \tag{5.135}$$

Proof. This follows verbatim from the modular curve computation treated by Nekovář [Nek95, II.2.10]. □

The previous lemma extends by linearity to a divisor D and a class map $\mathrm{cl} : \mathbb{Q}_p \to \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m)$. Fixing this data, consider now T a linear combination of Hecke operators acting by 0 on the p-adic Galois representation V of G_K, this gives us a diagram

(5.136)

$$\begin{array}{ccccccccc}
0 & \longrightarrow & V & \longrightarrow & H^1_{\text{ét}}((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle T=0} & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & V & \longrightarrow & H^1_{\text{ét}}((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & \bigoplus_{x \in D'} (\mathbb{L}_n)_x^{G_x}(m) & \longrightarrow & 0
\end{array}$$

and since the action of T on V is 0, this means that the central vertical maps factors through

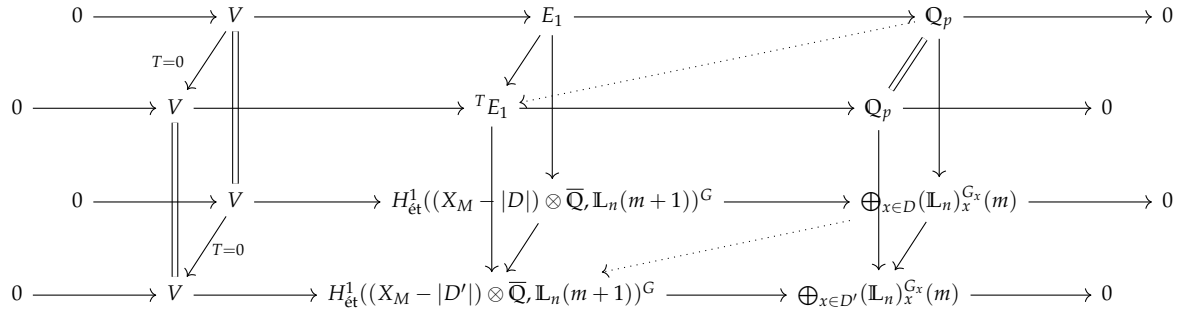
$$T : \bigoplus_{x \in D} (\mathbb{L}_n)_x^{G_x}(m) \to H^1_{\text{ét}}((X_M - |D'|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G. \tag{5.137}$$

Let e the extension given by pulling back the top exact row of the previous diagram along the class map cl

$$e : \qquad 0 \longrightarrow V \longrightarrow E_1 \longrightarrow \mathbb{Q}_p \longrightarrow 0 \tag{5.138}$$

then under the action T we can construct a split extension ${}^T e$ given by the pullback of the bottom row of the diagram (5.136) by the composition of the class map cl with the map T given by (5.137)

(5.139)



where then we have a canonical decomposition of p-adic Galois representations ${}^T E_1 = V \oplus \mathbb{Q}_p$. This picture promotes to the biextension setting. Fix now two reduced divisors D_1, D_2 with two corresponding class maps cl_i. Instead of using the divisor D_1, we consider the biextension given

by the divisors D_1' and D_2
(5.140)

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{x\in D_2}(\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H^1_{\text{ét},|D_2|}(X_M\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & H^1_{\text{ét}}(X_M\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \oplus_{x\in D_2}(\mathbb{L}_n)_x^{G_x}(m+1) & \longrightarrow & H^1_{\text{ét},|D_2|}((X_M-|D_1'|)\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & H^1_{\text{ét}}((X_M-|D_1'|)\otimes\overline{\mathbb{Q}},\mathbb{L}_n(m+1))^G & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \oplus_{x\in D_1'}(\mathbb{L}_n)_x^{G_x}(m) & = & \oplus_{x\in D_1'}(\mathbb{L}_n)_x^{G_x}(m) & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}.$$

We can now take the push-out along the dual of cl_2 and the pullback along the composition $T\circ\mathrm{cl}_1$ as described in the previous extension (5.139) obtaining

(5.141)
$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2^\vee(1) & \longrightarrow & V & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & {}^TE & \longrightarrow & {}^TE_1 & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \mathbb{Q}_p & = & \mathbb{Q}_p & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}.$$

Pulling back the central horizontal map under the dotted section gives us a simple extension

$$u: \quad 0\to\mathbb{Q}_p(1)\to U\to\mathbb{Q}_p\to 0. \tag{5.142}$$

We conclude observing that under the exact sequence of global Selmer group

$$0\to H^1_{\text{st}}(K,\mathbb{Q}_p(1))\to H^1_{\text{st}}(K,E_2^\vee(1))\to H^1_{\text{st}}(K,V)\to 0 \tag{5.143}$$

the class $[{}^TE]$ is sent to the class $[{}^TE_1]=[V\oplus\mathbb{Q}_p]=0$ that means that the class $[{}^TE]$ comes from $[U]\in H^1_{\text{st}}(K,\mathbb{Q}_p(1))$ as defined in equation (5.142).

Proposition 5.2. *Suppose that D_1 reduces to a unique vertex and $\sum_{z\in D_1}Q_{1,z}/\#\Gamma_z=0$. The local p-adic height of the biextension $[{}^TE]$ can be computed by*
(5.144)

$$h({}^Te_1,e_2)_{{}^TE,v}=\mathrm{Tr}_{K_v/\mathbb{Q}_p}\left(\int_{\pi_1\pi_2^{-1}D_1}(\Psi_{Q_2},TQ_1)dz\right)=\mathrm{Tr}_{K_v/\mathbb{Q}_p}\left(\sum_{z\in\pi_1\pi_2^{-1}D_1}\frac{(F_{\omega_{Q_2}}(z),(TQ_1)_z)_\Gamma}{\#\Gamma_z}\right).$$

Proof. The result follows from Theorem 5.2 observing that by linearity we can decompose the height along $D_1'=\pi_1\pi_2^{-1}D_1=D_1^{(1)}+\cdots D_1^{(r)}$ where $D_1^{(i)}$ reduce to a unique vertex and for each i we have $\sum_{z\in D_1^{(i)}}(TQ_1)_z/\#\Gamma_z=0$. □

6. Higher p-adic Green functions

6.1. **Poincaré series.** Following [BF25], we will define Poincaré series in this section, whose periods compute heights of Heegner cycles.

Fix $w_1, w_2 \in \mathcal{H}_p$ such that $\mathrm{Stab}_\Gamma(w_1) = \mathrm{Stab}_\Gamma(w_2)$ and $i, j \in \mathbb{N}$ such that $i + j = n$, define the Poincaré series

$$\Psi_\Gamma^{i,j}(z; w_1, w_2) := \sum_{\gamma \in \Gamma} \frac{1}{j(\gamma, z)^k} \cdot \frac{w_1 - w_2}{(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}}. \tag{6.1}$$

To prove convergence of this power series, we introduce some preliminary definitions. For $z \in \mathcal{H}_p$, define

$$\mathrm{Im}(z) = \inf\{|z - t|_p | t \in \mathbb{Q}_p\}.$$

Given two points $x, y \in \mathcal{H}_p$, let $d_{\mathcal{T}_p}(x, y)$ denote the distance on $\mathcal{T}_p$ between the reduction vertices $\mathrm{red}_p(x)$ and $\mathrm{red}_p(y)$.

Lemma 6.1. *If $x, y \in \mathcal{H}_p$ reduce to different vertices of $\mathcal{T}_p$, then*

$$\mathrm{d}_{\mathcal{T}_p}(x, y) = v_p\left(\frac{|x - y|_p^2}{\mathrm{Im}(x)\mathrm{Im}(y)}\right)$$

Proof. This is [Wei18, Lemma 2.4] □

Lemma 6.2. *If $x \in \mathcal{H}_p$, then for every $\gamma \in \mathrm{GL}_2(\mathbb{Q}_p)$, we have*

$$\mathrm{Im}(\gamma \cdot x) = \det(\gamma)\frac{\mathrm{Im}(x)}{|j(\gamma, x)|^2}$$

Proof. This is [GR96, (1.1.5) Lemma]. □

Proposition 6.1. *The infinite sum $\Psi_\Gamma^{i,j}(z; w_1, w_2)$ converges to a rigid meromorphic modular form of weight k for Γ.*

Proof. Fix $z \in \mathcal{H}_p$. The lemmas above imply that

$$v_p(j(\gamma, z)^k(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}) = \\ \frac{-1}{2}((i+1)\mathrm{d}_{\mathcal{T}_p}(\gamma z, w_1) + (j+1)\mathrm{d}_{\mathcal{T}_p}(\gamma z, w_2) + v_p(\mathrm{Im}(w_1)^{i+1} \cdot \mathrm{Im}(w_2)^{j+1}))$$

for all but finitely many $\gamma \in \Gamma$. Since Γ acts discretely on $\mathcal{H}_p$, it follows that for each $N \in \mathbb{N}$ there exist only finitely many $\gamma \in \Gamma$ such that

$$v_p(j(\gamma, z)^k(\gamma z - w_1)^{i+1}(\gamma z - w_2)^{j+1}) \geq -N.$$

Hence, the sum converges. The modularity of the resulting function is automatic by its definition as a Poincaré series. □

The calculation of the residues of the differential $\Psi_\Gamma^{i,j}(z)(X - zY)^n\, dz$ follows precisely as in [BF25, Corollary 3.11].

Lemma 6.3. *For all $i, j \geq 0$, such that $i + j = n$, let Γ_{w_1,w_2} be the stabilizer in Γ of the pair (w_1, w_2) and*

$$\kappa_{\Gamma,w_1,w_2}^{i,j} := \sum_{\gamma \in \Gamma_{w_1,w_2}} \frac{1}{j(\gamma, w_1)^i j(\gamma, w_2)^j}.$$

Then

$$\mathrm{Res}_{z=w_1}\left(\Psi_\Gamma^{i,j}(z; w_1, w_2)(X - zY)^k\, dz\right) = \frac{(-1)^i \kappa_{\Gamma,w}^{i,j}}{(w_1 - w_2)^n}\binom{n}{i}(X - w_2Y)^i(X - w_1Y)^j.$$

Remark 6.1. *The value $\kappa_{\Gamma,w_1,w_2}^{i,j}$ is either equal to 0 or $|\Gamma_{w_1,w_2}|$.*

Let $\tilde{w}_1, \tilde{w}_2$ be any two points belonging to $X_\Gamma(\mathbb{Q}_{p^2})$ and define the divisor $D_2 = (\tilde{w}_1) + (\tilde{w}_2)$. Let $U_2 = X_\Gamma - \{\tilde{w}_1, \tilde{w}_2\}$. Assume further that both $\tilde{w}_1$ and $\tilde{w}_2$ reduce modulo p to smooth points lying on the same irreducible component of the special fiber of $\mathfrak{X}_\Gamma$. It follows that there exists lifts $w_1, w_2 \in \mathcal{H}_p \cap \mathbb{Q}_{p^2}$ of $\tilde{w}_1$ and $\tilde{w}_2$, respectively, such that

$$\mathrm{red}_p(w_1) = \mathrm{red}_p(w_2)$$

are both equal to a vertex of $\mathcal{T}_p$. We fix such lifts.

The differential $\Psi_\Gamma^{i,j}(z;w_1,w_2)(X-zY)^n\,dz$ descends to a cohomology class in the filtration

$$F_{U_2}^{m+1} \subset H^1_{\mathrm{dR}}(U_2,\mathcal{V}_n).$$

Lemma 6.4. *The differential*

$$\Psi_\Gamma^{i,j}(z)(X-zY)^n\,dz$$

is a Coleman–Gross differential (Definition 5.2).

Proof. It suffices to show that

$$N\left((\Psi_\Gamma^{i,j}(z)(X-zY)^n\,dz\right)=0.$$

Observe that $\Psi_\Gamma^{i,j}$ can be written as a limit of truncations

$$\Psi_\Gamma^{i,j}(z;w_1,w_2)=\lim_{d\to\infty}\left(\sum_{\gamma\in\Gamma^{\le d}}\frac{1}{j(\gamma,z)^k}\frac{w_1-w_2}{(\gamma z-w_1)^{i+1}(\gamma z-w_2)^{j+1}}\right).$$

Each of the sums inside the limit is a finite sum. Note that

$$\frac{1}{j(\gamma,z)^k}\frac{1}{(\gamma z-w_1)^{i+1}(\gamma z-w_2)^{j+1}}=\frac{1}{j(\gamma^{-1},w_1)^{i+1}j(\gamma^{-1},w_2)^{j+1}}\frac{1}{(z-\gamma^{-1}w_1)^{i+1}(z-\gamma^{-1}w_2)^{j+1}}.$$

Since $\gamma^{-1}w_1$ and $\gamma^{-1}w_2$ both reduce to the same vertex of $\mathcal{T}_p$ and have opposite residues, it follows that for every edge $e\in\mathcal{T}_p^1$,

$$\mathrm{res}_e\left(\frac{1}{(z-\gamma^{-1}w_1)^{i+1}(z-\gamma^{-1}w_2)^{j+1}}(X-zY)^n\,dz\right)=0.$$

Therefore,

$$\mathrm{res}_e(\Psi_\Gamma^{i,j}(z)(X-zY)^n\,dz)=0,$$

and so the result follows. □

It follows from Lemmas 6.3 and 6.4 that

$$\Psi_\Gamma^{i,j}(z;w_1,w_2)(X-zY)^n dz=\Psi_{Q_2},$$

where

$$Q_2^{i,j}=\frac{(-1)^i\kappa_{\Gamma,w_1,w_2}^{i,j}}{(w_1-w_2)^n}\binom{n}{i}\left((X-w_2Y)^i(X-w_1Y)^j,-(X-w_2Y)^i(X-w_1Y)^j\right)\in(\mathcal{V}_n)_{w_1}^{\Gamma_{w_1}}\oplus(\mathcal{V}_n)_{w_2}^{\Gamma_{w_2}}.$$

6.2. **Periods of Poincaré series.** To obtain an explicit description of a primitive of $\Psi_\Gamma^{i,j}(z;w_1,w_2)(X-zY)^n\,dz$, we describe here how to relate such a primitive to a primitive of the weight k meromorphic modular form $\Psi_\Gamma^{i,j}(z)$.

Lemma 6.5. *For any locally meromorphic function, $F(z)$, on $\mathcal{H}_p$. Let $F^{(l)}(z)$ denote the l-th derivative of F, and let*

$$\beta(F)(z):=n!\sum_{l=0}^{n}\frac{F^{(l)}(z)}{l!}(X-zY)^lY^{n-l}.$$

It holds that

$$\frac{d}{dz}\beta(F)(z)=F^{(n+1)}(z)(X-zY)^n.$$

Proof. The left hand side is a telescoping sum that reduces to the right hand side. □

Remark 6.2. *Note that one can recover F from $\beta(F)$ by "plugging in $X=z,Y=1$," or equivalently through the pairing*

$$F(z)=\left\langle\beta(F)(x),(X-zY)^n\right\rangle.$$

Lemma 6.6. *For $i, j \geq 0$ and $n = i + j$, let*

$$F_{i,j}(z; w_1, w_2) = \frac{(z - w_1)^j (z - w_2)^i}{(w_1 - w_2)^n} \log_{\mathscr{L}} \left(\frac{z - w_1}{z - w_2} \right),$$

then

$$F_{i,j}^{(n+1)}(z; w_1, w_2) = (-1)^i i! j! \frac{w_1 - w_2}{(z - w_1)^{i+1} (z - w_2)^{j+1}}$$

Proof. This statement holds by induction on $i + j = n$. □

Remark 6.3. *It follows that a Coleman primitive of*

$$\frac{w_1 - w_2}{(z - w_1)^{i+1} (z - w_2)^{j+1}} (X - zY)^n \, dz$$

is given by $\frac{(-1)^i}{i!j!} \beta(F_{i,j}(z; w_1, w_2))$.

Proposition 6.2. *Let $\xi \in \mathcal{H}_p \backslash (\Gamma \cdot \{w_1, w_2\})$ be an arbitrary point, then the infinite sum*

$$F_{i,j}^{\Gamma}(z; w_1, w_2) = \frac{(-1)^i}{i! j!} \sum_{\gamma \in \Gamma} \left(j(\gamma, z)^n F_{i,j}(\gamma z; w_1, w_2) - \sum_{i=0}^{n} \frac{d^i}{dx^i} \left[j(\gamma, x) F_{i,j}(\gamma x; w_1, w_2) \right]_{x = \xi} \frac{(z - \xi)^n}{i!} \right)$$

converges and satisifies

$$\left(F_{i,j}^{\Gamma} \right)^{(n+1)} (z) = \Psi_{\Gamma}^{i,j}(z).$$

Hence $\beta(F_{i,j}^{\Gamma})$ is a Coleman primitive of $\Psi_{\Gamma}^{i,j}(z)(X - zY)^n \, dz$, i.e.

$$\beta(F_{i,j}^{\Gamma})(z) = \int_{\xi}^{z} \Psi_{\Gamma}^{i,j}(x; w_1, w_2)(X - xY)^n \, dz.$$

Proof. The terms of the infinite sum are equal to

$$\left\langle \int_{\xi}^{z} \frac{1}{j(\gamma, x)^k} \cdot \frac{w_1 - w_2}{(\gamma x - w_1)^{i+1} (\gamma x - w_2)^{j+1}} (X - xY)^n \, dx, \ (X - zY)^n \right\rangle.$$

A primitive of $\Psi_{\Gamma}^{i,j}(z; w_1, w_2)$ under iterated integration $(n+1)$-times is given by

$$\left\langle \int_{\xi}^{z} \Psi_{\Gamma}^{i,j}(x; w_1, w_2)(X - xY)^n \, dx, (X - zY)^n. \right\rangle$$

The result follows since Coleman integrals commute with uniformly convergent infinite sums. □

Proposition 6.3. *Given any two points $z_1, z_2 \in \mathcal{H}_p \backslash (\Gamma \cdot \{w_1, w_2\})$, and some $Q(X, Y) \in \mathcal{V}_n$, the pairing*

$$(\beta(F_{i,j}^{\Gamma}(z; w_1, w_2)|_{z=z_1}, Q(X, Y)) - (\beta(F_{i,j}^{\Gamma}(z; w_1, w_2)|_{z=z_2}, Q(X, Y))$$

is equal to

$$\frac{(-1)^i}{i! j!} \sum_{\gamma \in \Gamma} \left\langle \beta(F_{i,j}(z; w_1, w_2))|_{z=\gamma z_1} - \beta(F_{i,j}(z; w_1, w_2))|_{z=\gamma z_2}, Q(X, Y) \right\rangle.$$

Proof. Exactly as before, note that both sides are computing

$$\left\langle \int_{z_2}^{z_1} \Psi_{\Gamma}^{i,j}(x; w_1, w_2)(X - xY)^n \, dx, Q(X, Y) \right\rangle,$$

since integration commutes with uniformly convergent infinite sums. □

Definition 6.1. *Given any distinct points $w_1, w_2, z_1, z_2 \in \mathcal{H}_p$ and $i, j, r, s \in \mathbb{N}$ such that $i + j = r + s = n$, define*

$$g_{r,s}^{i,j}(z_1, z_2; w_1, w_2) = \left\langle \beta(F_{i,j}(z; w_1, w_2))|_{z=z_1} - \beta(F_{i,j}(z; w_1, w_2))|_{z=z_2}, * * \frac{(X - z_1 Y)^r (X - z_2 Y)^s}{} \right\rangle.$$

Definition 6.2. *Define the p-adic Legendre function of the second kind by*

$$Q^{\mathscr{L}}_{n/2}(x) = \left\langle \beta(F_x)|_{z=1} - \beta(F_x)|_{z=-1}\ , \frac{(X^2-Y^2)^{n/2}}{2^{n/2}} \right\rangle.$$

Lemma 6.7. *Given $\sigma_1, \sigma_2, w_1, w_2$ distinct in a finite extension $L/\mathbb{Q}_p$, there exists a unique element $\gamma \in \mathrm{GL}_2(L)$ that satisfies*

$$\gamma(\sigma_1) = -1,\ \gamma(\sigma_2) = 1, \text{ and } \gamma(w_1) = \infty.$$

The transformation by γ also satisfies

$$\gamma w_2 = 2\frac{(\sigma_2 - w_1)(\sigma_1 - w_2)}{(\sigma_2 - \sigma_1)(w_2 - w_1)} - 1.$$

Proof. This follows by equating the cross-ration $[\sigma_1, \sigma_2; w_1, w_2]$ and $[-1, 1; \infty, x]$. □

Let $x = \gamma w_2$ as above

$$\frac{(X^2-Y^2)^{n/2}}{2^n} = \frac{(X-Y)^{n/2}(X+Y)^{n/2}}{(1-(-1))^{n/2}}$$

satisfies

$$\frac{(X^2-Y^2)^{n/2}}{2^n}|_\gamma = \frac{(X-\sigma_1 Y)^{n/2}(X-\sigma_2 Y)^{n/2}}{(\sigma_1-\sigma_2)^{n/2}}.$$

Similarly

$$(\beta(F_x)|_{z=-1} - \beta(F_x)|_{z=1})\,|_\gamma = \beta(F_{n/2,n/2}(z; w_1, w_2))|_{z=\sigma_1} - \beta(F_{n/2,n/2})(z; w_1, w_2))|_{z=\sigma_2}.$$

Since the pairing on polynomials is GL_2-equivariant, it follows that

$$\left\langle \beta(F_{n/2,n/2}(z; w_1, w_2))|_{z=\sigma_1} - \beta(F_{n/2,n/2})(z; w_1, w_2))|_{z=\sigma_2}, \frac{(X-\sigma_1 Y)^{n/2}(X-\sigma_2 Y)^{n/2}}{(\sigma_1-\sigma_2)^{n/2}} \right\rangle$$
$$= Q^{\mathscr{L}}_{n/2}\left(2\frac{(\sigma_2 - w_1)(\sigma_1 - w_2)}{(\sigma_2 - \sigma_1)(w_2 - w_1)} - 1\right).$$

Let D_2 be the divisor $(\tilde{w_1}) + (w_2)$ and D_1 be the divisor $(\sigma_1) + (\sigma_2)$. Assume for simplicity that $w_1, w_2, \sigma_1, \sigma$ have trivial stabilizers. For any two elements $x, y \in \mathbb{C}_p$, define

$$c^{i,j}_{x,y} = \frac{(X-xY)^i(X-yY)^j}{(x-y)^n} \in V_n.$$

Let $Q_2 \in \mathcal{V}_{w_1} \oplus \mathcal{V}_{w_2} = V_n \oplus V_n$ be the residue class defined by

$$Q^{i,j}_2 = \left(c^{i,j}_{w_1,w_2}, -c^{i,j}_{w_1,w_2}\right).$$

Similarly, let $Q_2 \in (\mathcal{V}_n)_{\sigma_1} \oplus (\mathcal{V}_{\sigma_n})_{\sigma_2} = V_n \oplus V_n$ be the residue class defined by

$$Q^{r,s}_1 = \left(c^{r,s}_{\sigma_1,\sigma_2}, -c^{r,s}_{\sigma_1,\sigma_2}\right).$$

These choices of divisors lead to a bi-extension $E^{i,j}_{r,s}$ as described in ... . By ... the associated local p-adic height is given by

$$h_{[E^{i,j}_{r,s}]} = (F_{\Psi_{Q_2}}(\sigma_1) - F_{\Psi_{Q_2}}(\sigma_2), c^{r,s}_{\sigma_1,\sigma_2}).$$

It follows by Lemma 6.3 and 6.4 that

$$\Psi_{Q_2}(z) = \frac{(-1)^i \binom{n}{i}^{-1}}{2^{???}} \Psi^{j,i}_\Gamma(z; w_1, w_2)(X - zY)^n\, dz$$

To obtain an explicit description of the primitive $F_{\Psi_{Q_2}}$, we describe how to relate primitives of the weight k meromorphic modular form $\Psi^{i,j}_\Gamma(z)$ to primitives of the vector valued differential $\Psi^{i,j}_\Gamma(z)(X - zY)^n\, dz$.

Theorem 6.1. *The local height pairing for the bi-extension $E^{i,j}_{r,s}$ is given explicitly by*

$$\begin{aligned} h_{[E]^{i,j}_{r,s}} &= \frac{1}{2^{???}n!}\sum_{\gamma\in\Gamma} c^{r,s}_{\sigma_1,\sigma_2}\left(\beta(F_{i,j}(\gamma z;w_1,w_2))|_{z=\sigma_1} - \beta(F_{i,j}(\gamma z;w_1,w_2))|_{z=\sigma_2}\right) \\ &= \sum_{\gamma\in\Gamma}\left(\sum_{l=0}^{n}\frac{1}{l!}\left(\frac{d^l}{dz^l}\left[\frac{(\gamma z-w_1)^i(\gamma z-w_2)^j}{(w_1-w_2)^n}\log_{\mathscr{L}}\left(\frac{\gamma z-w_1}{\gamma z-w_2}\right)\right]_{z=\sigma_1}(X-\sigma_1 Y)^l\right.\right. \\ &\left.\left. - \frac{d^l}{dz^l}\left[\frac{(\gamma z-w_1)^i(\gamma z-w_2)^j}{(w_1-w_2)^n}\log_{\mathscr{L}}\left(\frac{\gamma z-w_1}{\gamma z-w_2}\right)\right]_{z=\sigma_2}(X-\sigma_2 Y)^l\right), \frac{(X-\sigma_1Y)^r(X-\sigma_2Y)^s}{(\sigma_1-\sigma_2)^n}\right). \end{aligned}$$

Proof. For every V_n-valued meromorphic differential, $\omega = f(z)(X-zY)^n dz$, on $\mathcal{H}_p$, let $F(\omega)$ denote a choice of Coleman primitive of ω. For every $P\in\mathcal{P}_n$, the value given by

$$\left(F(\sigma_1)-F(\sigma_2), P^\vee\right),$$

is equal to the value given by the Morita pairing

$$\int_{\mathbb{P}^1(\mathbb{Q}_p)} P(t,1)\log_{\mathscr{L}}\left(\frac{t-\sigma_1}{t-\sigma_2}\right)d\mu_f(t),$$

where μ_f is the measure on $\mathbb{P}^1(\mathbb{Q}_p)$ associated to f Cite Teitelbaum. The Morita pairing is a *topological* duality. It follows that the local height

$$\left(F_{\Psi_{\mathbb{Q}_2}}(\sigma_1) - F_{\Psi_{\mathbb{Q}_2}}(\sigma_2), \frac{(X-\sigma_1Y)^r(X-\sigma_2Y)^s}{(\sigma_1-\sigma_2)^n}\right)$$

can be computed term by term inside the convergent sum (6.1) by

$$\left(\sum_{\gamma\in\Gamma}\left[F\left(\frac{1}{j(\gamma,z)^k}\cdot\frac{w_1-w_2}{(\gamma z-w_1)^{i+1}(\gamma z-w_2)^{j+1}}(X-zY)^n dz\right)\Big|_{z=\sigma_1}\right.\right. \\ \left.\left. - F\left(\frac{1}{j(\gamma,z)^k}\cdot\frac{w_1-w_2}{(\gamma z-w_1)^{i+1}(\gamma z-w_2)^{j+1}}(X-zY)^n dz\right)\Big|_{z=\sigma_2}\right], \frac{(X-\sigma_1Y)^r(X-\sigma_2Y)^s}{(\sigma_1-\sigma_2)^n}\right).$$

The result follows by Remark 6.3. □

For $x\in\mathbb{C}_p$, let

$$F_x(z) = (-1)^{n/2}\left(\left(\frac{n}{2}\right)!\right)^2(z-x)^{n/2}\log_{\mathscr{L}}(z-x).$$

7. A p-adic Gross–Kohnen–Zagier result

In this section we present a p-adic analogue of the Gross–Kohnen–Zagier theorem [GKZ87] of the log-rationality of sums of higher Green functions evaluated at Heegner points. Let $\mathcal{G}_k$ denote the higher p-adic Green function constructed in Section 6, with the branch $\log_{\mathscr{L}}$ and the normalization conventions fixed there.

Theorem 7.1 (p-adic Gross–Kohnen–Zagier)**.** *Fix the Shimura curve X and the positive even integer $k\geq 4$. Consider a non-zero finite linear combination of Hecke operators*

$$T = \sum_{r\geq 1} c_r T_r, \qquad c_r\in\mathbb{Q},$$

which kills V. Let z_0, w_0 be CM points associated to imaginary quadratic fields K_1, K_2 with coprime discriminants satisfying the Heegner hypothesis and the local hypotheses of Sections 5 and 6. Consider the Heegner biextension described in Section 5.3. If H_i denotes the Hilbert class field of K_i, then the sum over the Galois orbits of Heegner points z_0, w_0 of the evaluation of the higher Green's function is equal to

$$\mathrm{Tr}_{H_v/\mathbb{Q}_p}\left(\sum_{\sigma,\tau}\mathcal{G}_k(T(\sigma\cdot z_0),\tau\cdot w_0)\right) = C_{k,K_1,K_2}\log_p(\alpha_T) \tag{7.1}$$

with $\alpha_T\in\mathbb{Q}$, $\sigma\in\mathrm{Gal}(H_1/K_1)$, $\tau\in\mathrm{Gal}(H_2/K_2)$ and an explicit constant C_{k,K_1,K_2}.

Proof. Let T be a combination of Hecke operators annihilating V, let $H = H_1H_2$, and let E be the biextension considered in Section 5. By Section 5.6, the transformed biextension $[{}^TE]$ reduces to an element

$$[U] \in \mathrm{Ext}^1_{\underline{\mathrm{Rep}}_{G_H}}(\mathbb{Q}_p, \mathbb{Q}_p(1)) = H^1(H, \mathbb{Q}_p(1)) \cong H^\times \hat{\otimes} \mathbb{Q}_p \tag{7.2}$$

where the last isomorphism is the Kummer isomorphism (5.21). Let $\alpha \in H^\times \hat{\otimes} \mathbb{Q}_p$ denote the class corresponding to $[U]$. With the local sign convention fixed in Section 5, the height identity is

$$\sum_{v|p} h({}^Te_1, e_2)_{E,v} = -\sum_{v|p} l_v(\alpha) = \sum_{v\nmid p} l_v(\alpha) = -\sum_{v\nmid p} \mathrm{ord}_v({}^TE)\log_p(N_{H/\mathbb{Q}}(v)), \tag{7.3}$$

where the second equality follows from the global relation among the local characters. Now observe that by the compatibility with the Galois action on the places we have

$$h(e_1^\sigma, e_2^\sigma)_{\sigma(E),\sigma(v)} = h(e_1, e_2)_{E,v} \quad \text{for all } \sigma \in \mathrm{Gal}(H/\mathbb{Q}). \tag{7.4}$$

Since the discriminants are coprime, we obtain that $H^1 \cap H^2 = \mathbb{Q}$ and the Galois group is given by the product of the two dihedral groups associated with H_1, H_2. Recall that p is inert in K_1, K_2 so it is totally split on the Hilbert class fields H_1, H_2 so there are $[H:\mathbb{Q}]/2$ places above p. Fixing a place v, the formula then becomes

$$\sum_{v|p} h({}^Te_1, e_2)_{E,v} = \frac{1}{2}\sum_{\sigma\in\mathrm{Gal}(H/\mathbb{Q})} h({}^Te_1^\sigma, e_2^\sigma)_{\sigma(E),v} \tag{7.5}$$

In particular, the right sum amounts of taking a sum over a Galois orbit of the Heegner cycles in $\mathrm{emb}(\mathcal{O}_{K_1}, R)$, $\mathrm{emb}(\mathcal{O}_{K_2}, R)$ under the compatibility of the Galois action described in Section 3.8.1. Now observe that by Theorem 5.4 the local contribution away from p is given by

$$\mathrm{ord}_v({}^TE) = \sum_{\underline{x}\in\underline{TD_1}\cap\underline{D_2}} (\underline{Q_1}, \underline{Q_2})_x + \alpha_{v,T}. \tag{7.6}$$

We can then rewrite the equation (7.3)

$$\mathrm{Tr}_{H_v/\mathbb{Q}_p}\left(\sum_{\sigma,\tau} \mathcal{G}_k(T(\sigma\cdot)z_0, \tau\cdot w_0)\right) = \sum_{\sigma\in\mathrm{Gal}(H/\mathbb{Q})} h({}^Te_1^\sigma, e_2^\sigma)_{E,v} = \tag{7.7}$$

$$= \frac{(-1)^{m+1}2^{m+1}}{\#\Gamma_{z_0}\#\Gamma_{w_0}\binom{2m}{m}} \sum_{v\nmid p}\log_p(N_{H/\mathbb{Q}}(v)) \left(\sum_{x\in\underline{TD_1}\cap\underline{D_2}} P_m\left(\frac{n_x}{\sqrt{D_{K_1}D_{K_2}}}\right) + \alpha_{v,T}\right) \tag{7.8}$$

concluding the proof.

□

Remark 7.1. *The explicit factorization of the rational number could be studied in terms of local deformation theory for the intersection pairing of the two Heegner divisors as done by* [Zha97].

Remark 7.2. *The motivic conjecture of Brown and Fonseca would give a second route. It predicts that α lies in*

$$\mathrm{Ext}^1_{DM}(\mathbb{Q}, \mathbb{Q}(1)) = H^1(H, \mathbb{Q}(1)) \cong H^\times \otimes \mathbb{Q}.$$

Appendix A. Legendre functions

Recall that

$$\beta(F) = \sum_{i=0}^{n} \frac{F^{(i)}(z)}{i!}(X - zY)^i,$$

and $\langle\cdot,\cdot\rangle$ is the pairing on P_n satisfies the identity

$$\langle (X - xY)^i Y^{n-i}, P(X,Y)\rangle = (-1)^{n-i}\frac{i!}{n!}\frac{d^{n-i}}{dx^{n-1}}P(x,1).$$

For $x \in \mathbb{C}_p$, define

$$F_x(z) = (-1)^{n/2}\left(\left(\frac{n}{2}\right)!\right)^2 (z-x)^{n/2}\log_{\mathscr{L}}(z-x).$$

Definition A.1. *Define the p-adic Legendre function of the second kind by*

$$Q^{\mathscr{L}}_{n/2}(x) = \left\langle \beta(F_x)|_{z=1} - \beta(F_x)|_{z=-1}\ , \frac{(X^2-Y^2)^{n/2}}{2^{n/2}} \right\rangle.$$

The complex Legendre function $Q_{n/2}(x)$ is given by

$$Q_{n/2}(x) = \mathcal{P}_{n/2}(x)\log\left(\frac{x+1}{x-1}\right) + W_{n/2-1},$$

where $\mathcal{P}_{n/2}$ and $W_{n/2-1}$ are polynomials of degrees $n/2$ and $n/2-1$, respectively. The polynomials $\mathcal{P}_{n/2}$ are the Legendre polynomials; they are given by

$$\mathcal{P}_{n/2}(x) = \frac{1}{2^{n/2}(n/2)!}\frac{d^{n/2}}{dx^{n/2}}\left((x^2-1)^{n/2}\right).$$

The polynomials $W_{n/2-1}$ are uniquely determined by the fact that $Q_{n/2}(x)$ is a locally analytic function that vanishes to order $n/2+1$ as $x \to \infty$. That is, $W_{n/2-1}$ is precisely equal to the polynomial part of the expansion of $\mathcal{P}_{n/2}(x)\log\left(\frac{x+1}{x-1}\right)$ in a neighborhood of ∞.

Proposition A.1. *We have*

$$Q^{\mathscr{L}}_{n/2} = \mathcal{P}_{n/2}(x)\log_{\mathscr{L}}\left(\frac{x+1}{x-1}\right) + W_{n/2-1}.$$

Proof. Since

$$d^{(n+1)}\left((z-x)^{n/2}\log_{\mathscr{L}}(z-x)\right) = \frac{1}{(z-x)^{n/2+1}}.$$

It follows that

$$\frac{d}{dx}(\beta(F_x)) = \frac{(X-zY)^n}{(z-x)^{n/2+1}}\,dz.$$

Integrating both sides yields the following equality (modulo polynomials constant in z)

$$\begin{aligned}\beta(F_x) =&(-1)^{n/2}\binom{n}{n/2}\log_{\mathscr{L}}(z-x)(X-xY)^{n/2}Y^{n/2}\\ &+\sum_{\substack{0\le i\le n\\ i\neq n/2}}(-1)^i\binom{n}{i}\frac{(z-x)^{i-n/2}}{i-n/2}(X-xY)^{n-i}Y^i\end{aligned}$$

So we get the following expression:

$$Q^{\mathscr{L}}_{n/2}(x) = \frac{1}{2^{n/2}(n/2)!}\frac{d^{n/2}}{dx^{n/2}}\left((x^2-1)^{n/2}\right)\log_{\mathscr{L}}\left(\frac{x-1}{x+1}\right) \tag{A.1}$$

$$+\sum_{\substack{0\le i\le n\\ i\neq n/2}}\frac{1}{2^{n/2}i!(i-n/2)}\left[(1-x)^{i-n/2}-(-1-x)^{i-n/2}\right]\frac{d^i}{dx^i}\left((x^2-1)^{n/2}\right). \tag{A.2}$$

Note that (A.2) is in fact a polynomial of degree at most $n/2-1$. It remains therefore to show that $Q^{\mathscr{L}}_{n/2}(x)$ is a locally analytic function in x that vanishes to order $n/2+1$ as x tends to ∞. This holds because $\frac{(X-zY)^n}{(z-x)^{n/2+1}}$ vanishes to order $n/2+1$ as x tends to ∞, as is argued below.

For x close enough to ∞, there exists an open neighborhood $U \subset \mathbb{C}_p$ containing $1, -1$, on which

$$\frac{(X-zY)^n}{(z-x)^{n/2+1}}$$

is analytic and can be written as

$$\frac{(X-zY)^n}{(z-x)^{n/2+1}} = \frac{1}{x^{n/2+1}}\left(\sum_{j=0}^{\infty}\binom{-(n/2+1)}{j}(-1)^{n/2+1-j}\frac{z^j}{x^j}\right)\left(\sum_{i=0}^{n}(-1)^i\binom{n}{i}(X)^{n-i}(zY)^i\right)$$

$$= \frac{1}{x^{n/2+1}} \sum_{j=0}^{n} \sum_{i=0}^{\infty} c_{i,j} z^i X^j Y^{n-j},$$

where the coefficients $c_{i,j}$ are bounded. By formally integrating this power series, we have

$$\beta(F_x) = \frac{1}{x^{n/2}+1} \sum \frac{c_{i,j}}{i+1} z^{i+1} X^j Y^{n-j}.$$

It follows that $\beta(F_x)|_{z=1} - \beta(F_x)|_{z=-1}$ vanishes to order $n/2+1$ as $x \to \infty$, and therefore so does $Q^{\mathscr{L}}_{n/2}(x)$.

□

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